

Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]

    cols = list(dict.fromkeys(target_col + features_X +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))

    df = df[cols]
    #df['campaign'] = "ALL"

    return df

def import_data_IR():
    if target_col[0] == "i_NH3":
```

```

        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/switched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

df = pd.read_csv(file_path, sep=";")
df["datetime"] = pd.to_datetime(df["datetime"])
df.sort_values("datetime").reset_index(drop=True)

if target_col[0] == "i_NH3":
    df["campaign"] = df["datetime"].apply(assign_campaign)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/switched/features_decomposition_list.json", "r") as f:
        dict_features = json.load(f)
elif target_col[0] == "i_CO2":
    df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
    df = df[df["campaign"] != "other"].copy()
    df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
    df = df[df["innova_point"] != "no"].copy()
    with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
        dict_features = json.load(f)
else:
    raise ValueError("The target col do not match any existing function")

df["hot"] = df.apply(assign_hot_season,axis=1)
df["umidity_range"] = df.apply(assign_humidity, axis = 1)
#Removing of rabbit 4
df["column"] = df.apply(assign_colonna, axis = 1)
df['height'] = df.apply(assign_altezza, axis = 1)

return df,dict_features

```

```

def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

```

```

# if camp == "2025_Jun":
#     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
# elif camp == "2025_Sep":
#     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
if camp == "2025_Apr":
    #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
    #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
    mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
elif camp == "2025_Jun":
    mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
    #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
elif camp == "2025_Sep":
    mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
    #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
elif camp == "2026_Gen":
    #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
    #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
    mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
else:
    mapping = {}

return mapping.get(rid, "no")

```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping : campaign, column

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

Grouping : campaign, column

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] $r_test < 0.95$ (under performance of the test)

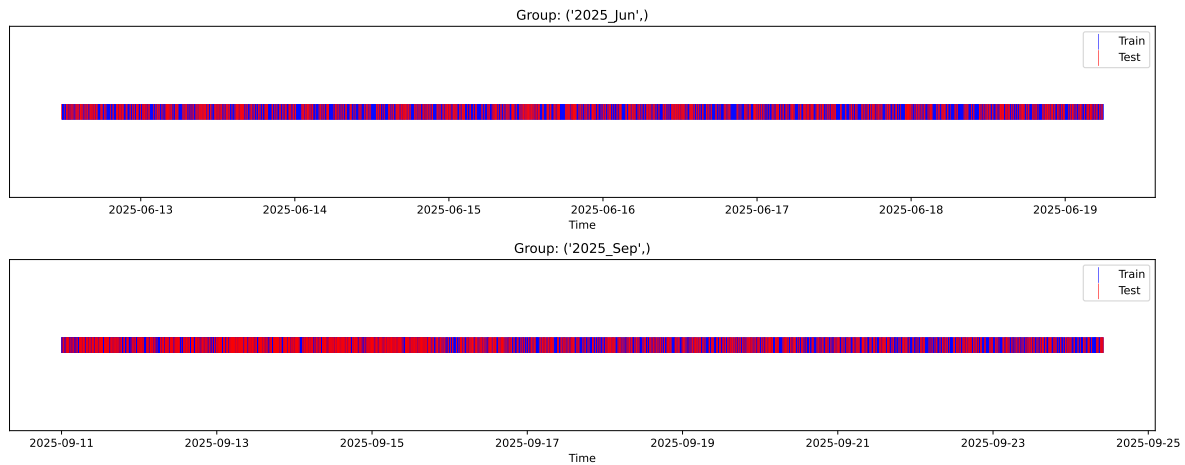
[Green] $r_test > 1.05$ (over performance of the test)

[Red] r_train not in $[0.95, 1.05]$ (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_{train} vs X_{test}

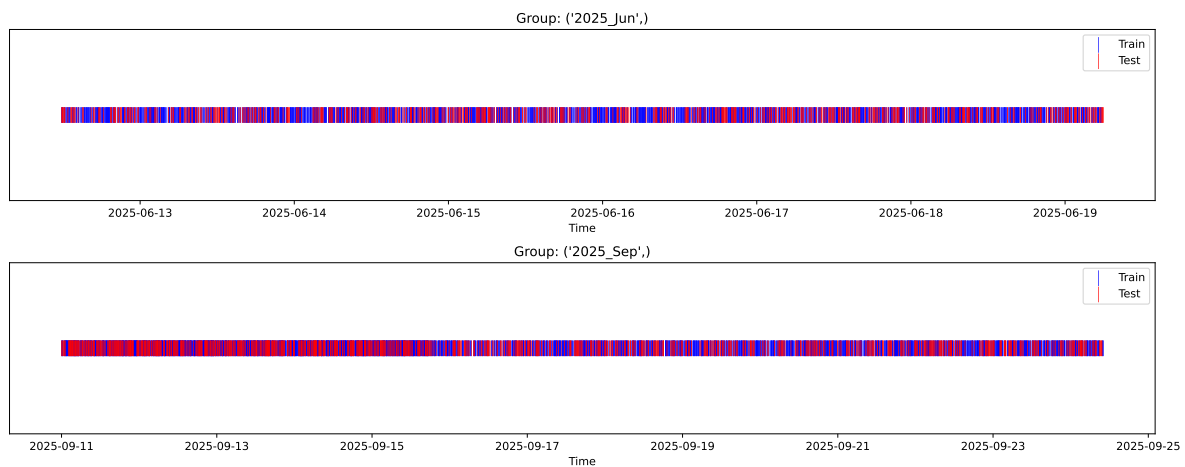
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

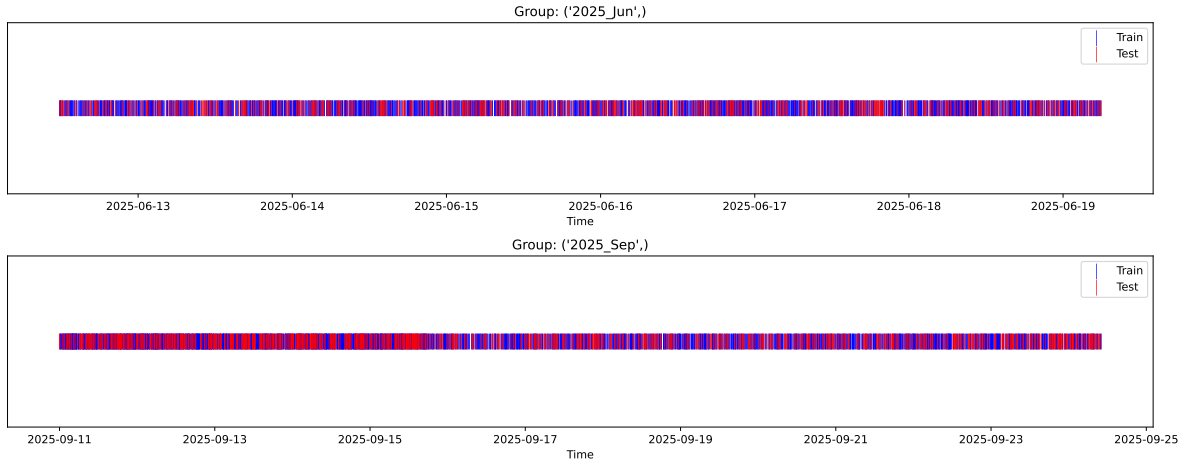
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



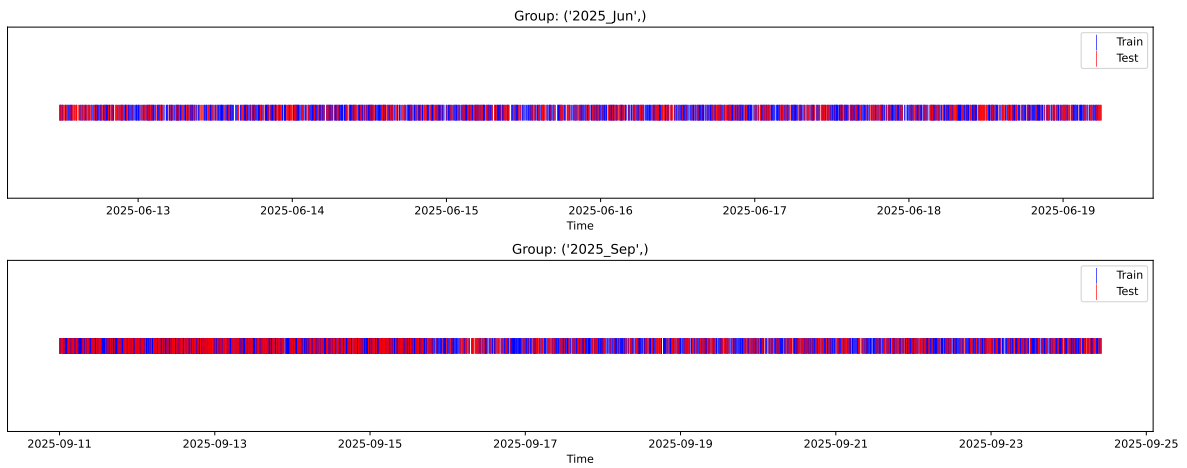
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



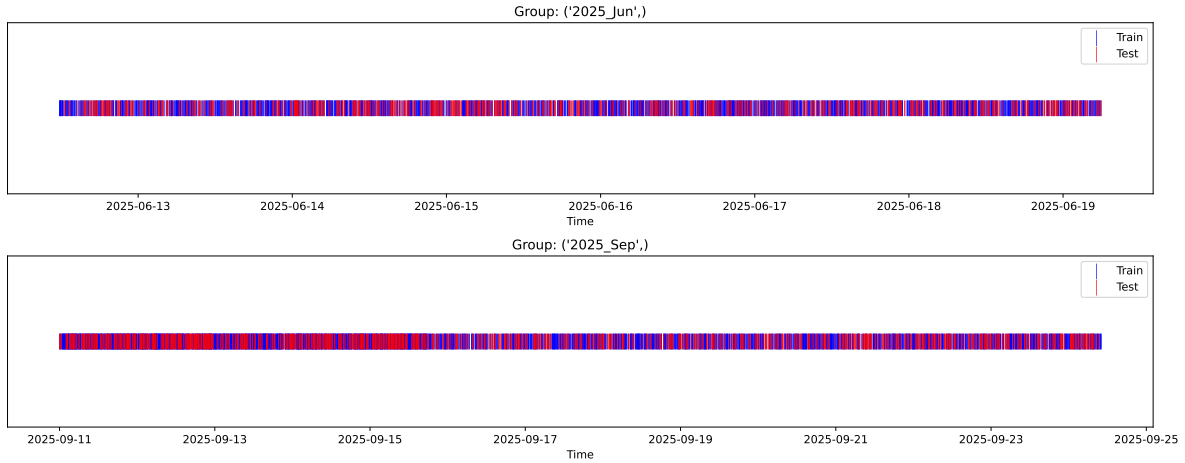
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



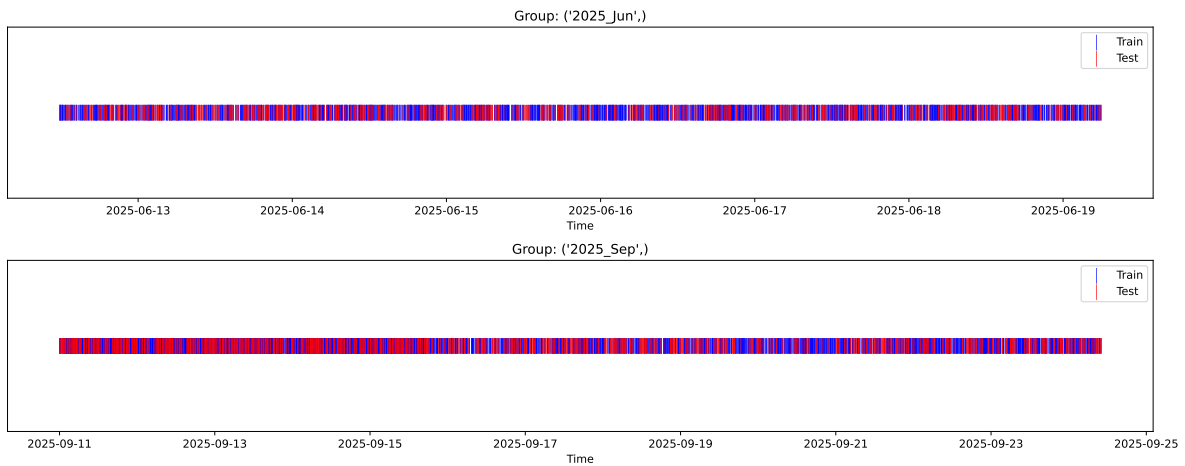
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch — Group: 2025_Jun_1

Len(group) : 1585, Len(training) : 1110, Len(test) : 475

Distribution y_train vs y_test: y_train: mean=2.772, std=0.805, min=1.432, max=13.845
y_test: mean=2.807, std=0.764, min=1.373, max=8.682

Fold 0 — y_test: mean=2.75, std=0.70, y_train: mean=2.78, std=0.85

Fold 1 — y_test: mean=2.79, std=0.69, y_train: mean=2.77, std=0.85

Fold 2 — y_test: mean=2.77, std=0.74, y_train: mean=2.77, std=0.83

Fold 3 — y_test: mean=2.80, std=0.70, y_train: mean=2.76, std=0.85

Fold 4 — y_test: mean=2.77, std=0.77, y_train: mean=2.77, std=0.82

train scores CV: [0.33880003 0.37113528 0.3409849 0.35093461 0.3337788]

test scores CV: [0.45309325 0.33756815 0.43639596 0.41255846 0.44433302]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3471	0.4168	0.0418	0.3937	0.396	-0.0208	0.9501	0.8329

Coefficients:

coef_r_CO2 = -0.0889

coef_r_NH3 = -0.2349

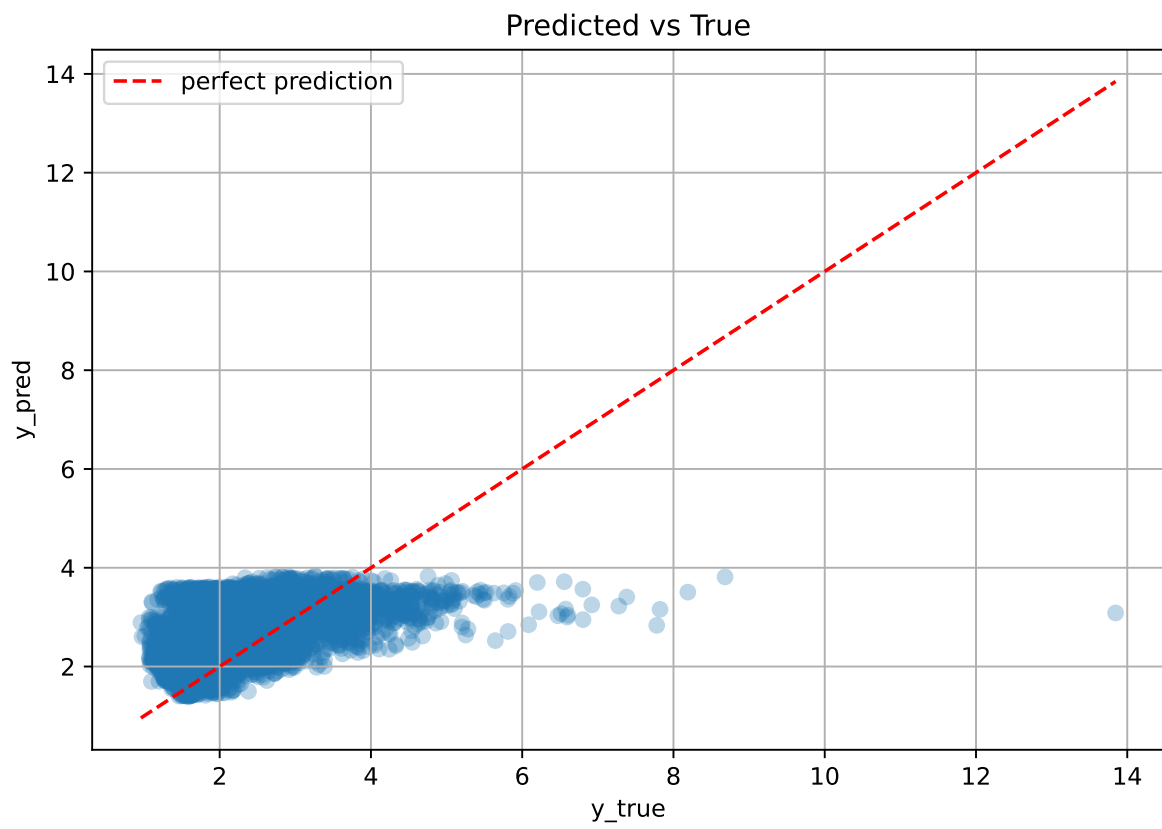
coef_r_THI = 0.8022

coef_r_temperature = -0.6449

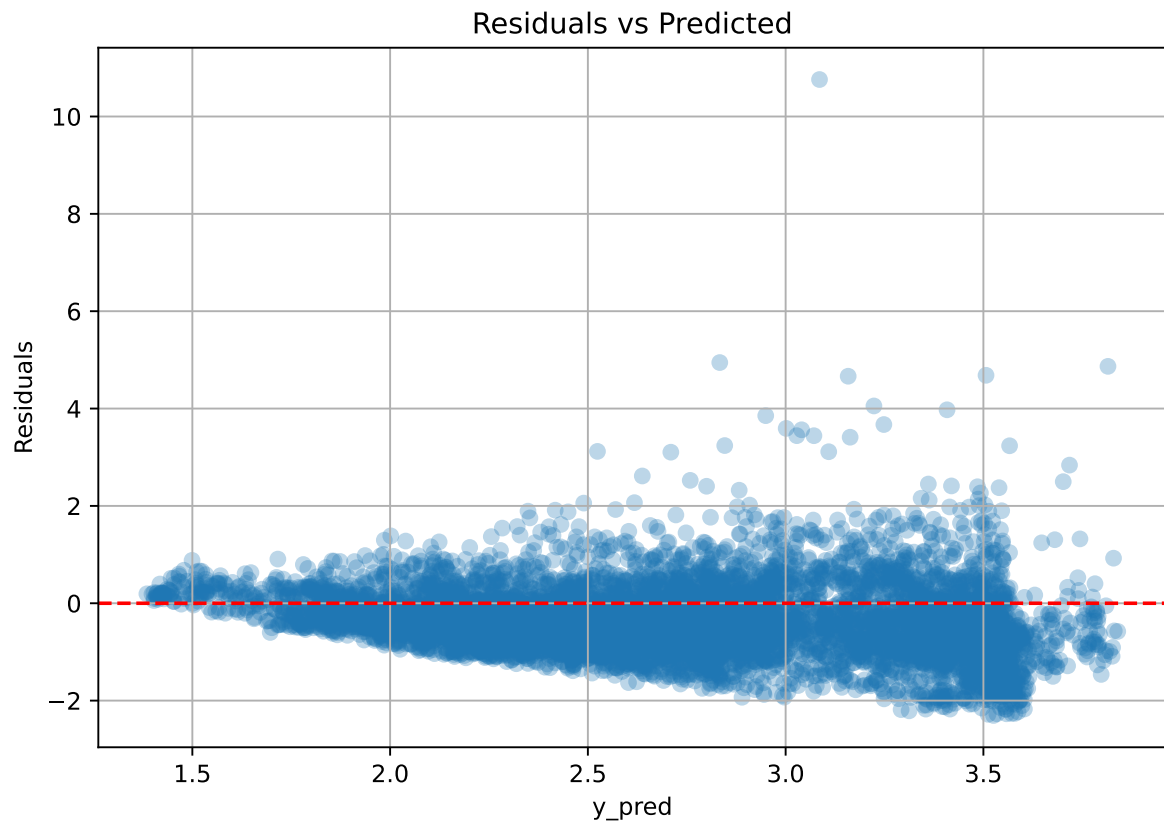
coef_r_umidity = -0.3798

intercept = 2.7723

2.3.2 Scatter



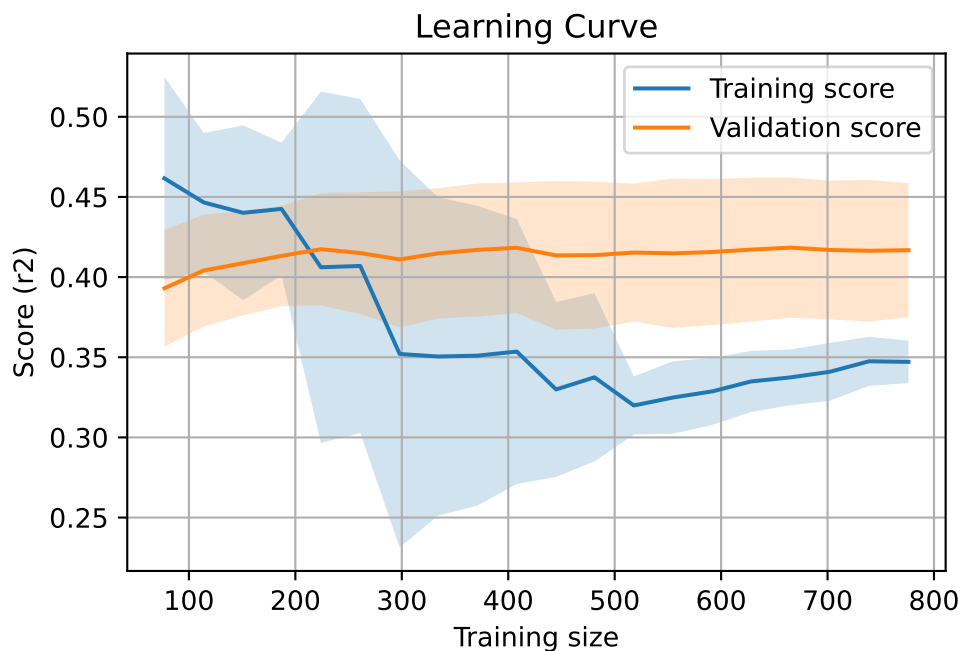
2.3.3 Scatter residuals



2.3.4 Learning curve — Group: 2025_Jun_1

`Len(group)` : 1585, `Len(training)` : 1110,

`Len(training_real)` : 777, `Len(test_of_training)` : 333, `Len(test)` : 475



2.3.5 Gridsearch — Group: 2025_Jun_2

Len(group) : 1587, Len(training) : 1111, Len(test) : 476

Distribution y_train vs y_test: y_train: mean=2.509, std=0.675, min=1.253, max=6.222
y_test: mean=2.528, std=0.697, min=1.305, max=7.383

Fold 0 — y_test: mean=2.46, std=0.68, y_train: mean=2.53, std=0.67

Fold 1 — y_test: mean=2.51, std=0.69, y_train: mean=2.51, std=0.67

Fold 2 — y_test: mean=2.56, std=0.67, y_train: mean=2.49, std=0.67

Fold 3 — y_test: mean=2.46, std=0.67, y_train: mean=2.53, std=0.67

Fold 4 — y_test: mean=2.50, std=0.68, y_train: mean=2.51, std=0.67

train scores CV: [0.22810899 0.20774064 0.22069569 0.22214819 0.21156576]

test scores CV: [0.17360107 0.23547914 0.19859637 0.1954499 0.22480749]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.2181	0.2056	0.0221	0.2163	0.2167	0.0111	1.054	1.0606

Coefficients:

$\text{coef_r_CO2} = -0.0258$

$\text{coef_r_NH3} = 0.0226$

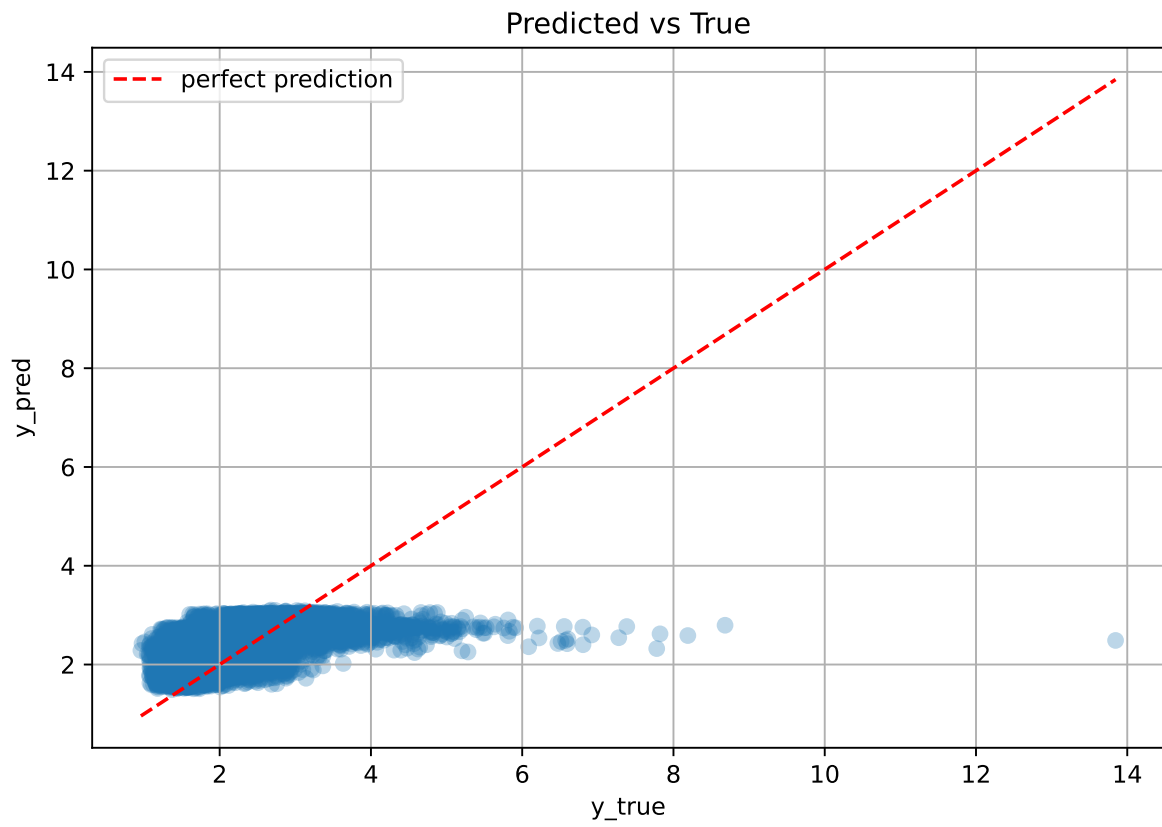
$\text{coef_r_THI} = 1.0313$

$\text{coef_r_temperature} = -1.2395$

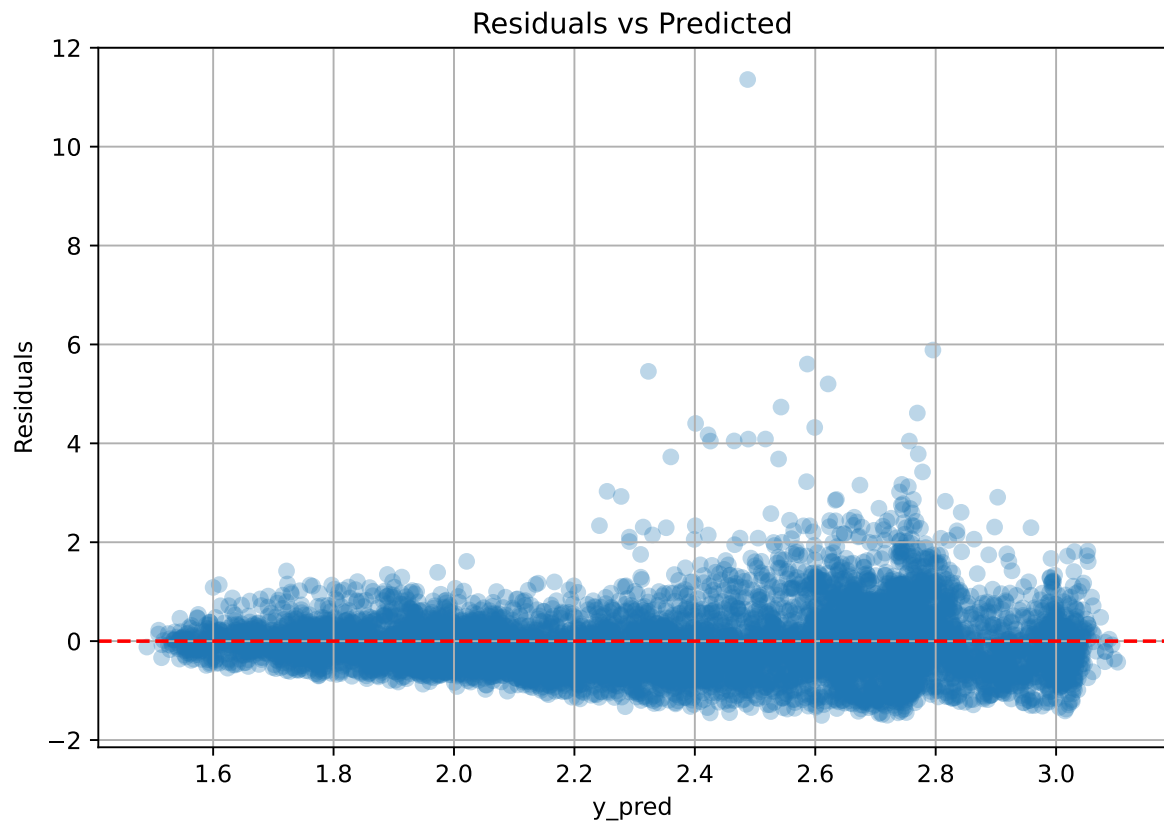
$\text{coef_r_umidity} = -0.5903$

$\text{intercept} = 2.5090$

2.3.6 Scatter



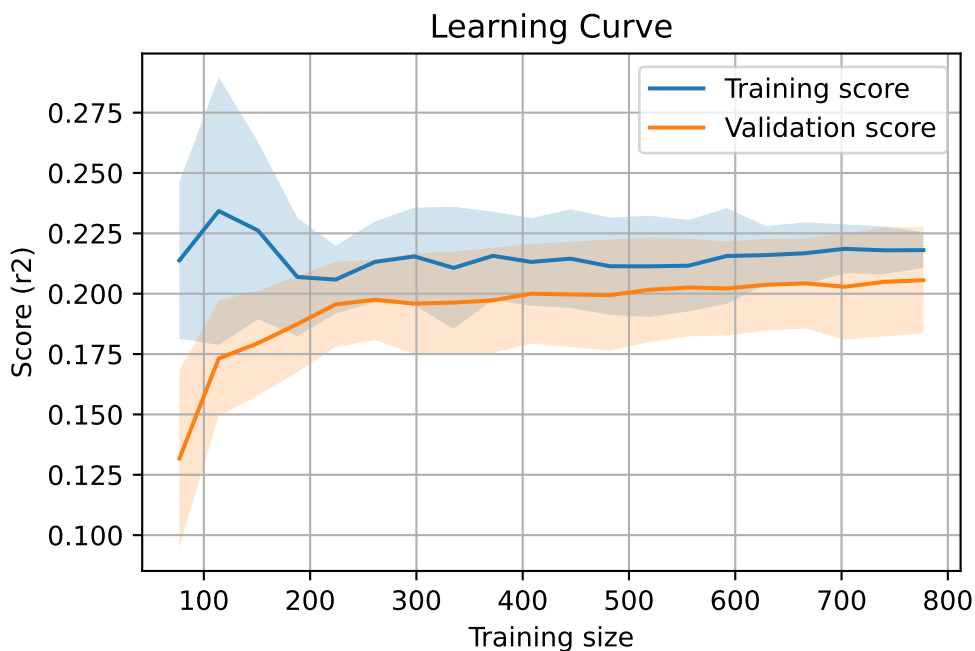
2.3.7 Scatter residuals



2.3.8 Learning curve — Group: 2025_Jun_2

`Len(group)` : 1587, `Len(training)` : 1111,

`Len(training_real)` : 778, `Len(test_of_training)` : 333, `Len(test)` : 476



2.3.9 Gridsearch — Group: 2025_Sep_1

Len(group) : 3867, Len(training) : 2707, Len(test) : 1160

Distribution y_train vs y_test: y_train: mean=2.231, std=0.761, min=0.973, max=8.190
y_test: mean=2.216, std=0.723, min=1.110, max=6.920

Fold 0 — y_test: mean=2.22, std=0.76, y_train: mean=2.23, std=0.76

Fold 1 — y_test: mean=2.24, std=0.79, y_train: mean=2.23, std=0.75

Fold 2 — y_test: mean=2.24, std=0.77, y_train: mean=2.23, std=0.76

Fold 3 — y_test: mean=2.20, std=0.71, y_train: mean=2.25, std=0.78

Fold 4 — y_test: mean=2.25, std=0.84, y_train: mean=2.22, std=0.73

train scores CV: [0.41726736 0.44420839 0.41962722 0.41860333 0.42885578]

test scores CV: [0.45204131 0.39303028 0.44590827 0.44993845 0.4231198]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.4257	0.4328	0.0224	0.4492	0.4498	0.017	1.0393	0.9836

Coefficients:

$\text{coef_r_CO2} = -0.0715$

$\text{coef_r_NH3} = -0.1378$

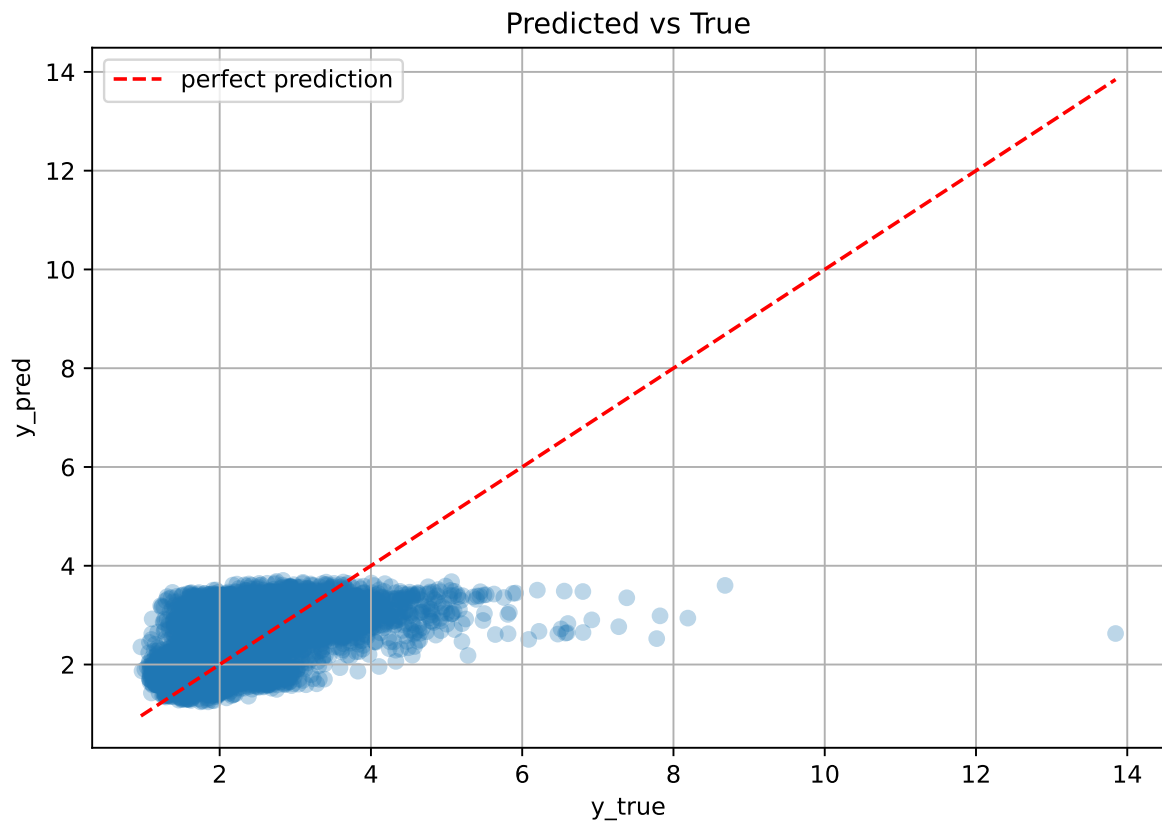
$\text{coef_r_THI} = 0.5493$

$\text{coef_r_temperature} = -0.6546$

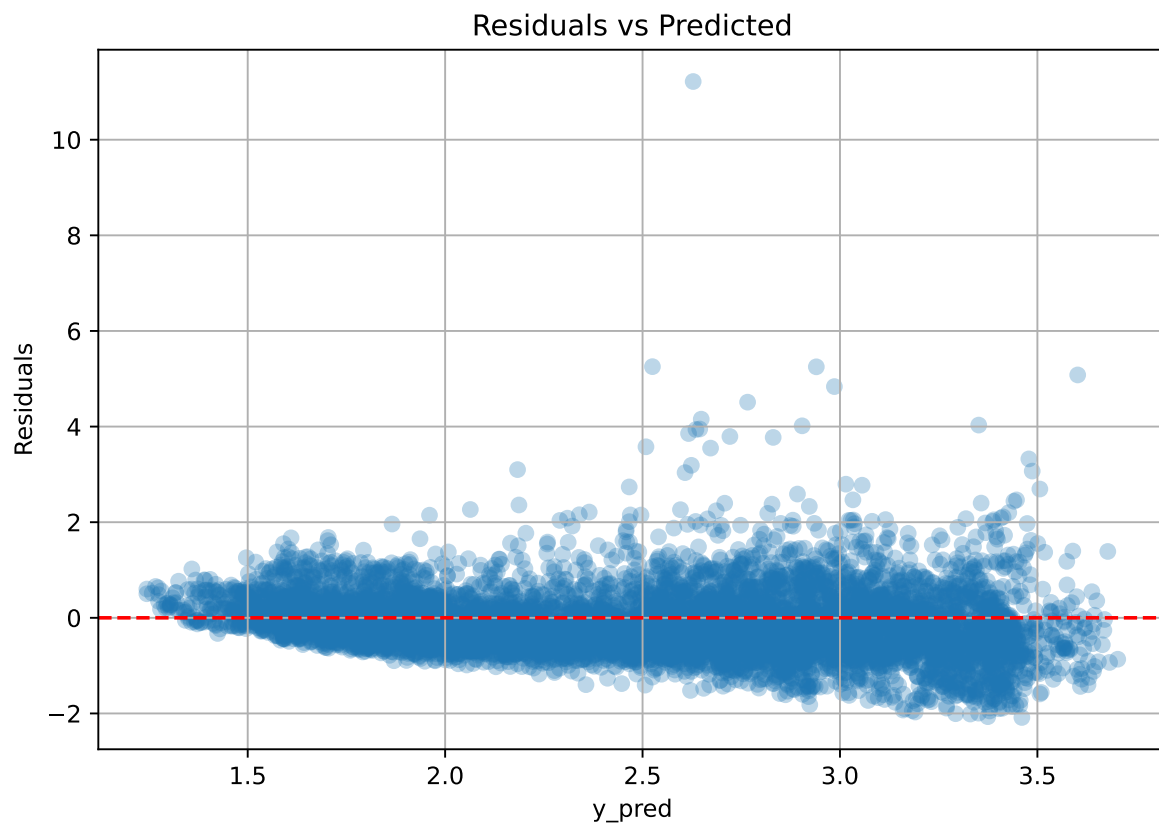
$\text{coef_r_umidity} = -0.6480$

$\text{intercept} = 2.2310$

2.3.10 Scatter



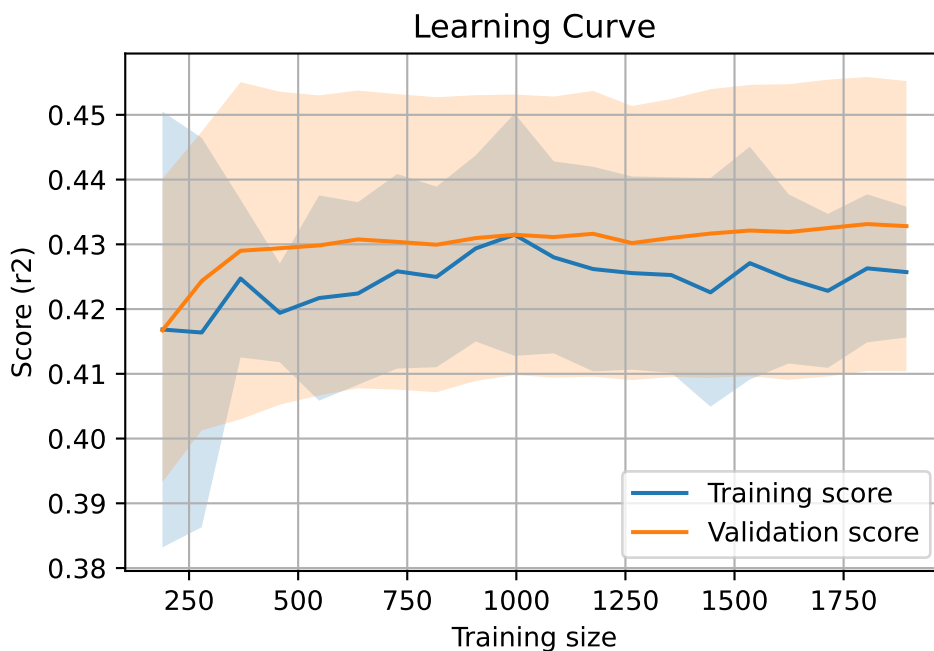
2.3.11 Scatter residuals



2.3.12 Learning curve — Group: 2025_Sep_1

`Len(group)` : 3867, `Len(training)` : 2707,

`Len(training_real)` : 1895, `Len(test_of_training)` : 812, `Len(test)` : 1160



2.3.13 Gridsearch — Group: 2025_Sep_2

Len(group) : 1129, Len(training) : 791, Len(test) : 338

Distribution y_train vs y_test: y_train: mean=1.926, std=0.742, min=0.959, max=6.804
y_test: mean=1.945, std=0.805, min=1.070, max=7.822

Fold 0 — y_test: mean=1.85, std=0.64, y_train: mean=1.96, std=0.78

Fold 1 — y_test: mean=1.95, std=0.82, y_train: mean=1.91, std=0.71

Fold 2 — y_test: mean=1.92, std=0.77, y_train: mean=1.93, std=0.73

Fold 3 — y_test: mean=1.88, std=0.76, y_train: mean=1.94, std=0.73

Fold 4 — y_test: mean=1.89, std=0.68, y_train: mean=1.94, std=0.77

train scores CV: [0.37483985 0.38789261 0.38559975 0.37728724 0.3613433]

test scores CV: [0.33651168 0.33696129 0.33543488 0.352859 0.39298824]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3774	0.351	0.022	0.3741	0.3745	0.0235	1.067	1.0753

Coefficients:

coef_r_CO2 = 0.0043

coef_r_NH3 = -0.0467

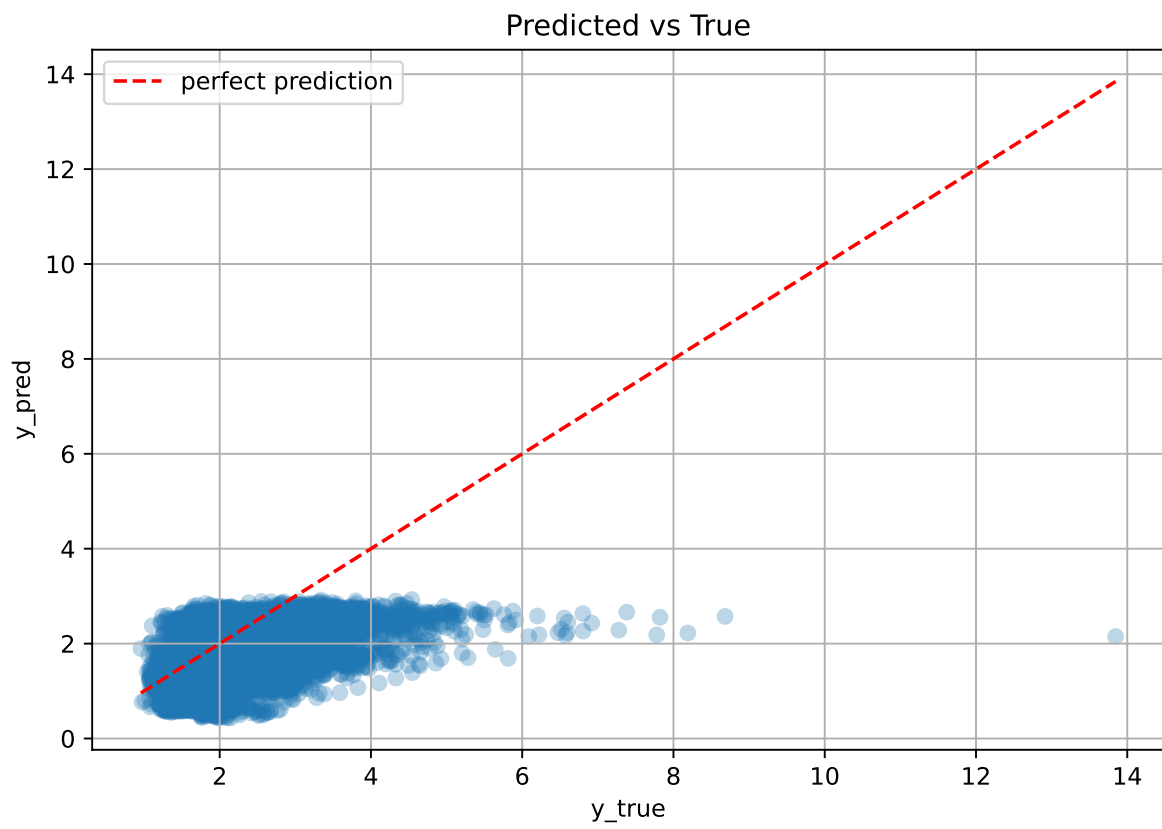
coef_r_THI = 1.3504

coef_r_temperature = -1.8574

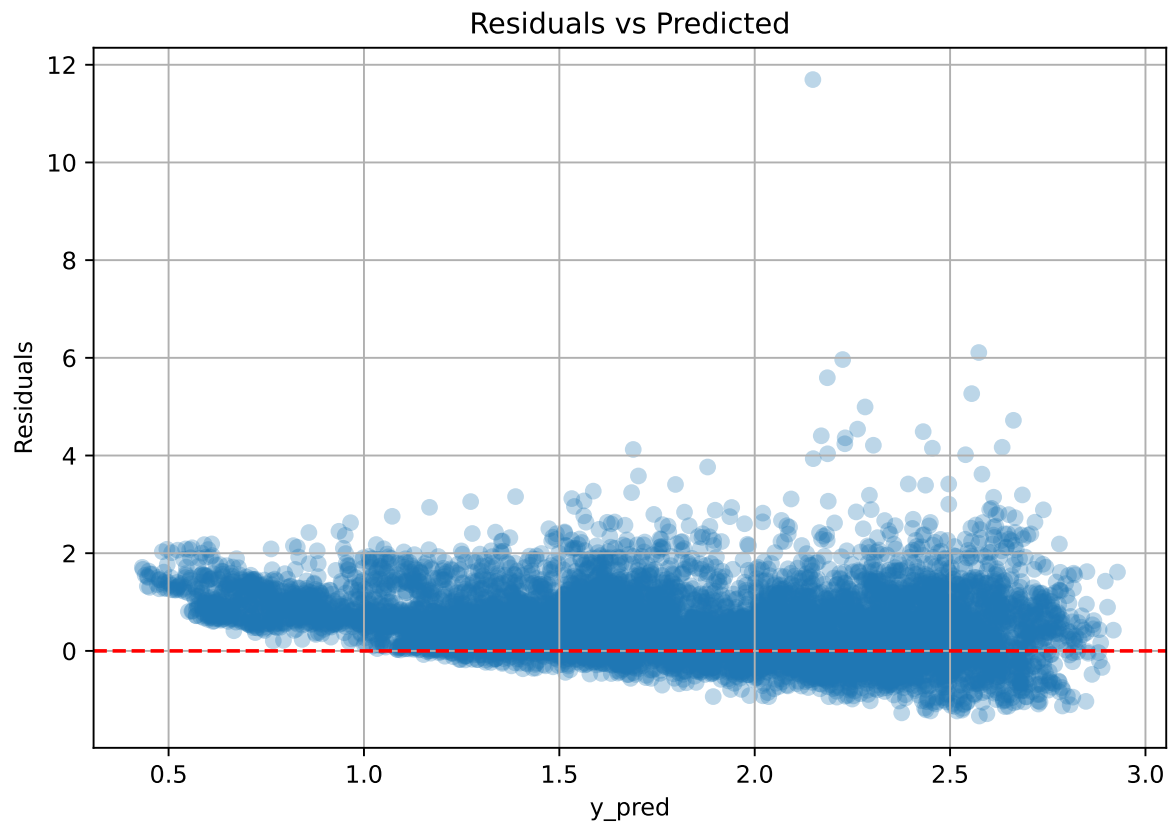
coef_r_umidity = -0.9884

intercept = 1.9256

2.3.14 Scatter



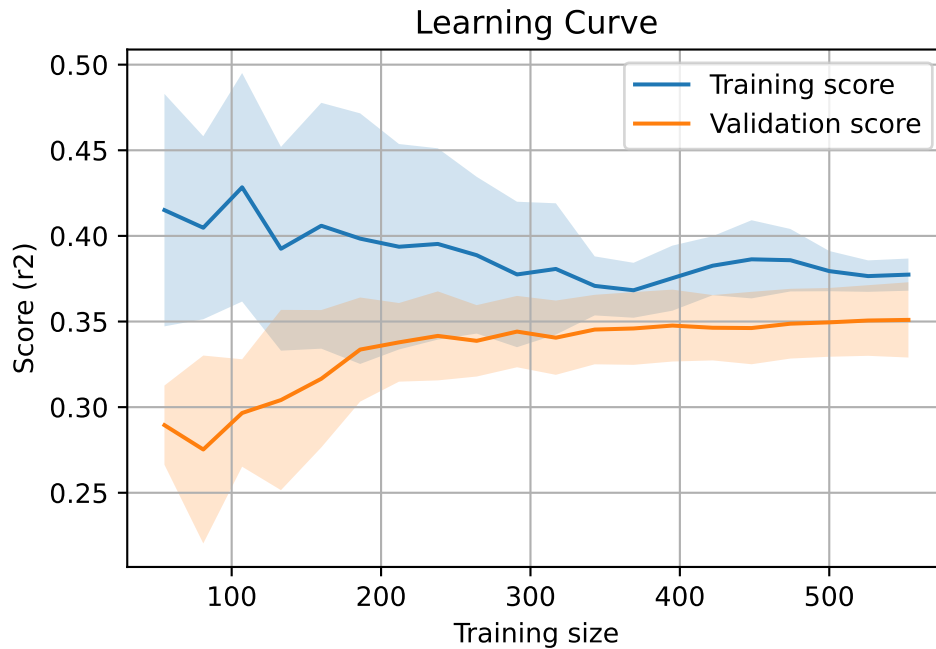
2.3.15 Scatter residuals



2.3.16 Learning curve — Group: 2025_Sep_2

`Len(group)` : 1129, `Len(training)` : 791,

`Len(training_real)` : 554, `Len(test_of_training)` : 237, `Len(test)` : 338



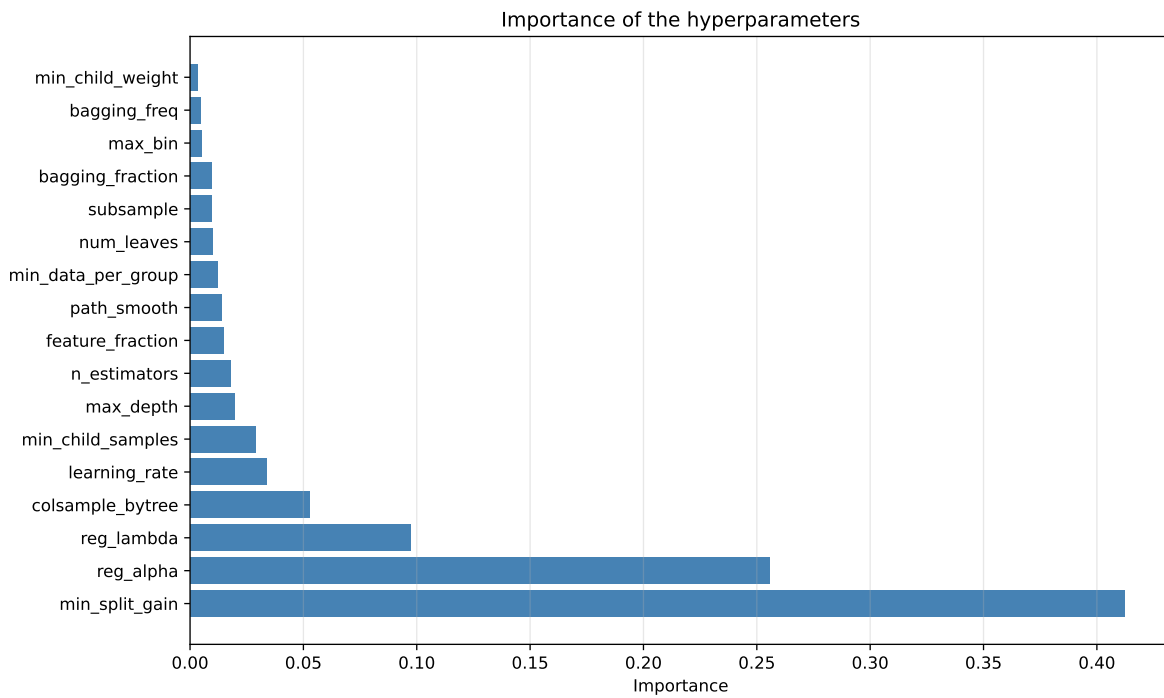
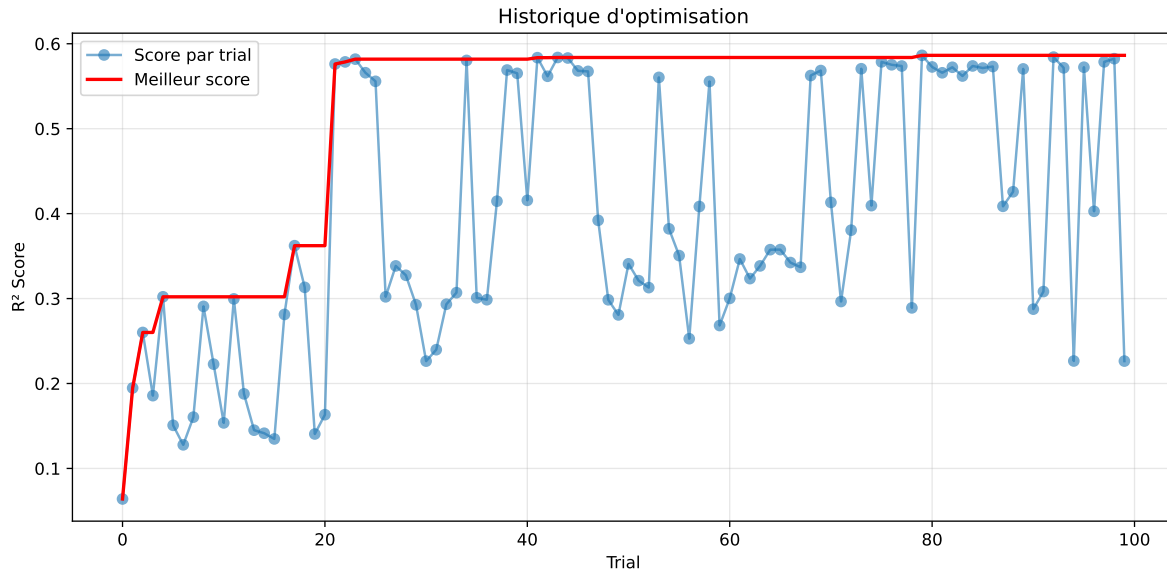
2.4 Model : LGBM

2.4.1 Gridsearch — Group: 2025_Jun_1

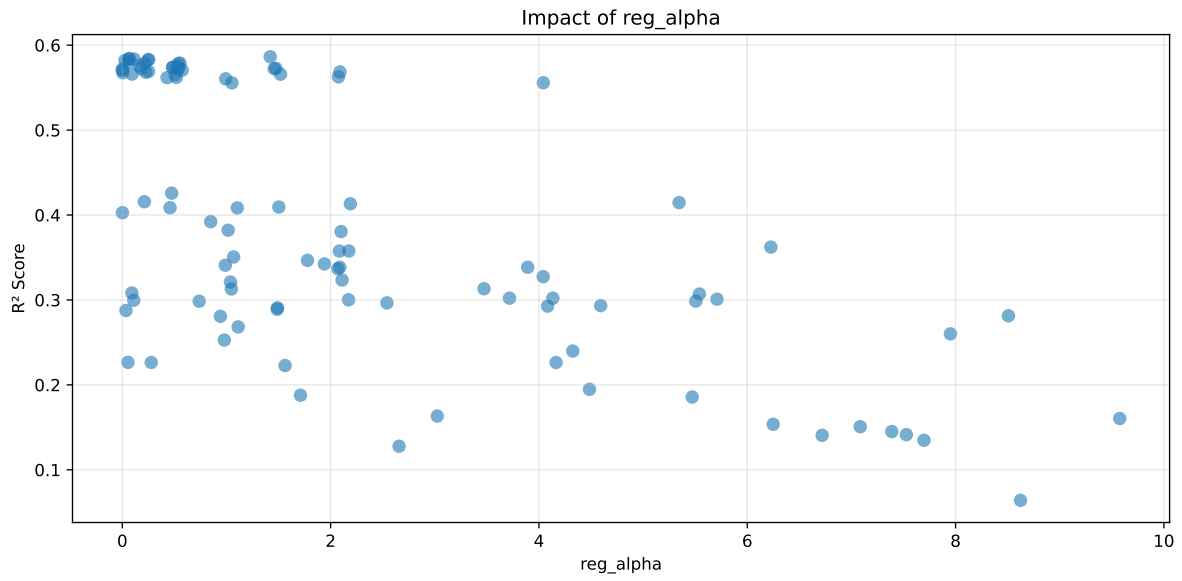
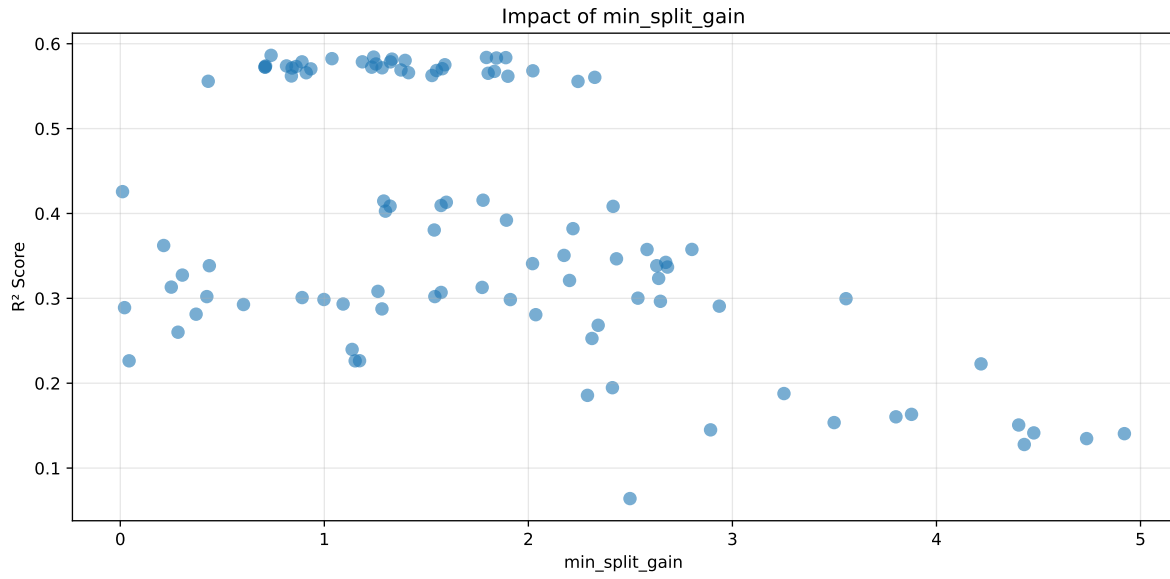
Len(group) : 1585, Len(training) : 1110, Len(test) : 475

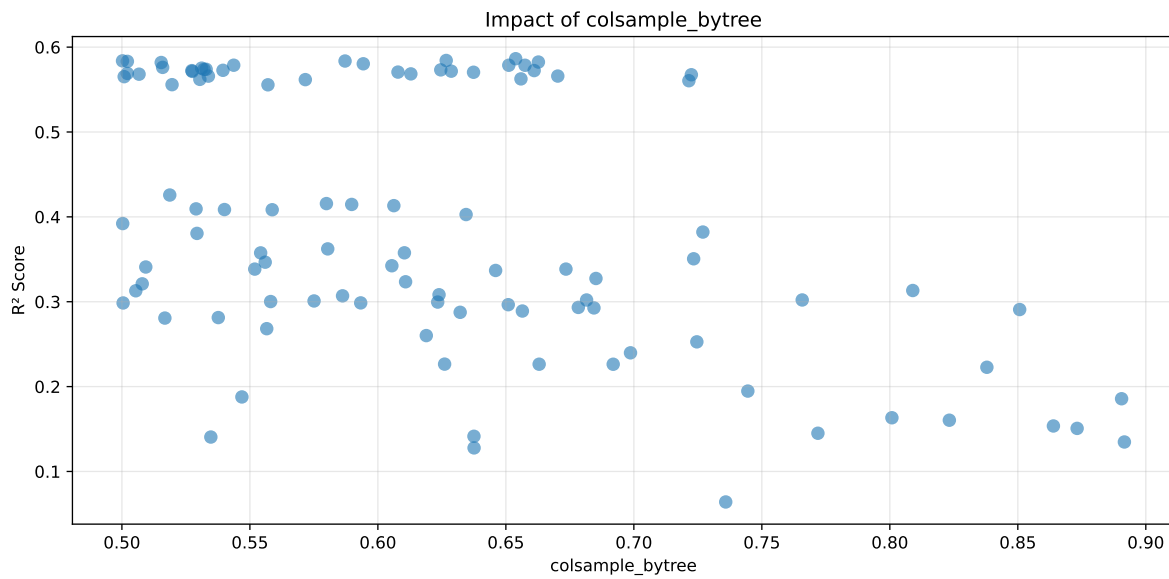
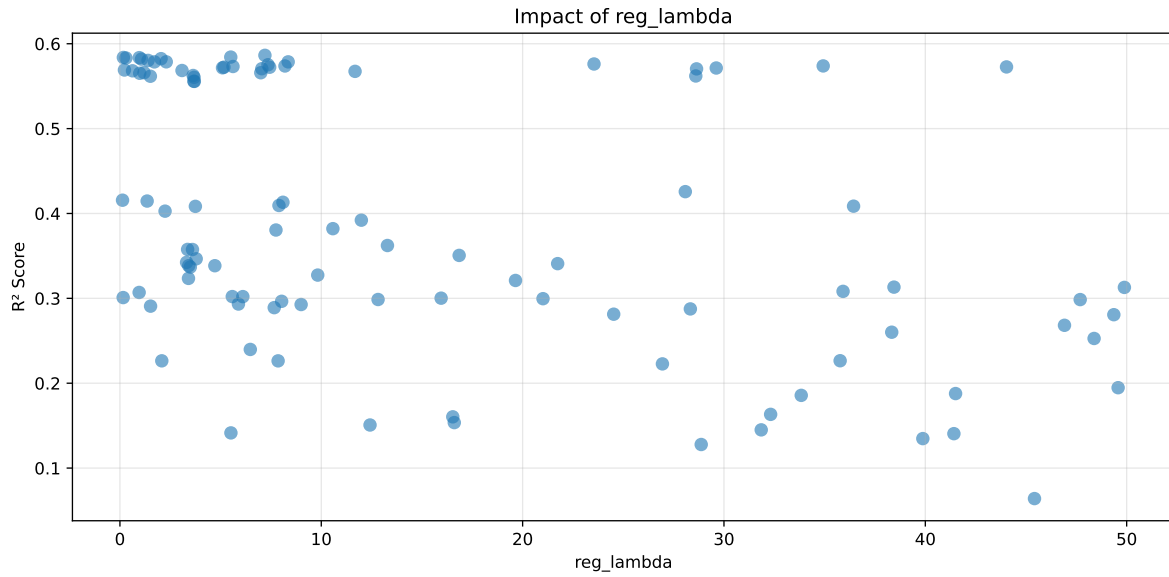
Distribution y_train vs y_test: y_train: mean=2.772, std=0.805, min=1.432, max=13.845
y_test: mean=2.807, std=0.764, min=1.373, max=8.682

===== TOP Importance FEATURES =====
feature importance r_NH3 791 r_umidity 639 r_CO2 510 r_temperature 400 r_THI 398



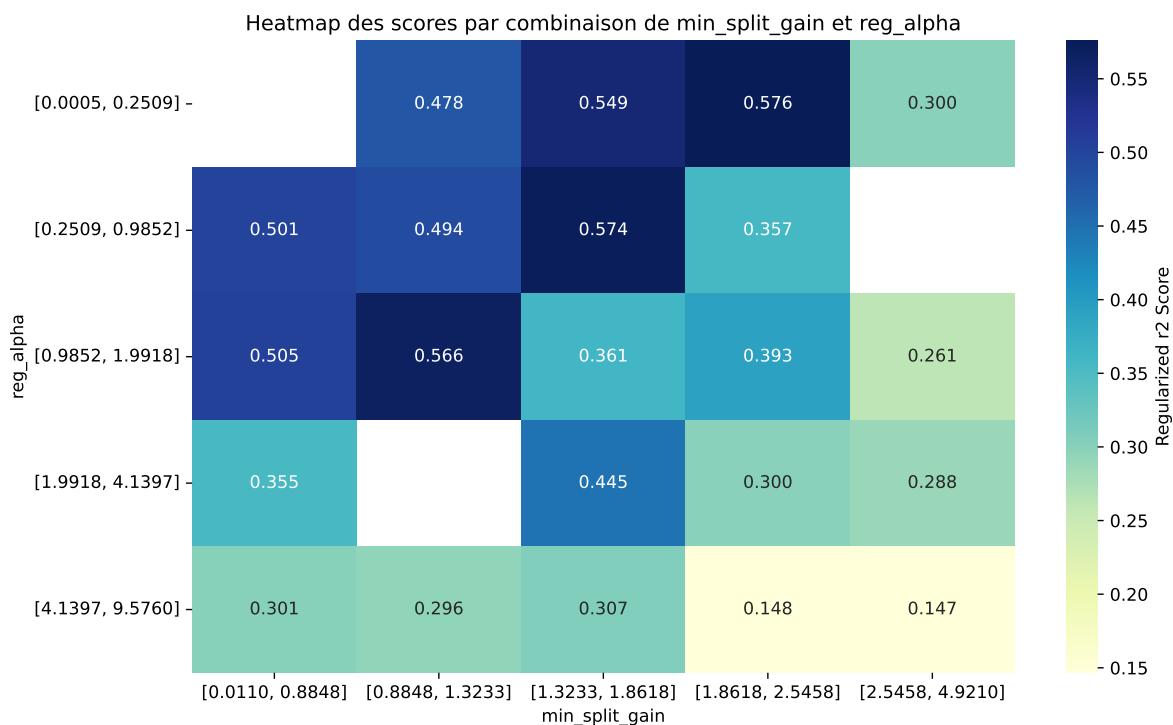
Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha', 'reg_lambda', 'colsample_bytree']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5869	0.5863	0.5717	0.0377	0.5863	0.5987	0.0124	1.0211	1.0009
2	0.571	0.5843	0.5683	0.0408	0.5843	0.5919	0.0076	1.013	0.9773
3	0.5967	0.5838	0.571	0.0377	0.5838	0.5943	0.0105	1.018	1.0221
4	0.591	0.5836	0.5686	0.0359	0.5836	0.5893	0.0056	1.0097	1.0126
5	0.5832	0.5832	0.5683	0.0389	0.5832	0.5887	0.0055	1.0094	0.9999
6	0.5801	0.5824	0.5693	0.0399	0.5824	0.5985	0.0161	1.0276	0.9961
7	0.5726	0.5819	0.5615	0.0351	0.5819	0.5848	0.0029	1.005	0.9841
8	0.5818	0.5804	0.5693	0.0389	0.5804	0.5837	0.0033	1.0058	1.0024
9	0.5704	0.5787	0.5626	0.04	0.5787	0.592	0.0133	1.0229	0.9856
10	0.5684	0.5787	0.5641	0.0335	0.5787	0.596	0.0174	1.03	0.9823
---	---	---	---	---	---	---	---	---	---
100	0.3701	0.4466	0.41	0.0149	0.0641	0.4361	-0.0105	0.9764	0.8287

Hyperparameters:

Rank 1: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.01202831506452764, 'reg__num_leaves': 41, 'reg__subsample': 0.7059773568421649, 'reg__colsample_bytree': 0.6538376353643165, 'reg__min_child_samples': 24, 'reg__reg_alpha': 1.419899488564711, 'reg__reg_lambda': 7.200668284784596, 'reg__min_split_gain': 0.7390492434871553, 'reg__path_smooth': 2.9873085272155313, 'reg__min_child_weight':

0.001699014935148473, 'reg__feature_fraction': 0.6795840221023394, 'reg__bagging_fraction': 0.8743101670942461, 'reg__bagging_freq': 1, 'reg__max_bin': 169, 'reg__min_data_per_group': 147}

Rank 2: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.01426826029202214, 'reg__num_leaves': 62, 'reg__subsample': 0.7269216465462175, 'reg__colsample_bytree': 0.6266956006370914, 'reg__min_child_samples': 20, 'reg__reg_alpha': 0.06878697841627356, 'reg__reg_lambda': 5.50453718170057, 'reg__min_split_gain': 1.2416461280647817, 'reg__path_smooth': 3.5404232522671046, 'reg__min_child_weight': 0.0016423349978695915, 'reg__feature_fraction': 0.618089769850996, 'reg__bagging_fraction': 0.8398656083027449, 'reg__bagging_freq': 1, 'reg__max_bin': 63, 'reg__min_data_per_group': 105}

Rank 3: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.015293767552449825, 'reg__num_leaves': 55, 'reg__subsample': 0.6802836393098651, 'reg__colsample_bytree': 0.5002449034989995, 'reg__min_child_samples': 13, 'reg__reg_alpha': 0.06018192154179814, 'reg__reg_lambda': 0.1707270635853655, 'reg__min_split_gain': 1.7944306331354802, 'reg__path_smooth': 3.1771672392728902, 'reg__min_child_weight': 0.0010282497041793732, 'reg__feature_fraction': 0.640692346834087, 'reg__bagging_fraction': 0.8513910502457233, 'reg__bagging_freq': 1, 'reg__max_bin': 127, 'reg__min_data_per_group': 141}

Rank 4: {'reg__n_estimators': 500, 'reg__max_depth': 7, 'reg__learning_rate': 0.014680223689601363, 'reg__num_leaves': 45, 'reg__subsample': 0.752864948054041, 'reg__colsample_bytree': 0.5871730622482865, 'reg__min_child_samples': 12, 'reg__reg_alpha': 0.11321319815962827, 'reg__reg_lambda': 0.9607850149492456, 'reg__min_split_gain': 1.8892869742135359, 'reg__path_smooth': 2.975903152385052, 'reg__min_child_weight': 0.0011794046389768916, 'reg__feature_fraction': 0.6334760219739354, 'reg__bagging_fraction': 0.8469091570751756, 'reg__bagging_freq': 1, 'reg__max_bin': 134, 'reg__min_data_per_group': 144}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 8, 'reg__learning_rate': 0.01628152793663406, 'reg__num_leaves': 56, 'reg__subsample': 0.687545639223453, 'reg__colsample_bytree': 0.5021383281825997, 'reg__min_child_samples': 13, 'reg__reg_alpha': 0.2532588952717125, 'reg__reg_lambda': 0.2981514899973825, 'reg__min_split_gain': 1.8435230200106472, 'reg__path_smooth': 3.1338729501713987, 'reg__min_child_weight': 0.0010780512287600083, 'reg__feature_fraction': 0.6426033520995532, 'reg__bagging_fraction': 0.8466840938555367, 'reg__bagging_freq': 1, 'reg__max_bin': 88, 'reg__min_data_per_group': 137}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.010827834655025282, 'reg__num_leaves': 43, 'reg__subsample': 0.7412633147515248, 'reg__colsample_bytree': 0.6627107751334883, 'reg__min_child_samples': 20, 'reg__reg_alpha': 0.24727278089525478, 'reg__reg_lambda': 2.0429974419813774, 'reg__min_split_gain': 1.0374396396739054, 'reg__path_smooth': 3.6581608655748625, 'reg__min_child_weight':

0.0031578126561551565, 'reg__feature_fraction': 0.6510761776629579, 'reg__bagging_fraction': 0.6850056385721269, 'reg__bagging_freq': 1, 'reg__max_bin': 91, 'reg__min_data_per_group': 103}

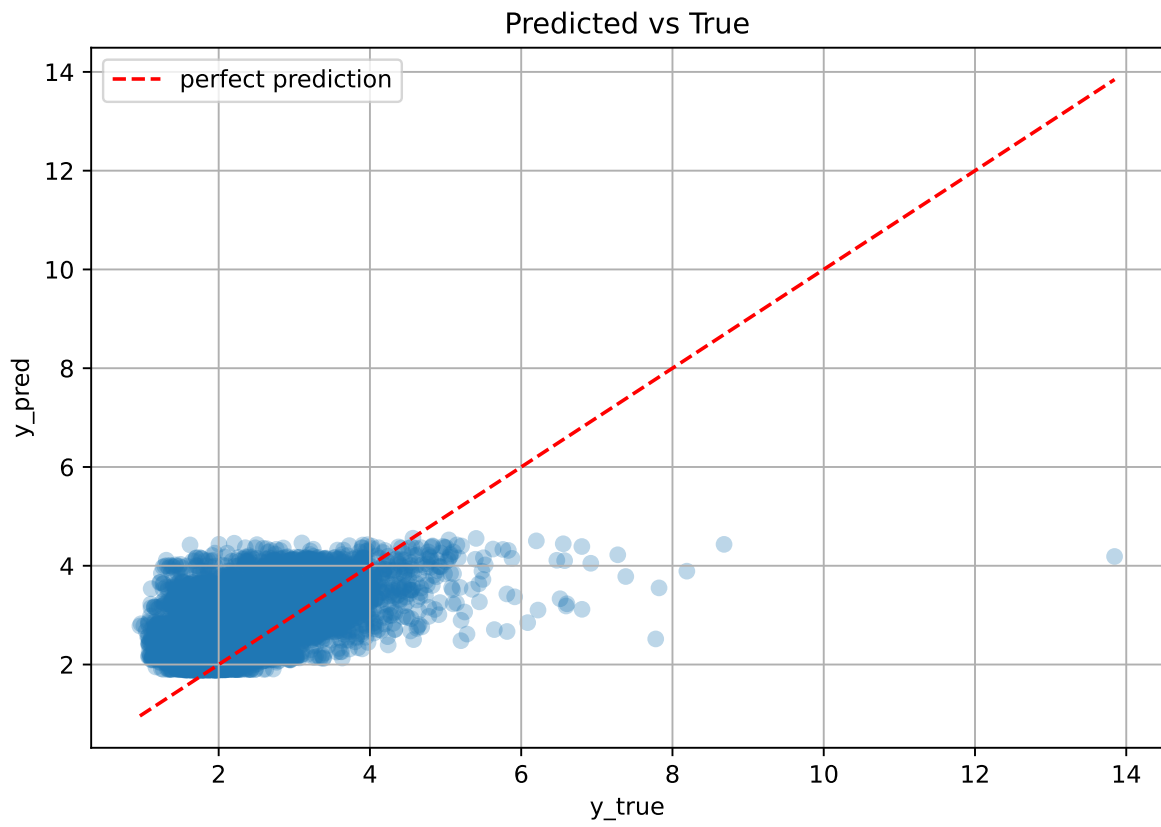
Rank 7: {'reg__n_estimators': 400, 'reg__max_depth': 4, 'reg__learning_rate': 0.01005904249223276, 'reg__num_leaves': 49, 'reg__subsample': 0.6877023828804517, 'reg__colsample_bytree': 0.515360805551612, 'reg__min_child_samples': 23, 'reg__reg_alpha': 0.024777400202917063, 'reg__reg_lambda': 1.0699201705241999, 'reg__min_split_gain': 1.3315127917941274, 'reg__path_smooth': 2.1733616313846387, 'reg__min_child_weight': 0.003845398746676307, 'reg__feature_fraction': 0.6342787447245801, 'reg__bagging_fraction': 0.8972504553950476, 'reg__bagging_freq': 1, 'reg__max_bin': 115, 'reg__min_data_per_group': 151}

Rank 8: {'reg__n_estimators': 300, 'reg__max_depth': 4, 'reg__learning_rate': 0.015442383296770178, 'reg__num_leaves': 62, 'reg__subsample': 0.8610365091940569, 'reg__colsample_bytree': 0.594298577497957, 'reg__min_child_samples': 11, 'reg__reg_alpha': 0.05710189134268734, 'reg__reg_lambda': 1.3968178612558582, 'reg__min_split_gain': 1.3960159725392258, 'reg__path_smooth': 2.852700172140654, 'reg__min_child_weight': 0.006982586874533505, 'reg__feature_fraction': 0.6262120720845802, 'reg__bagging_fraction': 0.6862209481896617, 'reg__bagging_freq': 1, 'reg__max_bin': 88, 'reg__min_data_per_group': 140}

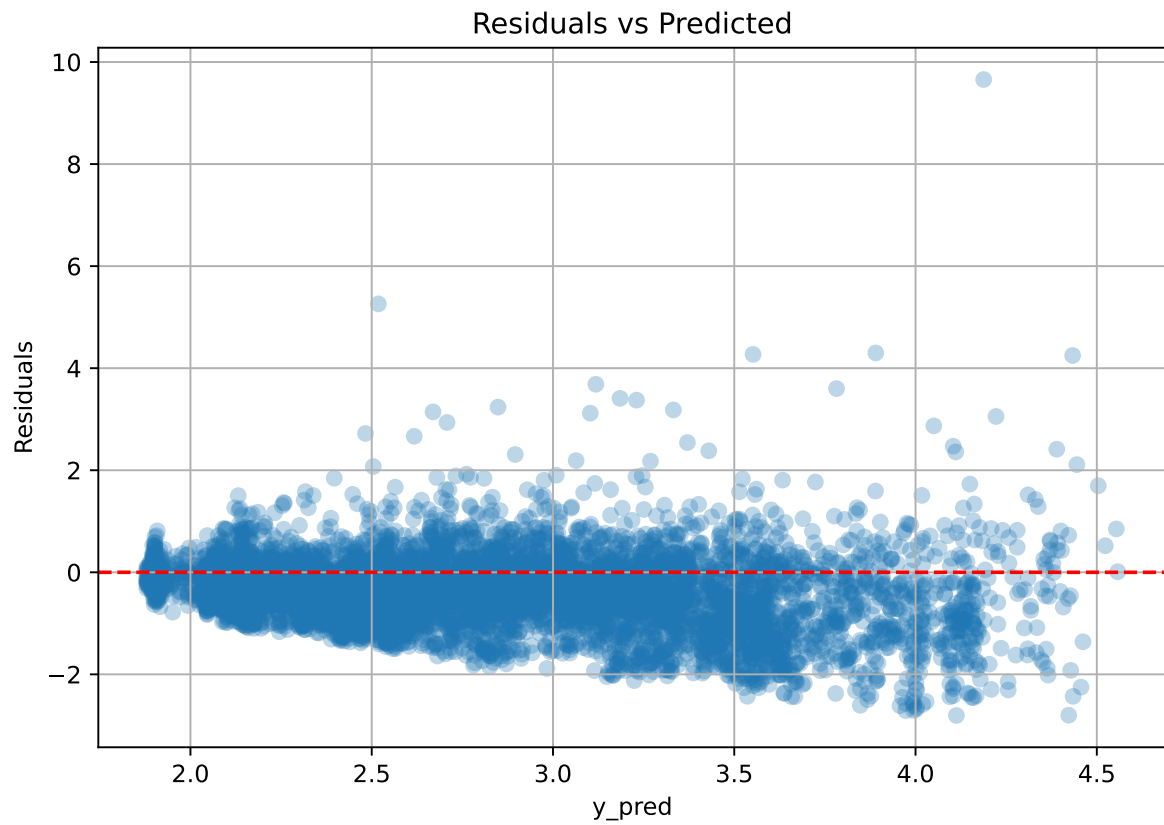
Rank 9: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.01075224614235402, 'reg__num_leaves': 37, 'reg__subsample': 0.8303028078310754, 'reg__colsample_bytree': 0.5436802224680153, 'reg__min_child_samples': 20, 'reg__reg_alpha': 0.21864424972496133, 'reg__reg_lambda': 2.300559183582301, 'reg__min_split_gain': 1.1864574662101381, 'reg__path_smooth': 3.2600193776023696, 'reg__min_child_weight': 0.0030695421170827642, 'reg__feature_fraction': 0.6521476128769832, 'reg__bagging_fraction': 0.6710121867430323, 'reg__bagging_freq': 1, 'reg__max_bin': 92, 'reg__min_data_per_group': 99}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.012331848938161678, 'reg__num_leaves': 59, 'reg__subsample': 0.712830126142805, 'reg__colsample_bytree': 0.6511395344550053, 'reg__min_child_samples': 24, 'reg__reg_alpha': 0.5540033387398189, 'reg__reg_lambda': 8.356378420510623, 'reg__min_split_gain': 0.8906012611091518, 'reg__path_smooth': 2.6014777514747047, 'reg__min_child_weight': 0.002343287709873661, 'reg__feature_fraction': 0.6819334406120037, 'reg__bagging_fraction': 0.8140778569570557, 'reg__bagging_freq': 1, 'reg__max_bin': 76, 'reg__min_data_per_group': 119}

2.4.2 Scatter



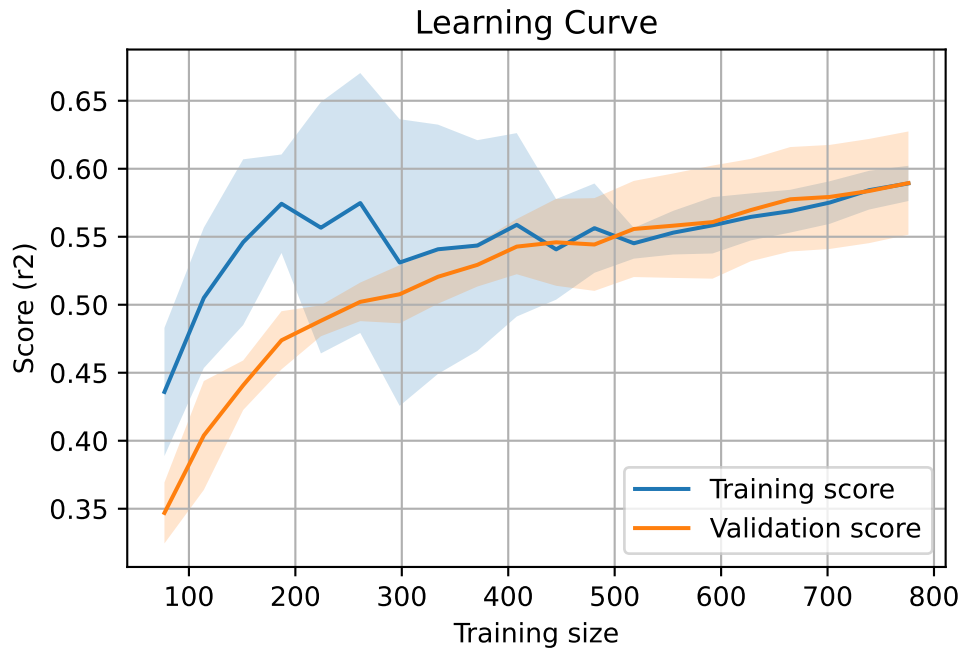
2.4.3 Scatter residuals



2.4.4 Learning curve — Group: 2025_Jun_1

`Len(group)` : 1585, `Len(training)` : 1110,

`Len(training_real)` : 777, `Len(test_of_training)` : 333, `Len(test)` : 475

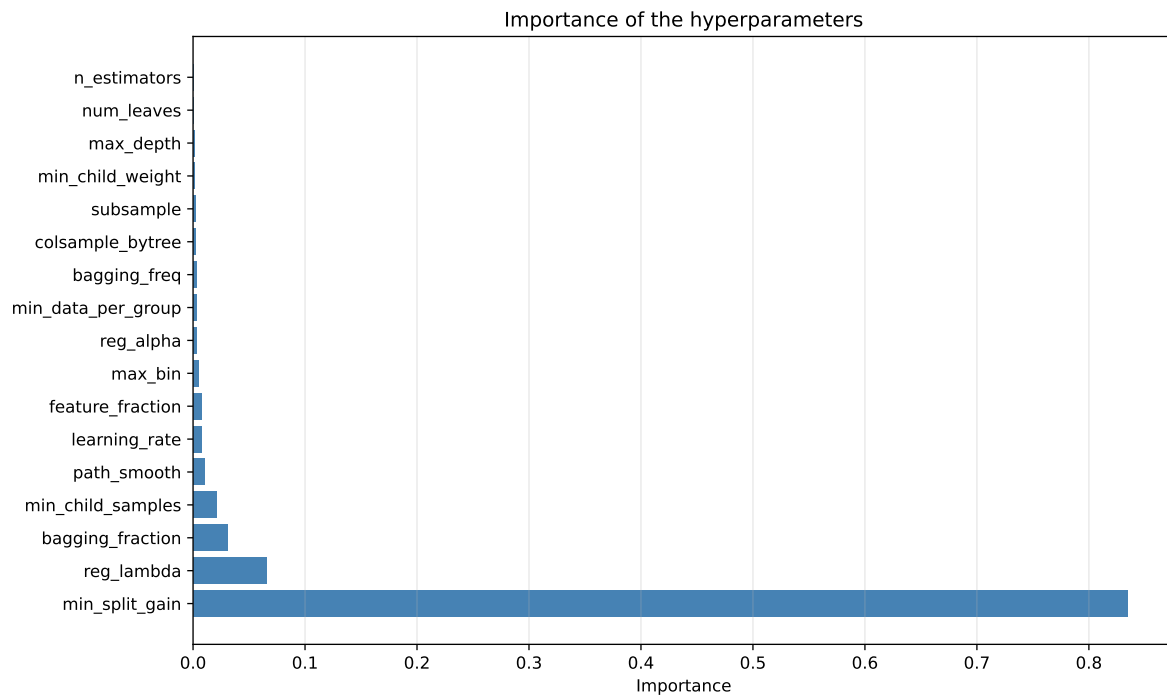
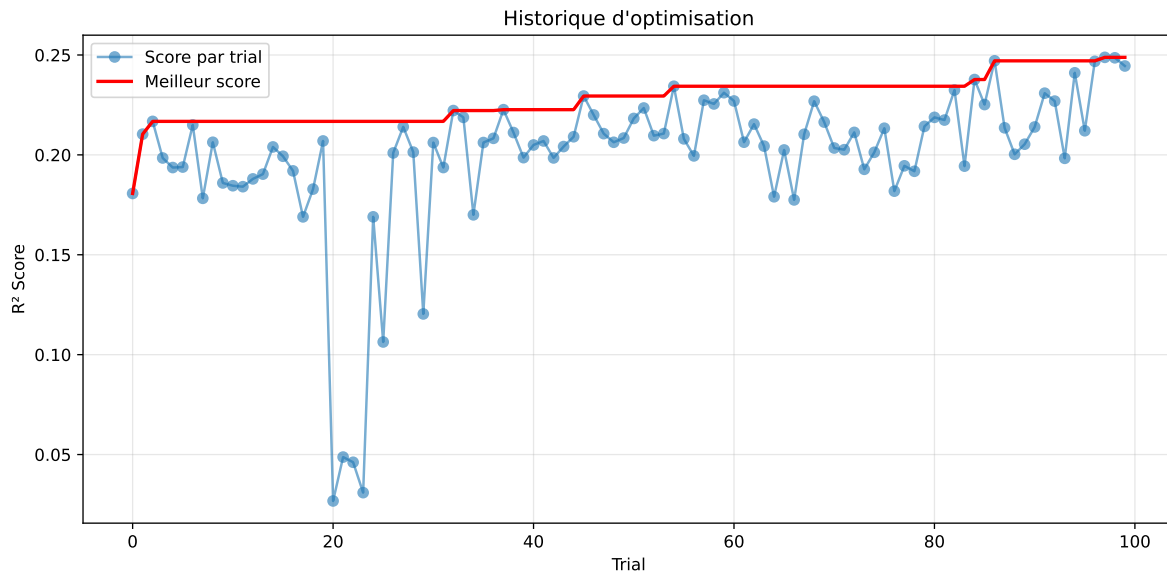


2.4.5 Gridsearch — Group: 2025_Jun_2

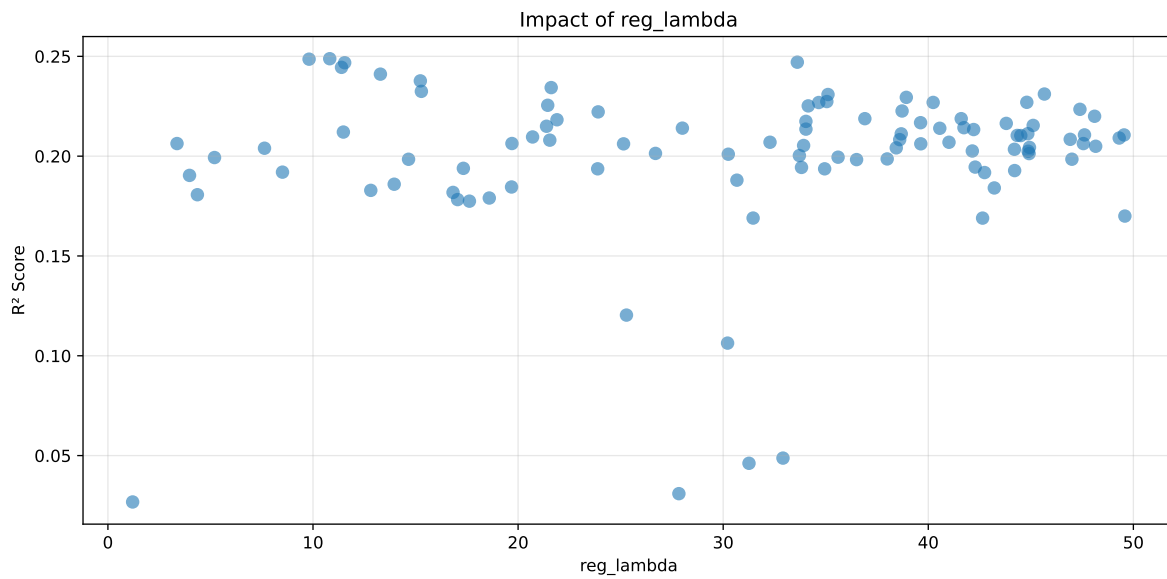
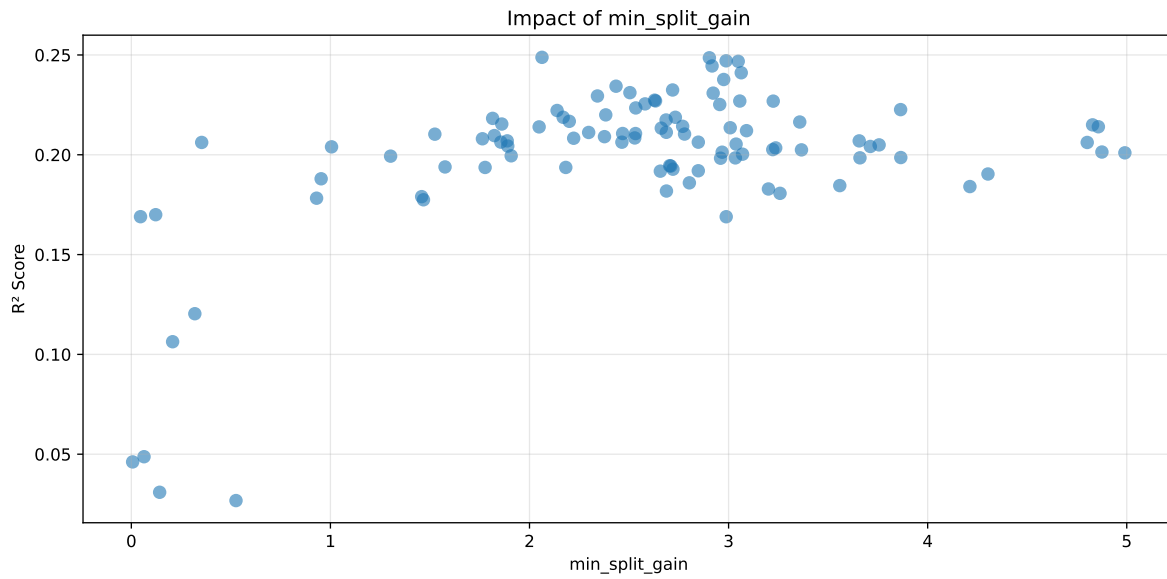
Len(group) : 1587, Len(training) : 1111, Len(test) : 476

Distribution y_train vs y_test: y_train: mean=2.509, std=0.675, min=1.253, max=6.222
y_test: mean=2.528, std=0.697, min=1.305, max=7.383

===== TOP Importance FEATURES =====
feature importance r_umidity 54 r_THI 53 r_temperature 44 r_NH3 38 r_CO2 26

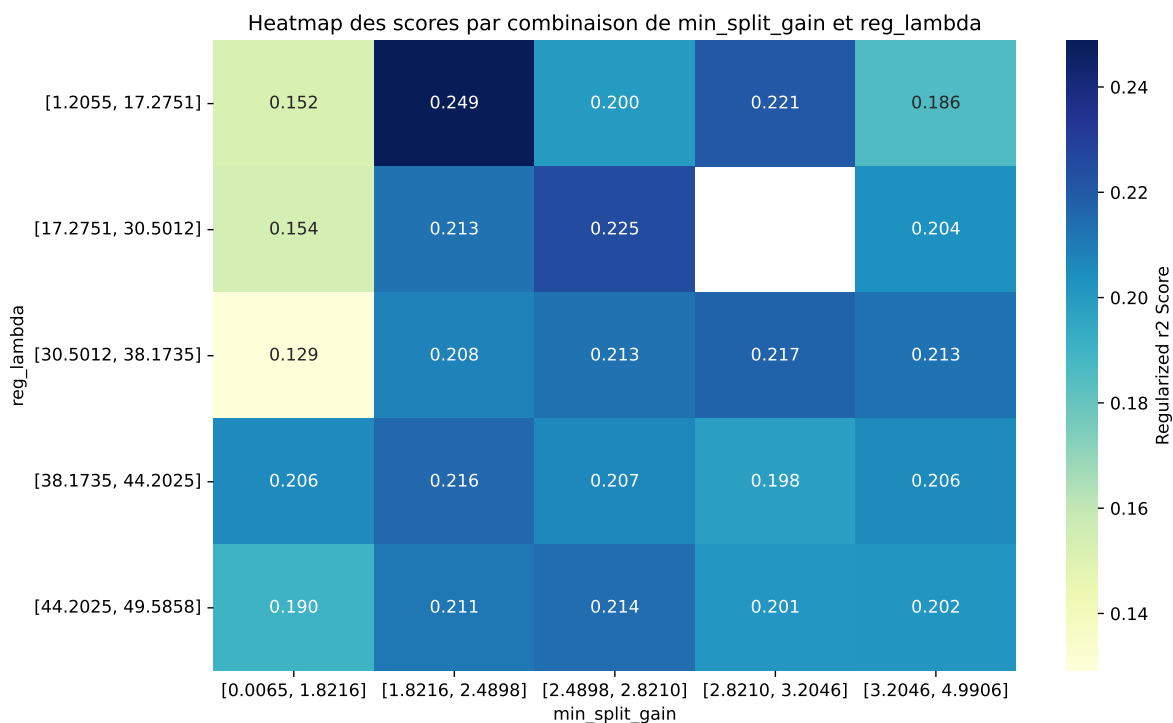


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_lambda']



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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4233	0.3942	0.3798	0.013	0.2488	0.4048	0.0106	1.0268	1.0738
2	0.4015	0.376	0.3604	0.0132	0.2486	0.3841	0.008	1.0214	1.0678
3	0.3946	0.37	0.3562	0.0146	0.2471	0.3836	0.0136	1.0366	1.0664
4	0.3794	0.3573	0.3412	0.0148	0.2468	0.3727	0.0154	1.043	1.0618
5	0.3958	0.3706	0.3554	0.0155	0.2445	0.3804	0.0098	1.0264	1.068
6	0.3777	0.355	0.3384	0.017	0.2411	0.3631	0.0081	1.0228	1.0642
7	0.3947	0.3685	0.3522	0.0179	0.2377	0.3778	0.0092	1.025	1.071
8	0.3863	0.3609	0.3472	0.0263	0.2344	0.387	0.0261	1.0723	1.0701
9	0.3941	0.3672	0.3523	0.0167	0.2325	0.3772	0.01	1.0272	1.0734
10	0.375	0.351	0.3373	0.0218	0.2311	0.3706	0.0196	1.0557	1.0683
---	---	---	---	---	---	---	---	---	---
100	0.5506	0.4633	0.4768	0.0101	0.0268	0.4989	0.0356	1.0769	1.1884

Hyperparameters:

Rank 1: {'reg__n_estimators': 800, 'reg__max_depth': 7, 'reg__learning_rate': 0.05066849072419819, 'reg__num_leaves': 55, 'reg__subsample': 0.5706135596209657, 'reg__colsample_bytree': 0.7441471396853395, 'reg__min_child_samples': 48, 'reg__reg_alpha': 2.8711999993088027, 'reg__reg_lambda': 10.815924850308228, 'reg__min_split_gain': 2.062734661809967, 'reg__path_smooth': 4.742592841136699, 'reg__min_child_weight':

0.00917603521972572, 'reg__feature_fraction': 0.595324694401319, 'reg__bagging_fraction': 0.560286815033806, 'reg__bagging_freq': 1, 'reg__max_bin': 68, 'reg__min_data_per_group': 86}

Rank 2: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.060716811483580435, 'reg__num_leaves': 56, 'reg__subsample': 0.5971557044712971, 'reg__colsample_bytree': 0.6937895853603658, 'reg__min_child_samples': 48, 'reg__reg_alpha': 1.210757651904363, 'reg__reg_lambda': 9.808719263165443, 'reg__min_split_gain': 2.903556440693272, 'reg__path_smooth': 4.677511363257955, 'reg__min_child_weight': 0.007732905326524087, 'reg__feature_fraction': 0.5955906289741301, 'reg__bagging_fraction': 0.5585259339789768, 'reg__bagging_freq': 1, 'reg__max_bin': 68, 'reg__min_data_per_group': 85}

Rank 3: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.048554650149304684, 'reg__num_leaves': 63, 'reg__subsample': 0.6116905286815711, 'reg__colsample_bytree': 0.6893611360277063, 'reg__min_child_samples': 49, 'reg__reg_alpha': 0.7904108799752797, 'reg__reg_lambda': 33.61465864738954, 'reg__min_split_gain': 2.98755519028344, 'reg__path_smooth': 4.708020697443666, 'reg__min_child_weight': 0.016334163389767273, 'reg__feature_fraction': 0.6897384252796162, 'reg__bagging_fraction': 0.5535206476297556, 'reg__bagging_freq': 1, 'reg__max_bin': 97, 'reg__min_data_per_group': 53}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 5, 'reg__learning_rate': 0.04876302084088423, 'reg__num_leaves': 55, 'reg__subsample': 0.7242529266845446, 'reg__colsample_bytree': 0.6930048678122395, 'reg__min_child_samples': 48, 'reg__reg_alpha': 3.212756227982392, 'reg__reg_lambda': 11.544380070264344, 'reg__min_split_gain': 3.048415309396047, 'reg__path_smooth': 4.692626849086808, 'reg__min_child_weight': 0.012970275194649059, 'reg__feature_fraction': 0.6513130219564914, 'reg__bagging_fraction': 0.5465646127489538, 'reg__bagging_freq': 1, 'reg__max_bin': 63, 'reg__min_data_per_group': 51}

Rank 5: {'reg__n_estimators': 300, 'reg__max_depth': 5, 'reg__learning_rate': 0.048603927243205475, 'reg__num_leaves': 36, 'reg__subsample': 0.5920705944157661, 'reg__colsample_bytree': 0.7414442877978391, 'reg__min_child_samples': 48, 'reg__reg_alpha': 1.2848551988583665, 'reg__reg_lambda': 11.390553304802499, 'reg__min_split_gain': 2.9170659292288383, 'reg__path_smooth': 4.666000237377158, 'reg__min_child_weight': 0.012187717737463372, 'reg__feature_fraction': 0.593159253030402, 'reg__bagging_fraction': 0.558483199235116, 'reg__bagging_freq': 1, 'reg__max_bin': 67, 'reg__min_data_per_group': 84}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.02673371139368416, 'reg__num_leaves': 53, 'reg__subsample': 0.8266754148327305, 'reg__colsample_bytree': 0.6944094771457315, 'reg__min_child_samples': 48, 'reg__reg_alpha': 3.0447752774848724, 'reg__reg_lambda': 13.286253385213879, 'reg__min_split_gain': 3.063565930872234, 'reg__path_smooth': 3.4076771202163925, 'reg__min_child_weight':

0.013760402645122903, 'reg__feature_fraction': 0.6506869844337364, 'reg__bagging_fraction': 0.5543715737269481, 'reg__bagging_freq': 1, 'reg__max_bin': 92, 'reg__min_data_per_group': 84}

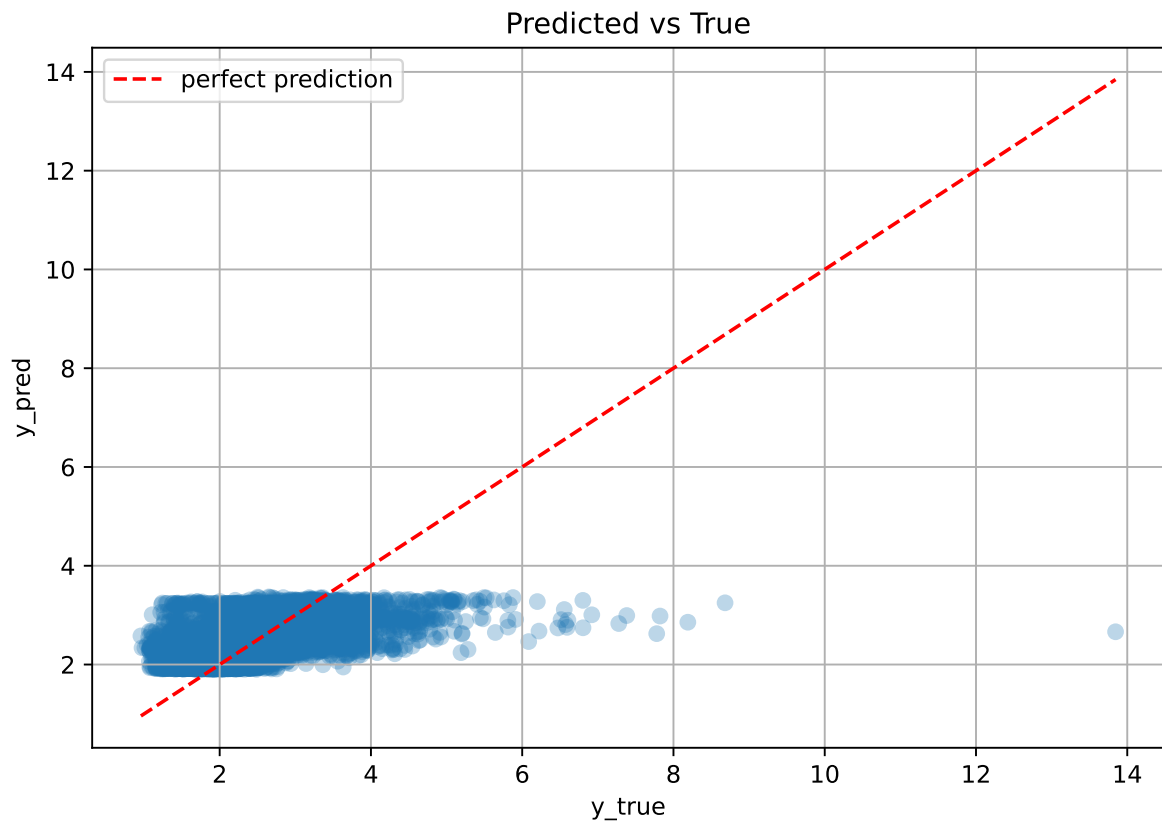
Rank 7: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.05041318620363714, 'reg__num_leaves': 63, 'reg__subsample': 0.6136929929322393, 'reg__colsample_bytree': 0.6924522034441655, 'reg__min_child_samples': 50, 'reg__reg_alpha': 0.7923194464709532, 'reg__reg_lambda': 15.233890676864938, 'reg__min_split_gain': 2.9754836236909292, 'reg__path_smooth': 3.5005015007503317, 'reg__min_child_weight': 0.015223483681486735, 'reg__feature_fraction': 0.6479734027279065, 'reg__bagging_fraction': 0.5610118601322744, 'reg__bagging_freq': 1, 'reg__max_bin': 95, 'reg__min_data_per_group': 57}

Rank 8: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.03164210544264113, 'reg__num_leaves': 52, 'reg__subsample': 0.7081169290956839, 'reg__colsample_bytree': 0.7097838822083415, 'reg__min_child_samples': 45, 'reg__reg_alpha': 0.12823265183903887, 'reg__reg_lambda': 21.613551019166373, 'reg__min_split_gain': 2.4346897759866915, 'reg__path_smooth': 4.541279524054681, 'reg__min_child_weight': 0.34129588591358884, 'reg__feature_fraction': 0.8497674616467071, 'reg__bagging_fraction': 0.598471688249477, 'reg__bagging_freq': 3, 'reg__max_bin': 75, 'reg__min_data_per_group': 78}

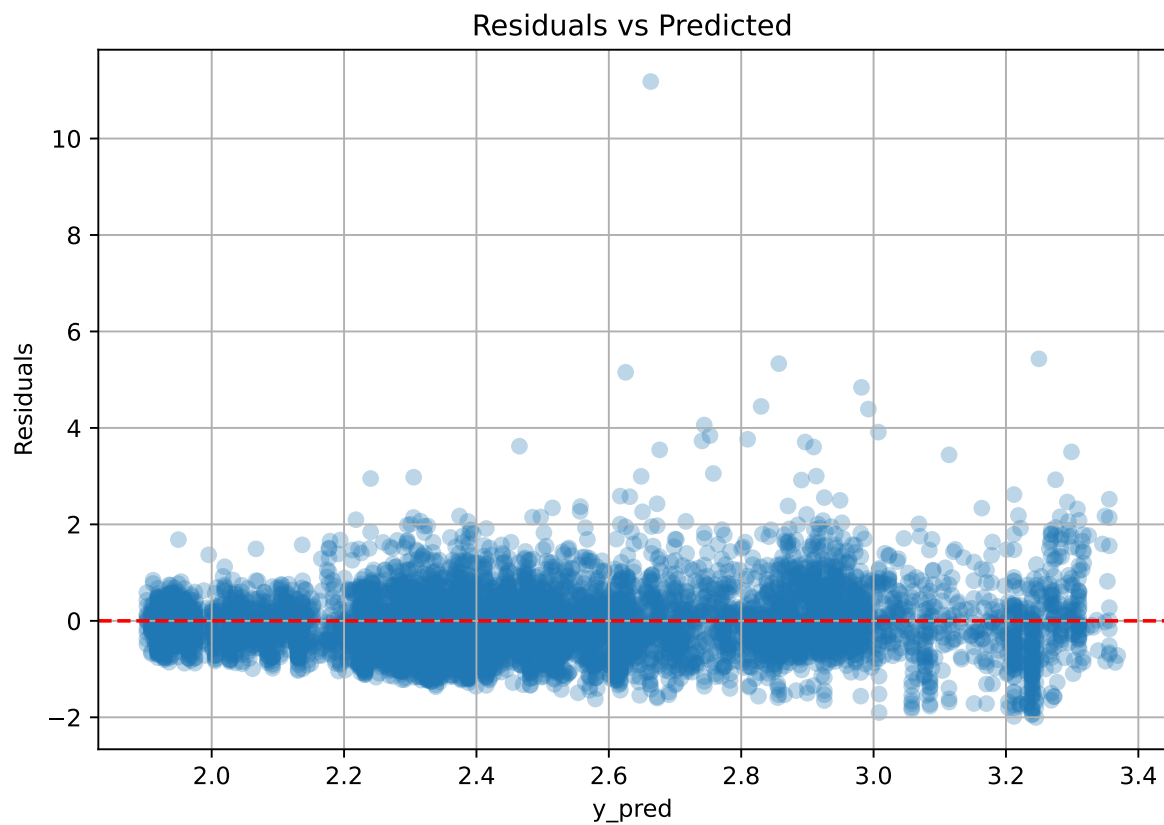
Rank 9: {'reg__n_estimators': 200, 'reg__max_depth': 6, 'reg__learning_rate': 0.027161406457723754, 'reg__num_leaves': 63, 'reg__subsample': 0.6337078221421627, 'reg__colsample_bytree': 0.6918944148437961, 'reg__min_child_samples': 48, 'reg__reg_alpha': 0.6699025827874394, 'reg__reg_lambda': 15.28400960101509, 'reg__min_split_gain': 2.718838676278421, 'reg__path_smooth': 3.463155289379471, 'reg__min_child_weight': 0.011623790595742341, 'reg__feature_fraction': 0.6364652062534526, 'reg__bagging_fraction': 0.5657415163463169, 'reg__bagging_freq': 1, 'reg__max_bin': 96, 'reg__min_data_per_group': 58}

Rank 10: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.030929604648394744, 'reg__num_leaves': 56, 'reg__subsample': 0.6717444455131925, 'reg__colsample_bytree': 0.709106691472642, 'reg__min_child_samples': 31, 'reg__reg_alpha': 0.0994930322429492, 'reg__reg_lambda': 45.661388443216254, 'reg__min_split_gain': 2.5043534309462965, 'reg__path_smooth': 4.458622026038931, 'reg__min_child_weight': 0.02797153118583578, 'reg__feature_fraction': 0.8567195521165623, 'reg__bagging_fraction': 0.5883840554531724, 'reg__bagging_freq': 3, 'reg__max_bin': 75, 'reg__min_data_per_group': 77}

2.4.6 Scatter



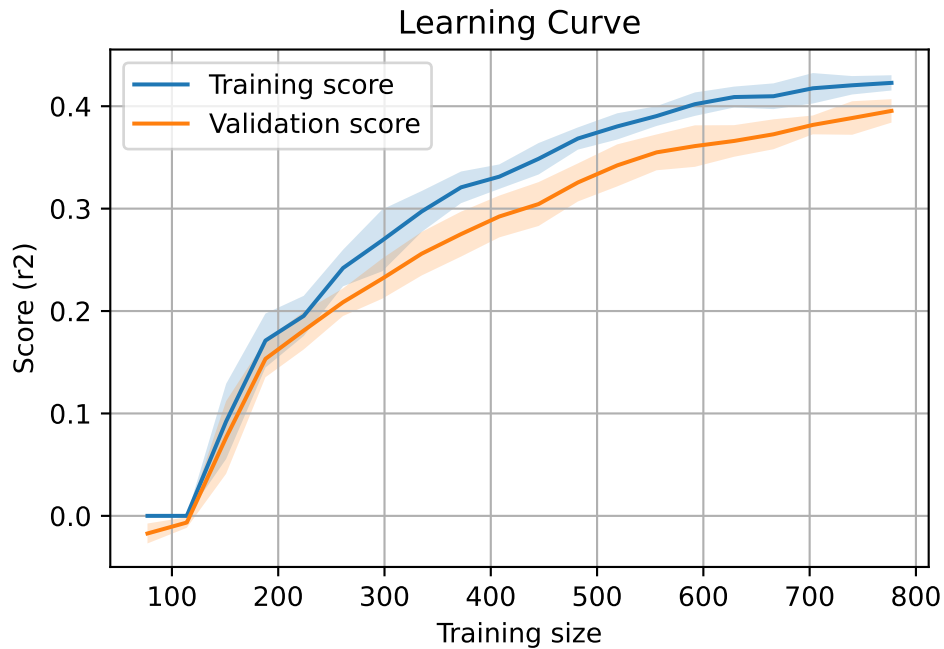
2.4.7 Scatter residuals



2.4.8 Learning curve — Group: 2025_Jun_2

`Len(group)` : 1587, `Len(training)` : 1111,

`Len(training_real)` : 778, `Len(test_of_training)` : 333, `Len(test)` : 476

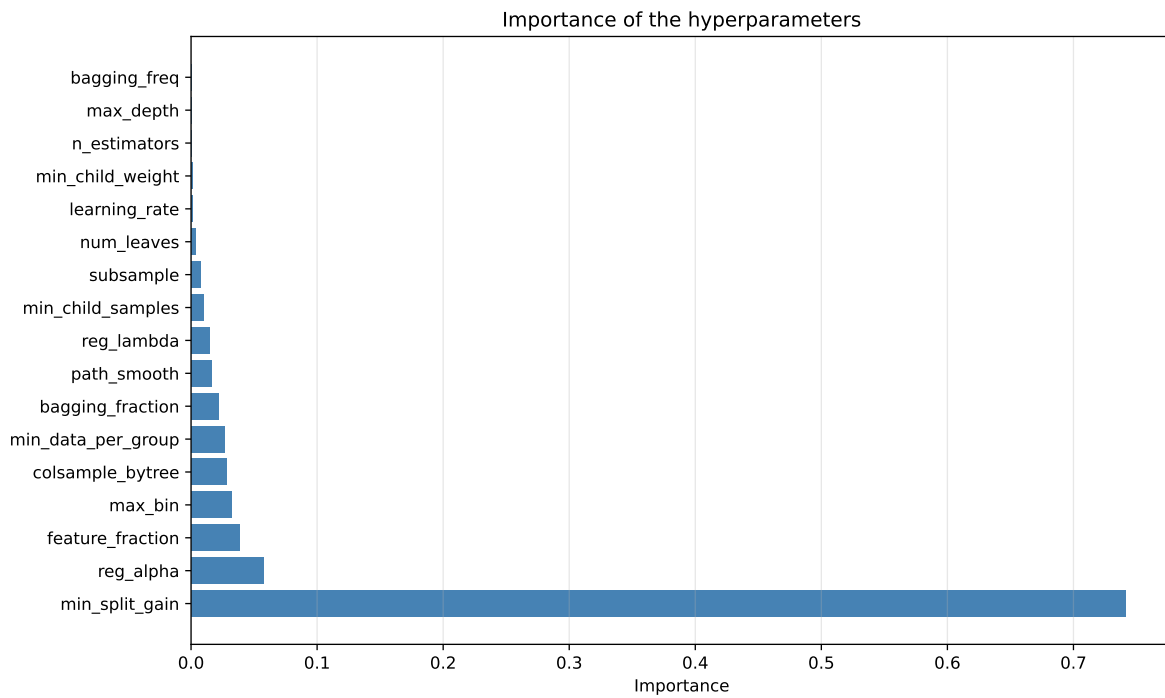
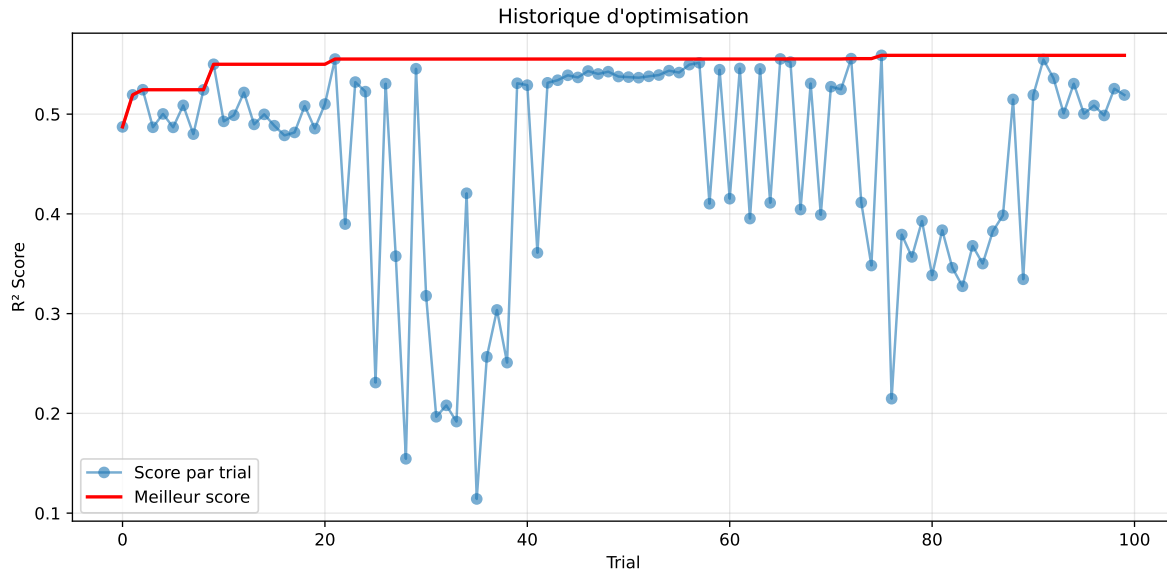


2.4.9 Gridsearch — Group: 2025_Sep_1

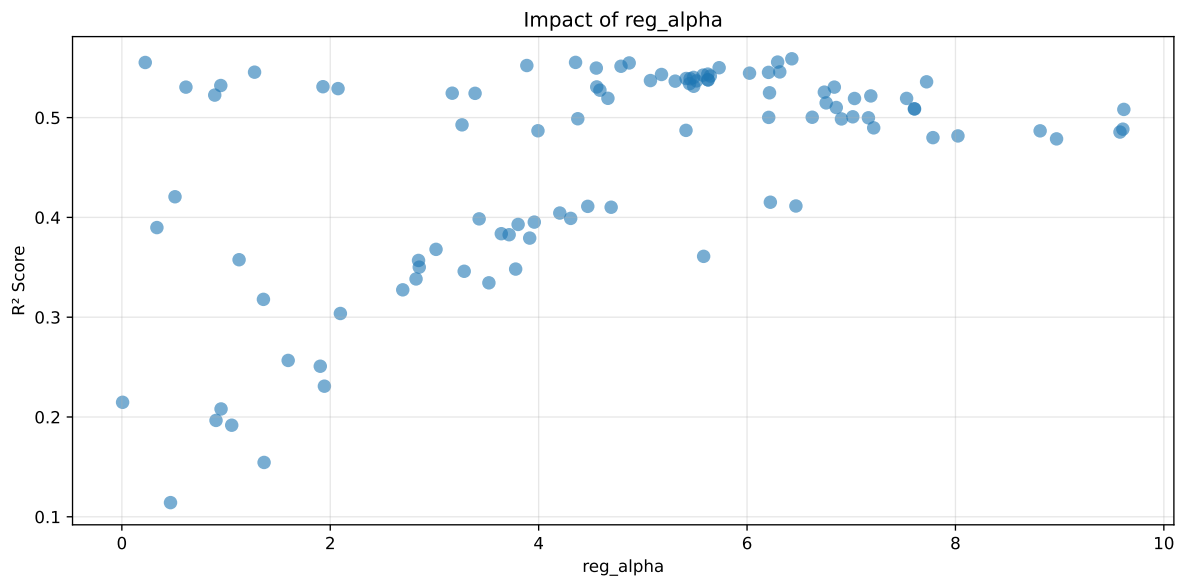
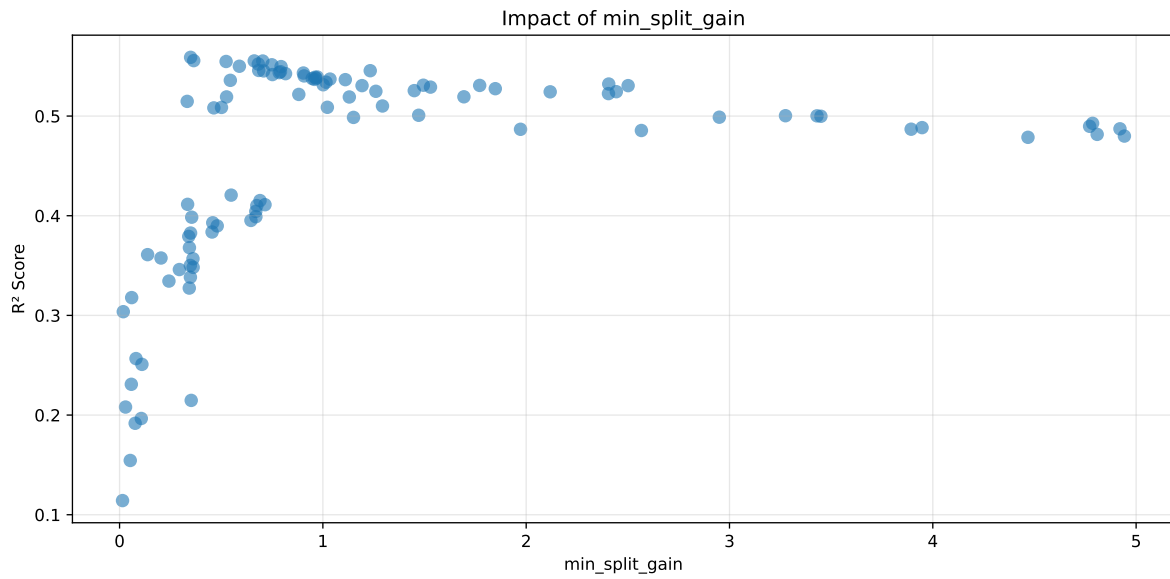
Len(group) : 3867, Len(training) : 2707, Len(test) : 1160

Distribution y_train vs y_test: y_train: mean=2.231, std=0.761, min=0.973, max=8.190
y_test: mean=2.216, std=0.723, min=1.110, max=6.920

===== TOP Importance FEATURES =====
feature importance r_NH3 256 r_CO2 215 r_umidity 202 r_THI 130 r_temperature 109

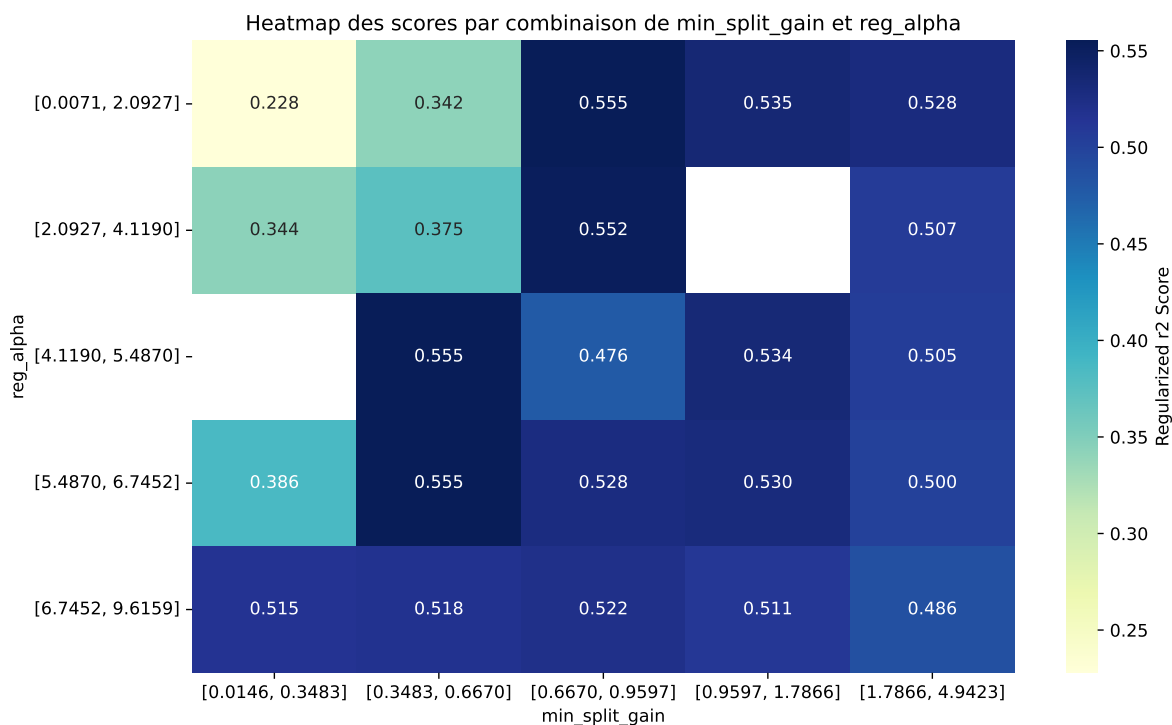


Paramètres avec >5% d'importance : ['min_split_gain', 'reg_alpha']



C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5847	0.5589	0.5614	0.0288	0.5589	0.5788	0.0199	1.0356	1.0462
2	0.5821	0.5556	0.5577	0.0302	0.5556	0.5799	0.0243	1.0437	1.0476
3	0.5818	0.5553	0.5531	0.0275	0.5553	0.5695	0.0142	1.0256	1.0477
4	0.581	0.5552	0.5644	0.028	0.5552	0.5789	0.0236	1.0426	1.0465
5	0.5814	0.5548	0.5602	0.0298	0.5548	0.5829	0.0281	1.0507	1.048
6	0.575	0.5522	0.5504	0.0273	0.5522	0.5699	0.0177	1.0321	1.0413
7	0.5775	0.5515	0.5489	0.0266	0.5515	0.5721	0.0207	1.0374	1.0472
8	0.5729	0.55	0.5506	0.0279	0.55	0.5724	0.0224	1.0407	1.0417
9	0.5704	0.5496	0.5484	0.027	0.5496	0.5613	0.0117	1.0212	1.0378
10	0.5689	0.5458	0.5467	0.0289	0.5458	0.5626	0.0168	1.0308	1.0424
---	---	---	---	---	---	---	---	---	---
100	0.7124	0.6127	0.6042	0.0232	0.1142	0.6191	0.0064	1.0105	1.1627

Hyperparameters:

Rank 1: {'reg__n_estimators': 300, 'reg__max_depth': 7, 'reg__learning_rate': 0.04586961481524148, 'reg__num_leaves': 44, 'reg__subsample': 0.8792046887463042, 'reg__colsample_bytree': 0.8067772005495943, 'reg__min_child_samples': 32, 'reg__reg_alpha': 6.429714520496289, 'reg__reg_lambda': 26.760036484709143, 'reg__min_split_gain': 0.34944763860742517, 'reg__path_smooth': 1.246289179875447, 'reg__min_child_weight':

0.001802686532202279, 'reg__feature_fraction': 0.7170582296042923, 'reg__bagging_fraction': 0.875921412205204, 'reg__bagging_freq': 3, 'reg__max_bin': 135, 'reg__min_data_per_group': 141}

Rank 2: {'reg__n_estimators': 300, 'reg__max_depth': 7, 'reg__learning_rate': 0.01838619274467115, 'reg__num_leaves': 45, 'reg__subsample': 0.7783063030242985, 'reg__colsample_bytree': 0.5963367304239072, 'reg__min_child_samples': 33, 'reg__reg_alpha': 6.2935495394980565, 'reg__reg_lambda': 27.827366241620826, 'reg__min_split_gain': 0.3647982569543269, 'reg__path_smooth': 1.2629906599204126, 'reg__min_child_weight': 0.001620215051881482, 'reg__feature_fraction': 0.7150632477784933, 'reg__bagging_fraction': 0.8764720235681505, 'reg__bagging_freq': 3, 'reg__max_bin': 175, 'reg__min_data_per_group': 156}

Rank 3: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.018174212833247053, 'reg__num_leaves': 50, 'reg__subsample': 0.8376542209118542, 'reg__colsample_bytree': 0.5639475406002377, 'reg__min_child_samples': 18, 'reg__reg_alpha': 4.35419622089144, 'reg__reg_lambda': 9.191893021722676, 'reg__min_split_gain': 0.6624530416229466, 'reg__path_smooth': 1.346442953777776, 'reg__min_child_weight': 0.0023490389234422627, 'reg__feature_fraction': 0.5533621232906564, 'reg__bagging_fraction': 0.89510286546781, 'reg__bagging_freq': 3, 'reg__max_bin': 177, 'reg__min_data_per_group': 161}

Rank 4: {'reg__n_estimators': 200, 'reg__max_depth': 3, 'reg__learning_rate': 0.03576036522379145, 'reg__num_leaves': 60, 'reg__subsample': 0.6046547689132914, 'reg__colsample_bytree': 0.5192706723807549, 'reg__min_child_samples': 43, 'reg__reg_alpha': 0.2245153030461302, 'reg__reg_lambda': 0.169446723413349, 'reg__min_split_gain': 0.7042319686587137, 'reg__path_smooth': 2.39388161018156, 'reg__min_child_weight': 0.04269700829138959, 'reg__feature_fraction': 0.661522008936092, 'reg__bagging_fraction': 0.7686347333122134, 'reg__bagging_freq': 5, 'reg__max_bin': 67, 'reg__min_data_per_group': 196}

Rank 5: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.05287396839316615, 'reg__num_leaves': 36, 'reg__subsample': 0.8931986084184732, 'reg__colsample_bytree': 0.5882403193817589, 'reg__min_child_samples': 38, 'reg__reg_alpha': 4.868886759497217, 'reg__reg_lambda': 23.782492129199923, 'reg__min_split_gain': 0.5239818065504132, 'reg__path_smooth': 0.49872335585851935, 'reg__min_child_weight': 0.007454704937637996, 'reg__feature_fraction': 0.7352847659184015, 'reg__bagging_fraction': 0.7902759600304261, 'reg__bagging_freq': 1, 'reg__max_bin': 148, 'reg__min_data_per_group': 150}

Rank 6: {'reg__n_estimators': 400, 'reg__max_depth': 6, 'reg__learning_rate': 0.018527471941118704, 'reg__num_leaves': 51, 'reg__subsample': 0.8779873413987997, 'reg__colsample_bytree': 0.639521622644681, 'reg__min_child_samples': 17, 'reg__reg_alpha': 3.88653060897204, 'reg__reg_lambda': 20.379033287871597, 'reg__min_split_gain': 0.682443912716599, 'reg__path_smooth': 1.247570046914802, 'reg__min_child_weight':

0.00195543491597943, 'reg__feature_fraction': 0.5637164350912695, 'reg__bagging_fraction': 0.8678540038923934, 'reg__bagging_freq': 3, 'reg__max_bin': 186, 'reg__min_data_per_group': 161}

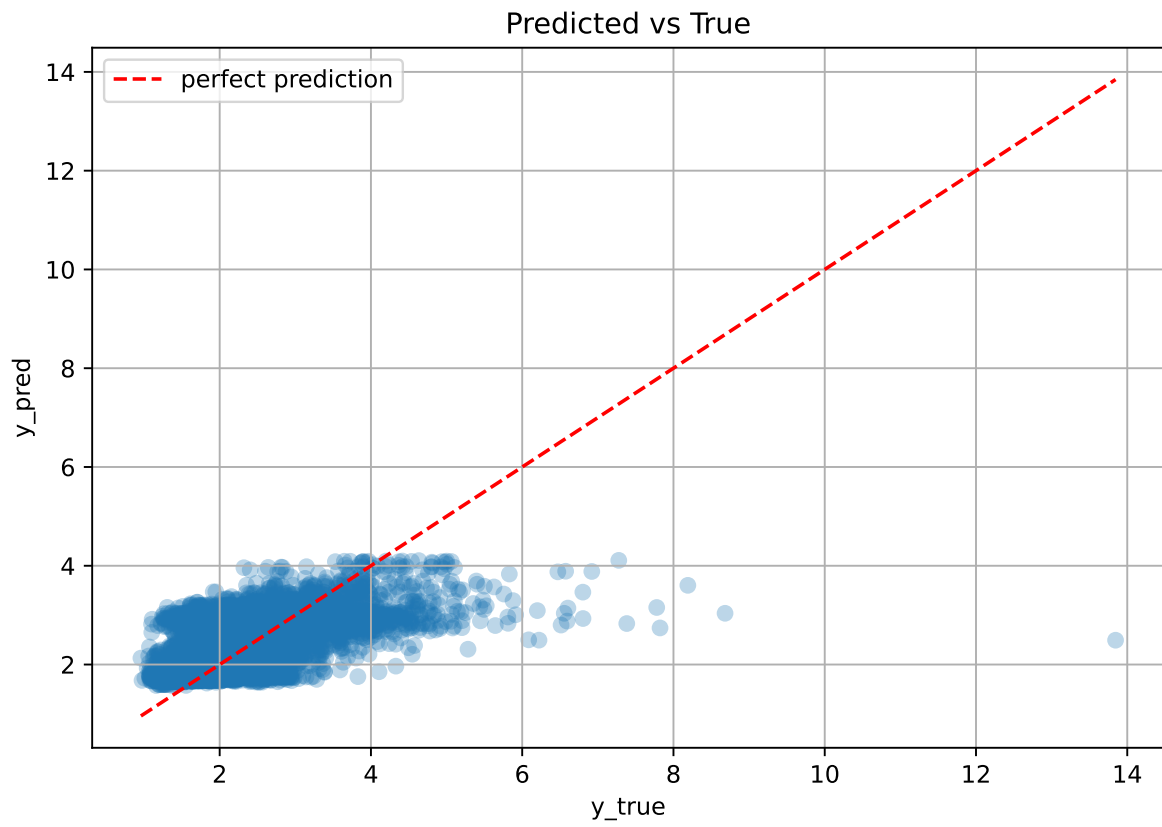
Rank 7: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.023131784194713907, 'reg__num_leaves': 50, 'reg__subsample': 0.8533948624682515, 'reg__colsample_bytree': 0.5385048427980043, 'reg__min_child_samples': 15, 'reg__reg_alpha': 4.790299511639633, 'reg__reg_lambda': 4.277494134579829, 'reg__min_split_gain': 0.7488484624470162, 'reg__path_smooth': 2.460711457939598, 'reg__min_child_weight': 0.002628478896510891, 'reg__feature_fraction': 0.6763989561127554, 'reg__bagging_fraction': 0.8576039057659635, 'reg__bagging_freq': 2, 'reg__max_bin': 159, 'reg__min_data_per_group': 164}

Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.0198409335437428, 'reg__num_leaves': 54, 'reg__subsample': 0.7921947380340804, 'reg__colsample_bytree': 0.8961439468652022, 'reg__min_child_samples': 24, 'reg__reg_alpha': 5.732090279855942, 'reg__reg_lambda': 2.6528884739884964, 'reg__min_split_gain': 0.5893336686845335, 'reg__path_smooth': 1.3188100602296386, 'reg__min_child_weight': 0.006045324308738134, 'reg__feature_fraction': 0.5720360757070723, 'reg__bagging_fraction': 0.8051780319727397, 'reg__bagging_freq': 4, 'reg__max_bin': 136, 'reg__min_data_per_group': 51}

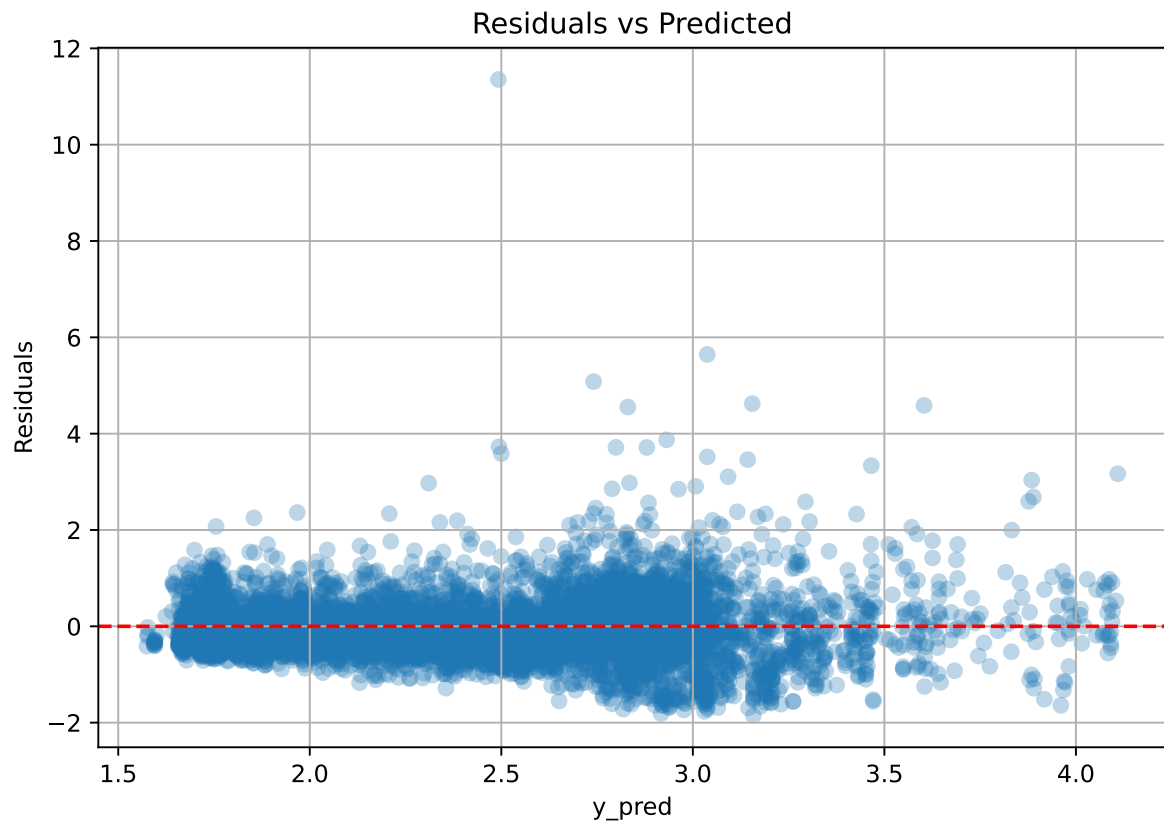
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 4, 'reg__learning_rate': 0.0480929828178055, 'reg__num_leaves': 46, 'reg__subsample': 0.8556717770166526, 'reg__colsample_bytree': 0.5957307810520711, 'reg__min_child_samples': 15, 'reg__reg_alpha': 4.5533227050089655, 'reg__reg_lambda': 5.575283994278957, 'reg__min_split_gain': 0.7944508980463263, 'reg__path_smooth': 1.3225035921326913, 'reg__min_child_weight': 0.0027845430635191006, 'reg__feature_fraction': 0.6834672241349478, 'reg__bagging_fraction': 0.856035176038416, 'reg__bagging_freq': 2, 'reg__max_bin': 159, 'reg__min_data_per_group': 167}

Rank 10: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.018380005312660194, 'reg__num_leaves': 50, 'reg__subsample': 0.7759737741324897, 'reg__colsample_bytree': 0.8201499443596025, 'reg__min_child_samples': 16, 'reg__reg_alpha': 6.3140488723603205, 'reg__reg_lambda': 3.3676159695668106, 'reg__min_split_gain': 0.6844350529648583, 'reg__path_smooth': 1.2931519795801454, 'reg__min_child_weight': 0.0019038829775721944, 'reg__feature_fraction': 0.5631150831583918, 'reg__bagging_fraction': 0.8981513693873859, 'reg__bagging_freq': 3, 'reg__max_bin': 188, 'reg__min_data_per_group': 133}

2.4.10 Scatter



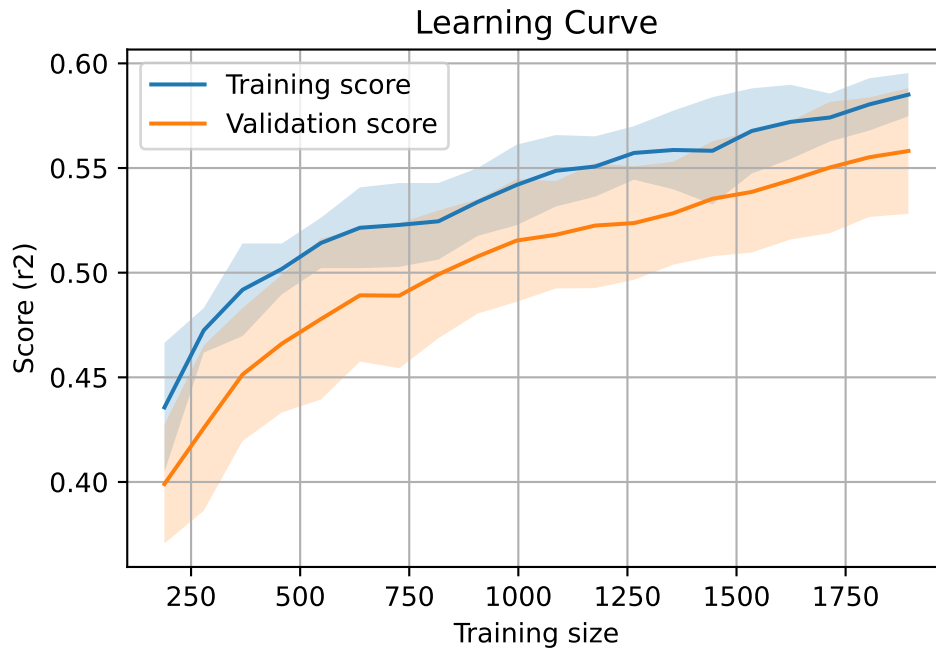
2.4.11 Scatter residuals



2.4.12 Learning curve — Group: 2025_Sep_1

`Len(group)` : 3867, `Len(training)` : 2707,

`Len(training_real)` : 1895, `Len(test_of_training)` : 812, `Len(test)` : 1160

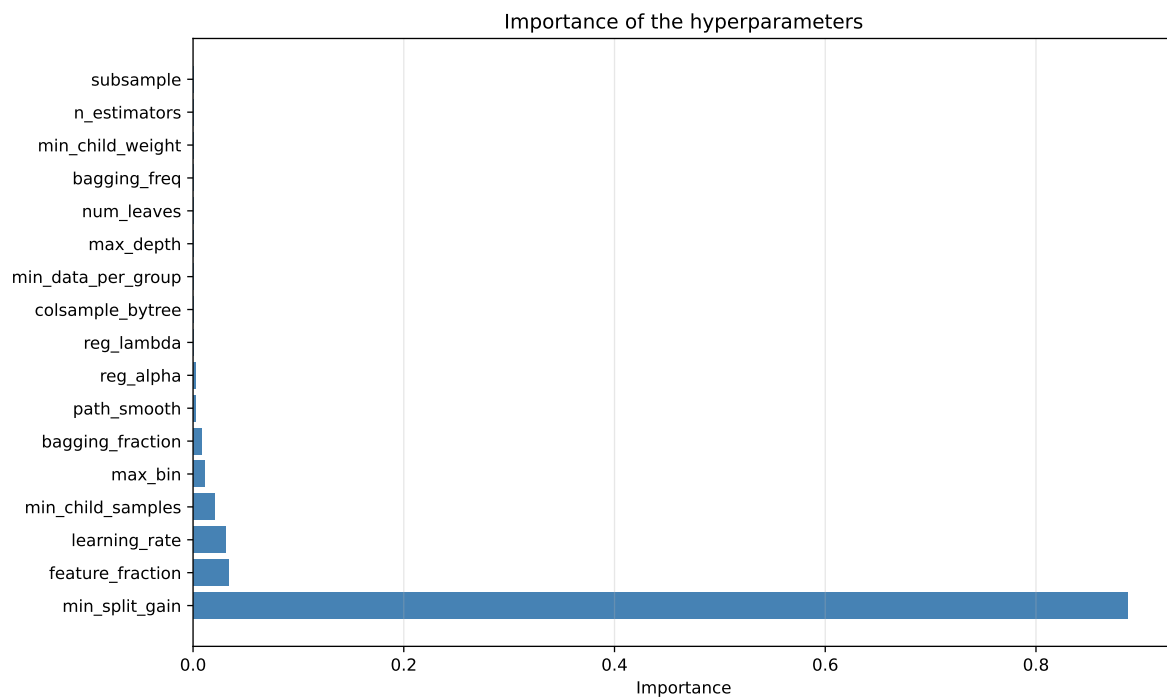
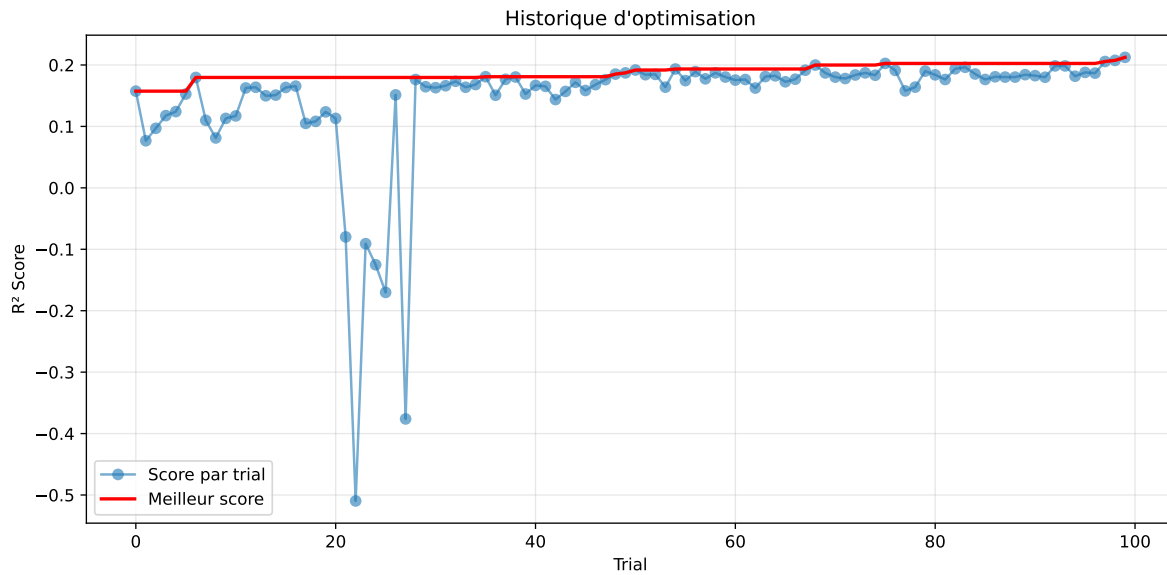


2.4.13 Gridsearch — Group: 2025_Sep_2

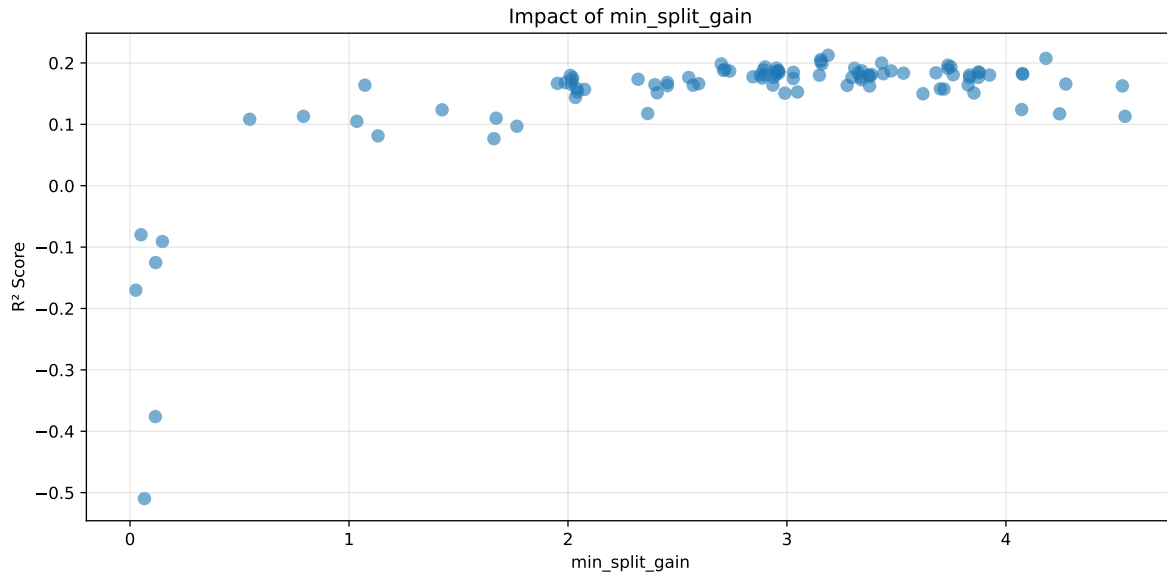
Len(group) : 1129, Len(training) : 791, Len(test) : 338

Distribution y_train vs y_test: y_train: mean=1.926, std=0.742, min=0.959, max=6.804
y_test: mean=1.945, std=0.805, min=1.070, max=7.822

===== TOP Importance FEATURES =====
feature importance r_umidity 74 r_THI 19 r_temperature 10 r_CO2 7 r_NH3 5



Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.3644	0.3391	0.3463	0.0321	0.2124	0.345	0.0059	1.0175	1.0747
2	0.3391	0.3172	0.321	0.0291	0.2075	0.3253	0.0082	1.0257	1.0691
3	0.3523	0.3279	0.3358	0.0306	0.2057	0.3432	0.0153	1.0466	1.0746
4	0.3517	0.3269	0.3337	0.0282	0.2025	0.3529	0.026	1.0795	1.0761
5	0.3497	0.3247	0.3296	0.0305	0.1999	0.3384	0.0137	1.0422	1.0769
6	0.3685	0.3401	0.3516	0.0311	0.1985	0.3571	0.0169	1.0497	1.0833
7	0.3686	0.3403	0.3522	0.0354	0.1985	0.3414	0.0011	1.0034	1.0833
8	0.3502	0.3246	0.3325	0.0316	0.1963	0.3408	0.0163	1.0501	1.079
9	0.3359	0.3123	0.3202	0.0296	0.1938	0.3357	0.0234	1.075	1.0759
10	0.3467	0.3211	0.3218	0.0243	0.1935	0.35	0.0289	1.0899	1.0795
---	---	---	---	---	---	---	---	---	---
100	0.6704	0.4737	0.5668	0.027	-0.5097	0.5899	0.1161	1.2451	1.4152

Hyperparameters:

Rank 1: {'reg__n_estimators': 800, 'reg__max_depth': 5, 'reg__learning_rate': 0.031693005441565364, 'reg__num_leaves': 33, 'reg__subsample': 0.6703171705115504, 'reg__colsample_bytree': 0.8797452573539859, 'reg__min_child_samples': 50, 'reg__reg_alpha': 4.192680554896904, 'reg__reg_lambda': 8.192105751992184, 'reg__min_split_gain': 3.1879572488937193, 'reg__path_smooth': 4.361836772449315, 'reg__min_child_weight': 0.01673526637320184, 'reg__feature_fraction': 0.8227193097608887, 'reg__bagging_fraction': 0.720959722246986, 'reg__bagging_freq': 6, 'reg__max_bin': 81, 'reg__min_data_per_group': 93}

Rank 2: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.032651133709328234, 'reg__num_leaves': 33, 'reg__subsample': 0.7674590586515657, 'reg__colsample_bytree': 0.896222662719978, 'reg__min_child_samples': 50, 'reg__reg_alpha': 4.427423397033791, 'reg__reg_lambda': 14.582595287713215, 'reg__min_split_gain': 4.182757618473184, 'reg__path_smooth': 4.364188385467188, 'reg__min_child_weight': 0.015777988490782432, 'reg__feature_fraction': 0.8181980797669735, 'reg__bagging_fraction': 0.7208372199599064, 'reg__bagging_freq': 6, 'reg__max_bin': 78, 'reg__min_data_per_group': 96}

Rank 3: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.03457398157568554, 'reg__num_leaves': 33, 'reg__subsample': 0.6676682491048079, 'reg__colsample_bytree': 0.8862985246814155, 'reg__min_child_samples': 39, 'reg__reg_alpha': 5.620486485758806, 'reg__reg_lambda': 14.573005409794586, 'reg__min_split_gain': 3.1546741872944293, 'reg__path_smooth': 4.37330769932996, 'reg__min_child_weight': 0.014604713160344875, 'reg__feature_fraction': 0.8180146585246668, 'reg__bagging_fraction': 0.7213048040841497, 'reg__bagging_freq': 6, 'reg__max_bin': 76, 'reg__min_data_per_group': 93}

Rank 4: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.049574058913694916, 'reg__num_leaves': 42, 'reg__subsample': 0.7165447635695416, 'reg__colsample_bytree': 0.7160721294572463, 'reg__min_child_samples': 43, 'reg__reg_alpha': 5.161545050132349, 'reg__reg_lambda': 6.72147890381704, 'reg__min_split_gain': 3.154266313410653, 'reg__path_smooth': 3.929334377912763, 'reg__min_child_weight': 0.18023271764915536, 'reg__feature_fraction': 0.8617944338241813, 'reg__bagging_fraction': 0.6605787608831705, 'reg__bagging_freq': 5, 'reg__max_bin': 92, 'reg__min_data_per_group': 103}

Rank 5: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.05087998128156428, 'reg__num_leaves': 44, 'reg__subsample': 0.6880511895068195, 'reg__colsample_bytree': 0.8265820572685583, 'reg__min_child_samples': 41, 'reg__reg_alpha': 4.6991651265707235, 'reg__reg_lambda': 6.06332372016581, 'reg__min_split_gain': 3.432385409112421, 'reg__path_smooth': 4.913165952419669, 'reg__min_child_weight': 0.0828763201798909, 'reg__feature_fraction': 0.8816639086681983, 'reg__bagging_fraction': 0.6772345281523973, 'reg__bagging_freq': 4, 'reg__max_bin': 71, 'reg__min_data_per_group': 89}

Rank 6: {'reg__n_estimators': 700, 'reg__max_depth': 6, 'reg__learning_rate': 0.04475821432676306, 'reg__num_leaves': 37, 'reg__subsample': 0.7561936847325361, 'reg__colsample_bytree': 0.8666785016782205, 'reg__min_child_samples': 39, 'reg__reg_alpha': 4.366864407636709, 'reg__reg_lambda': 9.400075634443494, 'reg__min_split_gain': 2.6996230346615926, 'reg__path_smooth': 3.6594207607539273, 'reg__min_child_weight': 0.11329327269756091, 'reg__feature_fraction': 0.816649415035641, 'reg__bagging_fraction': 0.6918045977847437, 'reg__bagging_freq': 6, 'reg__max_bin': 74, 'reg__min_data_per_group': 73}

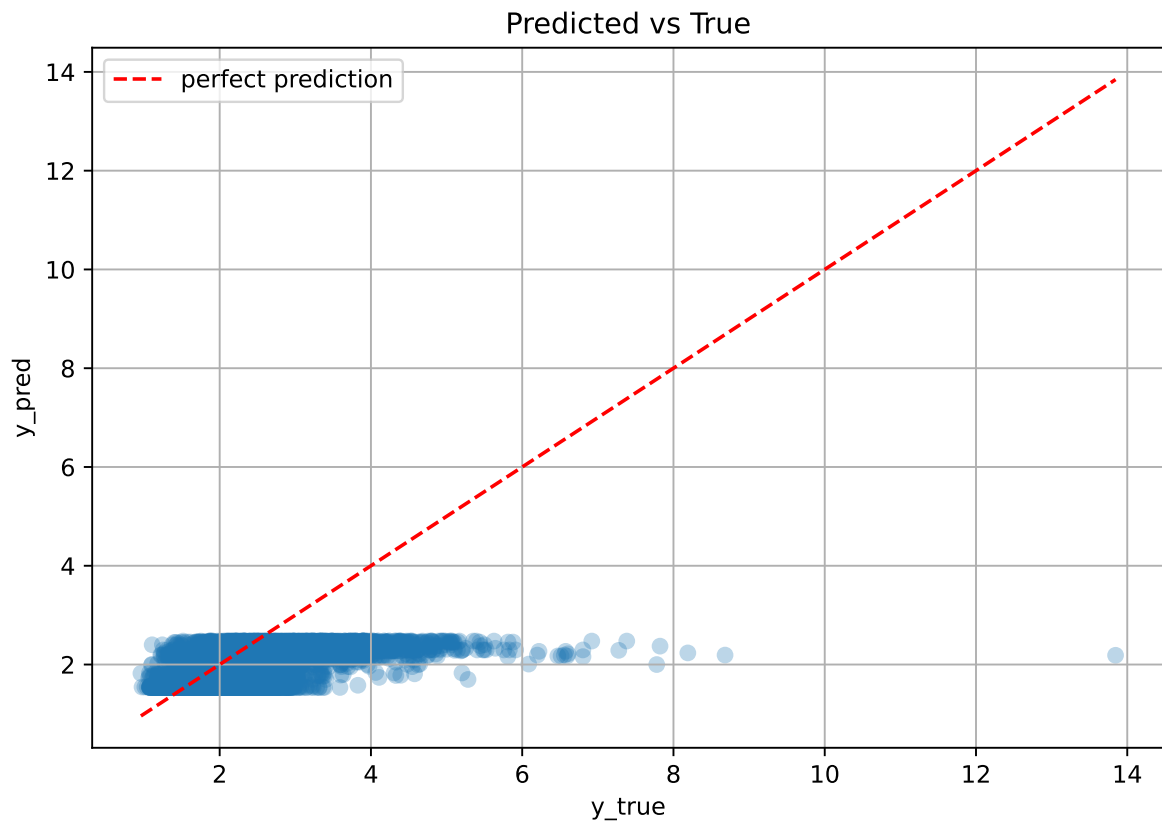
Rank 7: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.030775061216995546, 'reg__num_leaves': 39, 'reg__subsample': 0.7583181907821999, 'reg__colsample_bytree': 0.8716627496677377, 'reg__min_child_samples': 39, 'reg__reg_alpha': 4.343268596269939, 'reg__reg_lambda': 9.018657156544403, 'reg__min_split_gain': 3.1597106291727677, 'reg__path_smooth': 3.8940123939308244, 'reg__min_child_weight': 0.01821853151379943, 'reg__feature_fraction': 0.8472507120707342, 'reg__bagging_fraction': 0.7237925750901503, 'reg__bagging_freq': 6, 'reg__max_bin': 76, 'reg__min_data_per_group': 95}

Rank 8: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.029835337046875965, 'reg__num_leaves': 37, 'reg__subsample': 0.738103784609455, 'reg__colsample_bytree': 0.8776116256668902, 'reg__min_child_samples': 49, 'reg__reg_alpha': 4.332616271751154, 'reg__reg_lambda': 9.122473815321069, 'reg__min_split_gain': 3.7352164695672694, 'reg__path_smooth': 4.02227158278261, 'reg__min_child_weight': 0.050952470272790235, 'reg__feature_fraction': 0.8521709574937426, 'reg__bagging_fraction': 0.673256541128407, 'reg__bagging_freq': 5, 'reg__max_bin': 74, 'reg__min_data_per_group': 101}

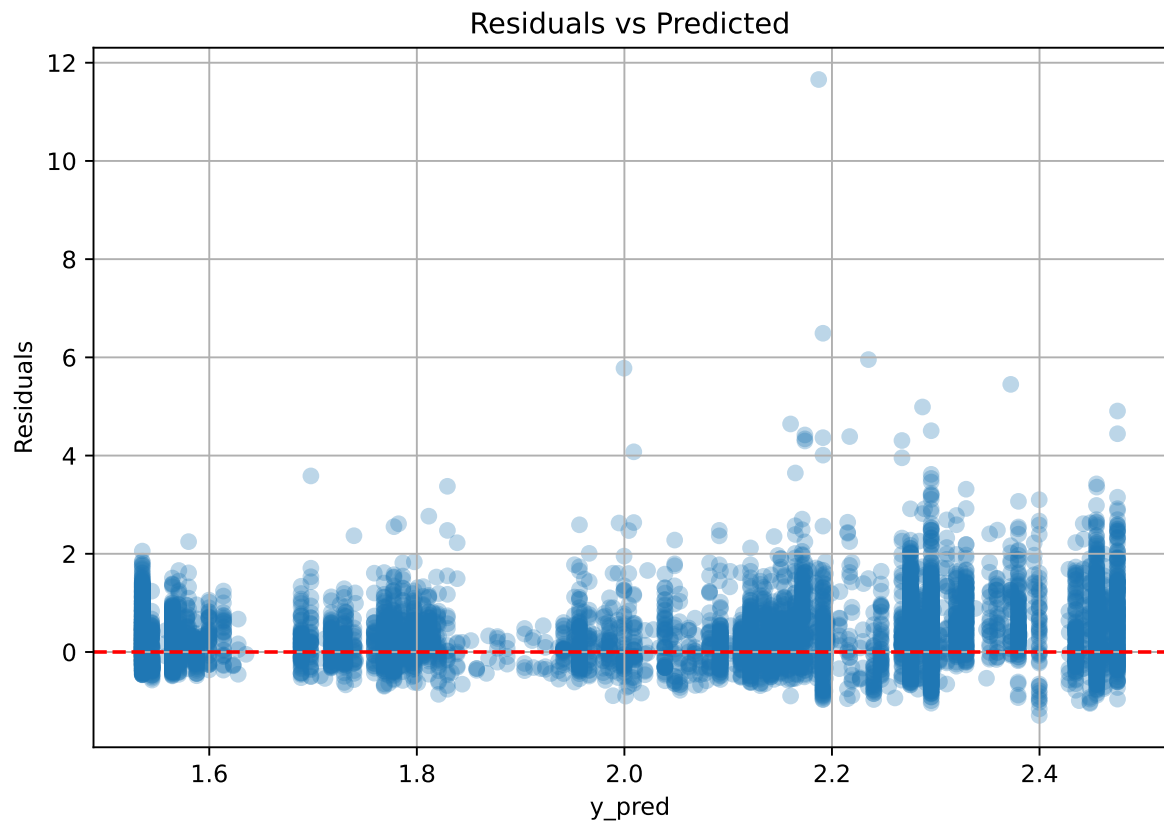
Rank 9: {'reg__n_estimators': 800, 'reg__max_depth': 6, 'reg__learning_rate': 0.0454010091508288, 'reg__num_leaves': 36, 'reg__subsample': 0.718490021532365, 'reg__colsample_bytree': 0.8613220086722003, 'reg__min_child_samples': 49, 'reg__reg_alpha': 5.485562052491754, 'reg__reg_lambda': 5.9309080369665494, 'reg__min_split_gain': 3.748427867583576, 'reg__path_smooth': 4.7607154333433135, 'reg__min_child_weight': 0.03921849425414797, 'reg__feature_fraction': 0.8522749815627781, 'reg__bagging_fraction': 0.6835494448478991, 'reg__bagging_freq': 5, 'reg__max_bin': 74, 'reg__min_data_per_group': 101}

Rank 10: {'reg__n_estimators': 700, 'reg__max_depth': 7, 'reg__learning_rate': 0.059227221324848686, 'reg__num_leaves': 47, 'reg__subsample': 0.5002696428414776, 'reg__colsample_bytree': 0.7814717090050911, 'reg__min_child_samples': 48, 'reg__reg_alpha': 4.754834937535451, 'reg__reg_lambda': 9.627700330848308, 'reg__min_split_gain': 2.9007976138363882, 'reg__path_smooth': 4.697201992038786, 'reg__min_child_weight': 0.3419569269231291, 'reg__feature_fraction': 0.8777215500852203, 'reg__bagging_fraction': 0.5951883433511084, 'reg__bagging_freq': 4, 'reg__max_bin': 110, 'reg__min_data_per_group': 110}

2.4.14 Scatter



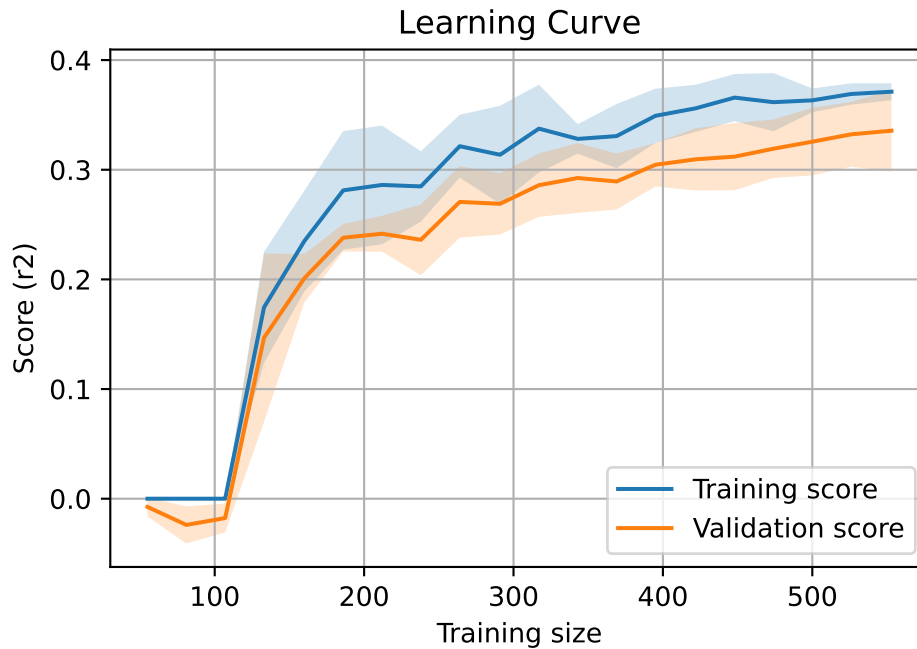
2.4.15 Scatter residuals



2.4.16 Learning curve — Group: 2025_Sep_2

`Len(group)` : 1129, `Len(training)` : 791,

`Len(training_real)` : 554, `Len(test_of_training)` : 237, `Len(test)` : 338



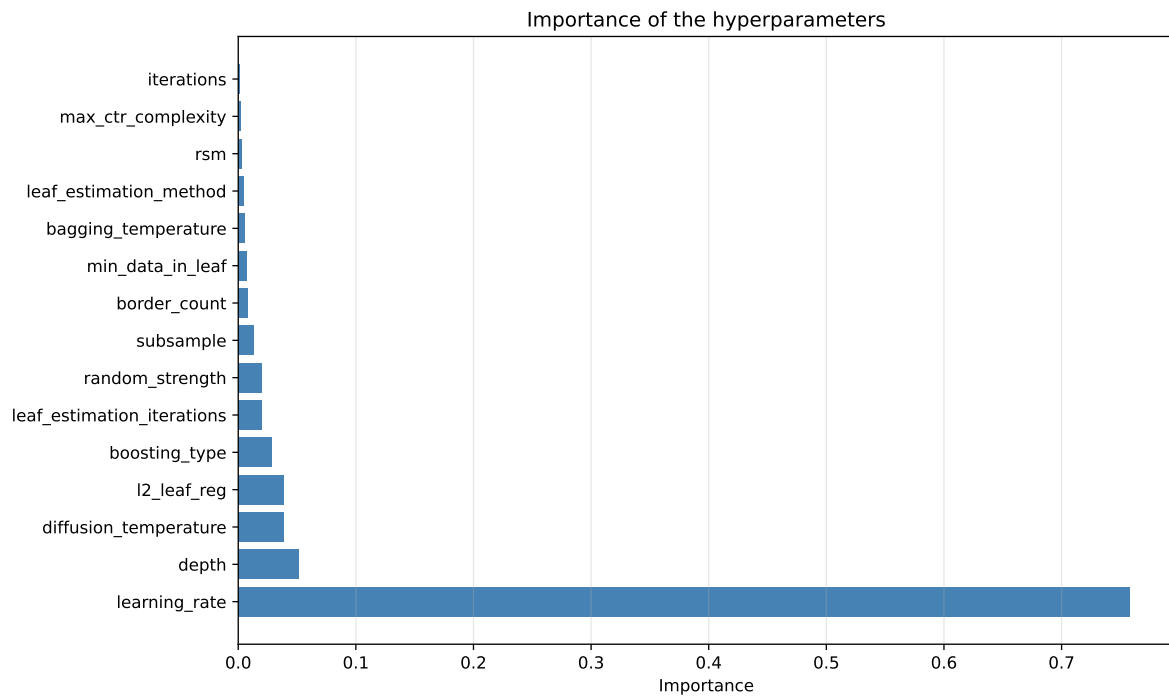
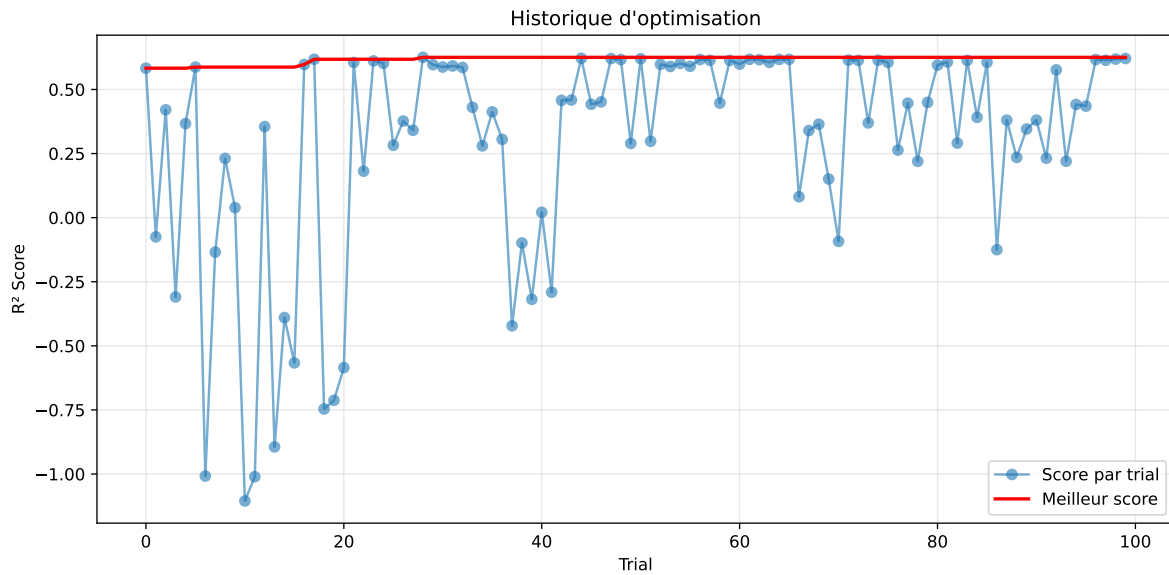
2.5 Model : CatBoost

2.5.1 Gridsearch — Group: 2025_Jun_1

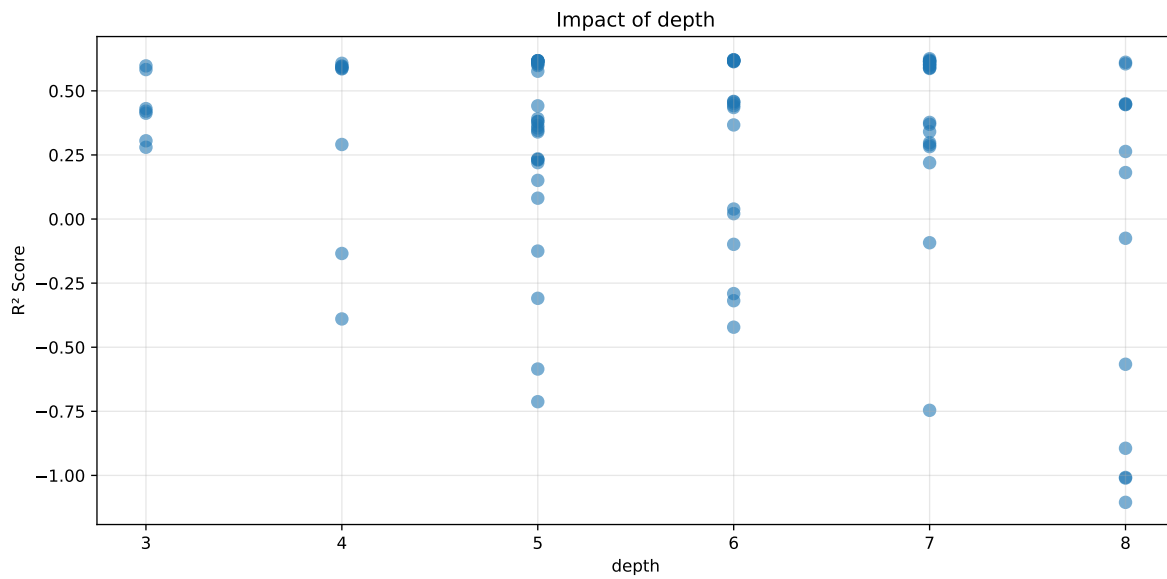
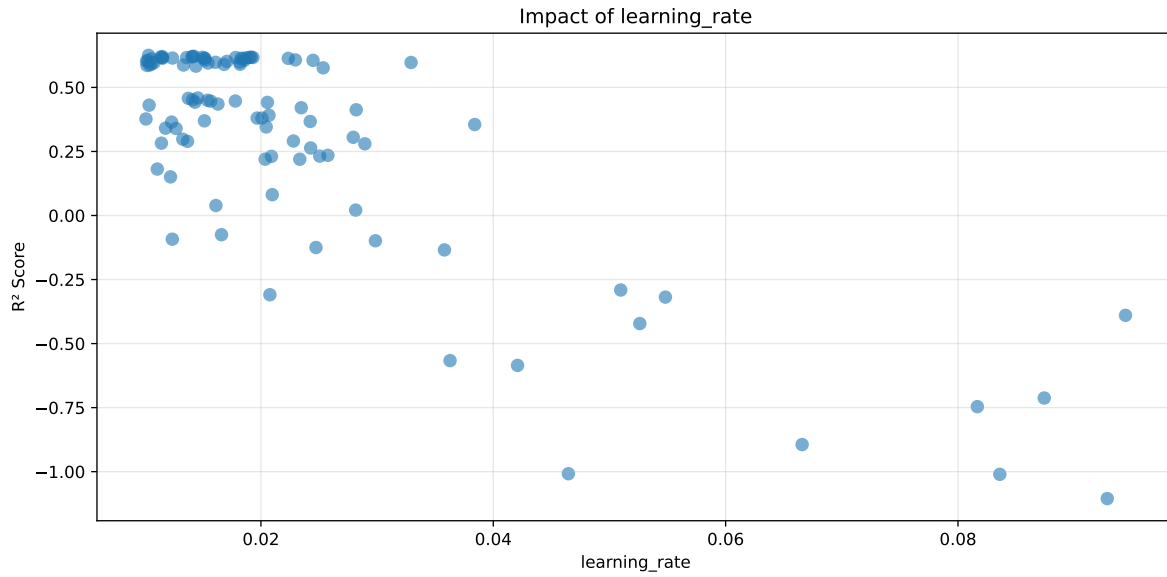
Len(group) : 1585, Len(training) : 1110, Len(test) : 475

Distribution y_train vs y_test: y_train: mean=2.772, std=0.805, min=1.432, max=13.845
y_test: mean=2.807, std=0.764, min=1.373, max=8.682

===== TOP Importance FEATURES =====
feature importance r_NH3 29.314369 r_CO2 19.073909 r_umidity 18.073284 r_THI 17.348601
r_temperature 16.189836

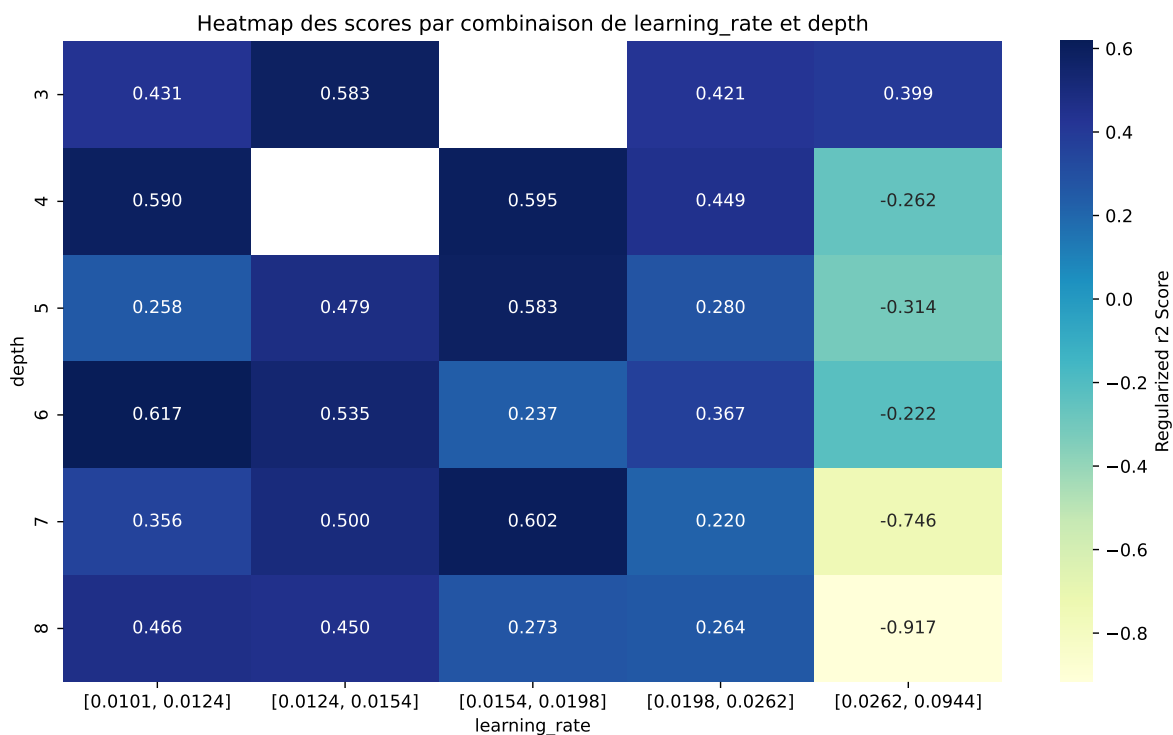


Paramètres avec >5% d'importance : ['learning_rate', 'depth']



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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6564	0.6253	0.6463	0.0409	0.6253	0.6693	0.044	1.0704	1.0498
2	0.6526	0.6216	0.6465	0.0393	0.6216	0.6674	0.0458	1.0737	1.05
3	0.6378	0.6207	0.6371	0.0432	0.6207	0.6561	0.0354	1.057	1.0276
4	0.6477	0.6204	0.6384	0.0392	0.6204	0.6599	0.0396	1.0638	1.0441
5	0.6503	0.6198	0.6352	0.0371	0.6198	0.6577	0.0379	1.0611	1.0492
6	0.6374	0.6185	0.6371	0.0412	0.6185	0.6588	0.0404	1.0653	1.0306
7	0.6481	0.6178	0.6391	0.0391	0.6178	0.6636	0.0458	1.0741	1.049
8	0.6327	0.6176	0.6395	0.0419	0.6176	0.6628	0.0452	1.0732	1.0244
9	0.6404	0.6176	0.6413	0.0402	0.6176	0.662	0.0445	1.072	1.0369
10	0.6402	0.6171	0.6411	0.0417	0.6171	0.6579	0.0408	1.0661	1.0374
---	---	---	---	---	---	---	---	---	---
100	0.9425	0.6013	0.6598	0.0398	-1.1051	0.7043	0.103	1.1713	1.5676

Hyperparameters:

Rank 1: {'reg__iterations': 800, 'reg__depth': 7, 'reg__learning_rate': 0.010324614454215161, 'reg__l2_leaf_reg': 31.848886318806652, 'reg__bagging_temperature': 3.162879577759222, 'reg__random_strength': 1.806396537645935, 'reg__subsample': 0.7298954706942375, 'reg__min_data_in_leaf': 40, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7467118636063802, 'reg__border_count': 75, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 2135.5709600254236}

Rank 2: {'reg__iterations': 700, 'reg__depth': 6, 'reg__learning_rate': 0.014221822613157666, 'reg__l2_leaf_reg': 46.4512846482168, 'reg__bagging_temperature': 3.7373942526313737, 'reg__random_strength': 1.1819117973398297, 'reg__subsample': 0.6994528940720028, 'reg__min_data_in_leaf': 41, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7110244131272412, 'reg__border_count': 105, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 1253.8218396155798}

Rank 3: {'reg__iterations': 700, 'reg__depth': 6, 'reg__learning_rate': 0.011505802370269344, 'reg__l2_leaf_reg': 47.82406560001001, 'reg__bagging_temperature': 0.21985893313733446, 'reg__random_strength': 0.7290979412226768, 'reg__subsample': 0.5620960240696916, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.7409377495503329, 'reg__border_count': 77, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 2364.756646314921}

Rank 4: {'reg__iterations': 700, 'reg__depth': 6, 'reg__learning_rate': 0.014109447860779502, 'reg__l2_leaf_reg': 46.01708062907609, 'reg__bagging_temperature': 0.8630921611288285, 'reg__random_strength': 2.529012806305018, 'reg__subsample': 0.5375388623048861, 'reg__min_data_in_leaf': 42, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7791484321592506, 'reg__border_count': 103, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 1221.501531877722}

Rank 5: {'reg__iterations': 600, 'reg__depth': 7, 'reg__learning_rate': 0.014066274972452171, 'reg__l2_leaf_reg': 47.13455720513827, 'reg__bagging_temperature': 0.9808711803726329, 'reg__random_strength': 2.4608524869881485, 'reg__subsample': 0.5398393813476237, 'reg__min_data_in_leaf': 43, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7883914663425028, 'reg__border_count': 107, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 2698.0445682771247}

Rank 6: {'reg__iterations': 700, 'reg__depth': 6, 'reg__learning_rate': 0.011369714242516718, 'reg__l2_leaf_reg': 47.74602401547126, 'reg__bagging_temperature': 2.1882821138921233, 'reg__random_strength': 0.7336293034679462, 'reg__subsample': 0.5600234939820646, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.7409022561274136, 'reg__border_count': 93, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 2322.6687670541055}

Rank 7: {'reg__iterations': 700, 'reg__depth': 5, 'reg__learning_rate': 0.01494046708558781, 'reg__l2_leaf_reg': 48.68298431611268, 'reg__bagging_temperature': 0.4046086485706263, 'reg__random_strength': 0.8494409623573401, 'reg__subsample': 0.542634164180009,

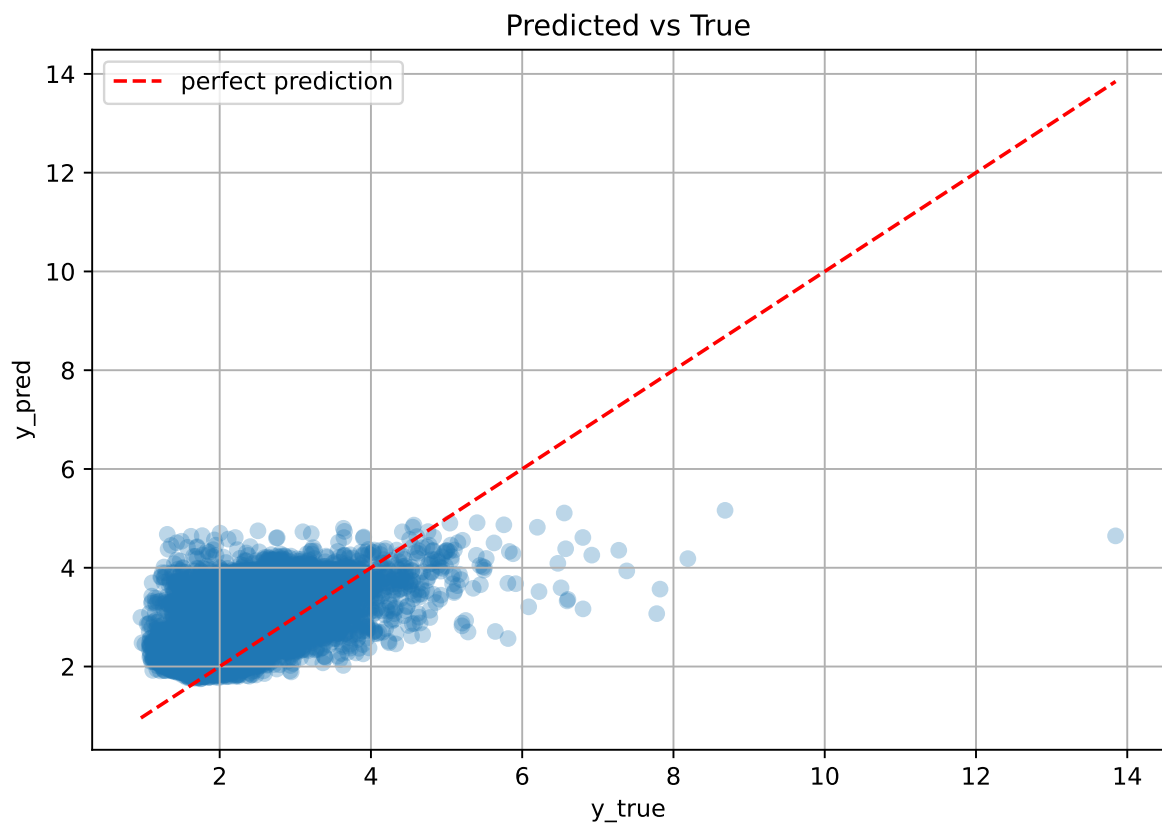
'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 8, 'reg__rsm': 0.77748385929358, 'reg__border_count': 112, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 811.8131248575012}

Rank 8: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.01905861499979225, 'reg__l2_leaf_reg': 49.79903536777679, 'reg__bagging_temperature': 0.8952661752223019, 'reg__random_strength': 0.9625553542455476, 'reg__subsample': 0.609563611389638, 'reg__min_data_in_leaf': 44, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.8005028842208738, 'reg__border_count': 98, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 296.21804457345706}

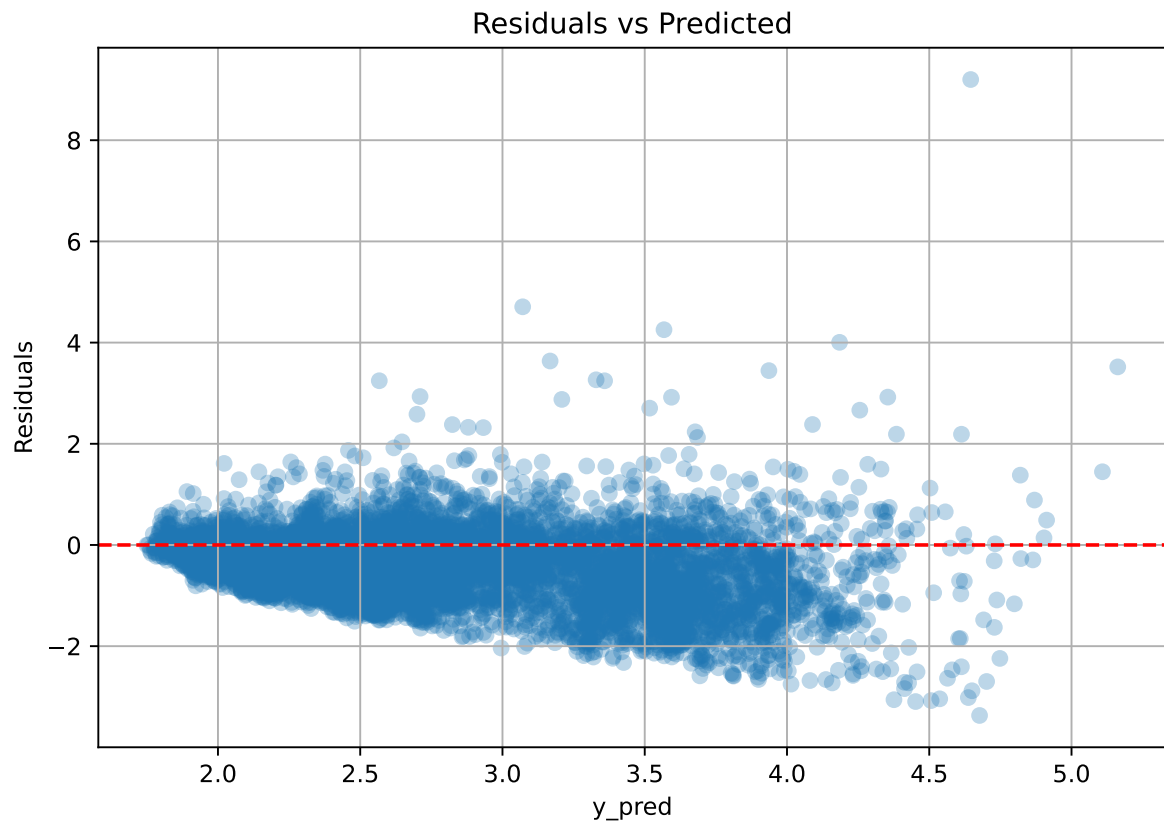
Rank 9: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.0191587394585492, 'reg__l2_leaf_reg': 49.1558661505424, 'reg__bagging_temperature': 0.1281046306384206, 'reg__random_strength': 0.9139162056890422, 'reg__subsample': 0.5957166449683455, 'reg__min_data_in_leaf': 45, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.8402325920823117, 'reg__border_count': 98, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 1625.1267578009183}

Rank 10: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.019313375043841403, 'reg__l2_leaf_reg': 49.603242911457514, 'reg__bagging_temperature': 0.03690889094562044, 'reg__random_strength': 0.9410492872493605, 'reg__subsample': 0.6088538560361691, 'reg__min_data_in_leaf': 45, 'reg__leaf_estimation_iterations': 4, 'reg__rsm': 0.8399400743809163, 'reg__border_count': 98, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 595.0968285367953}

2.5.2 Scatter



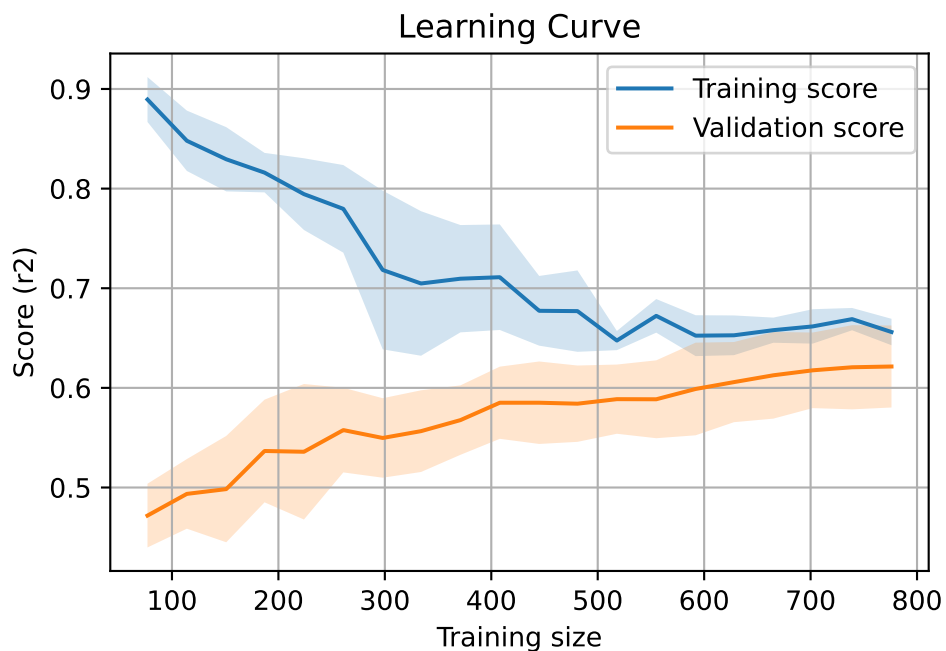
2.5.3 Scatter residuals



2.5.4 Learning curve — Group: 2025_Jun_1

`Len(group)` : 1585, `Len(training)` : 1110,

`Len(training_real)` : 777, `Len(test_of_training)` : 333, `Len(test)` : 475

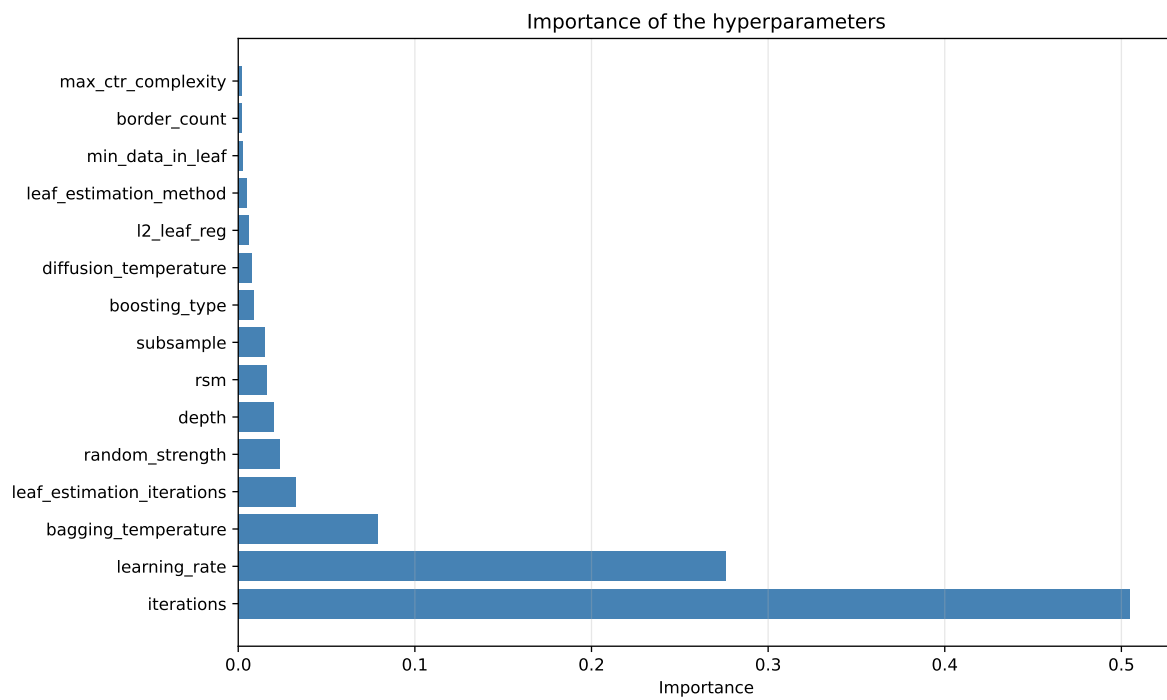
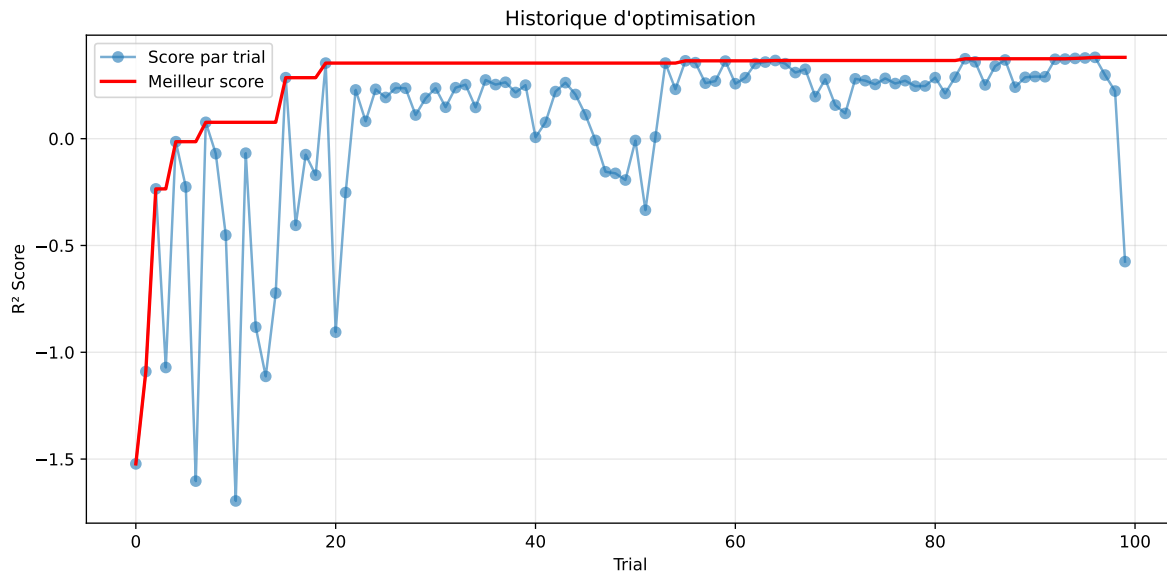


2.5.5 Gridsearch — Group: 2025_Jun_2

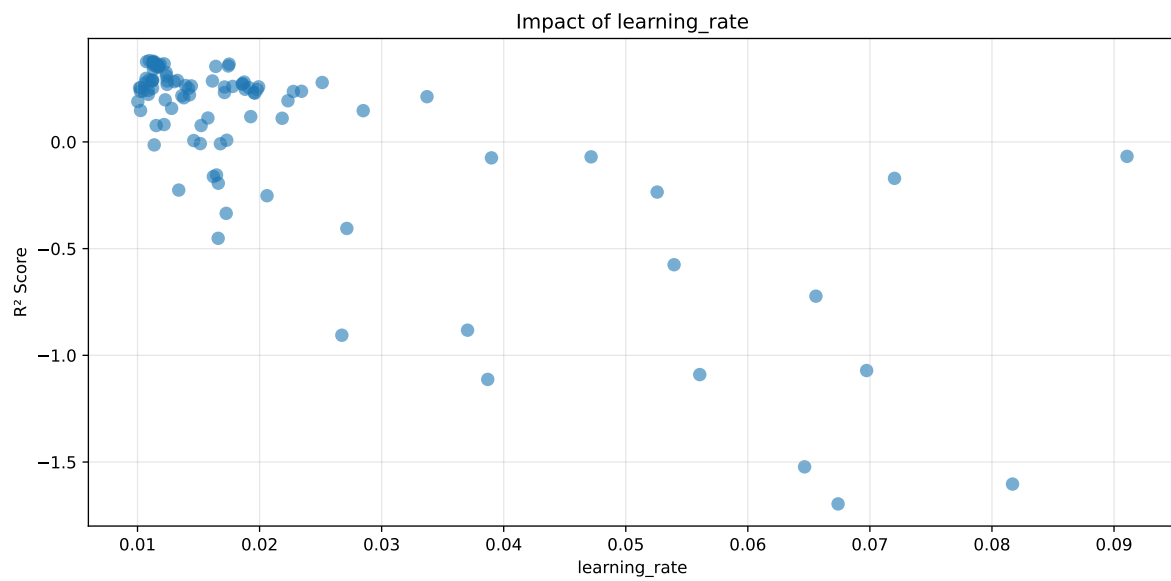
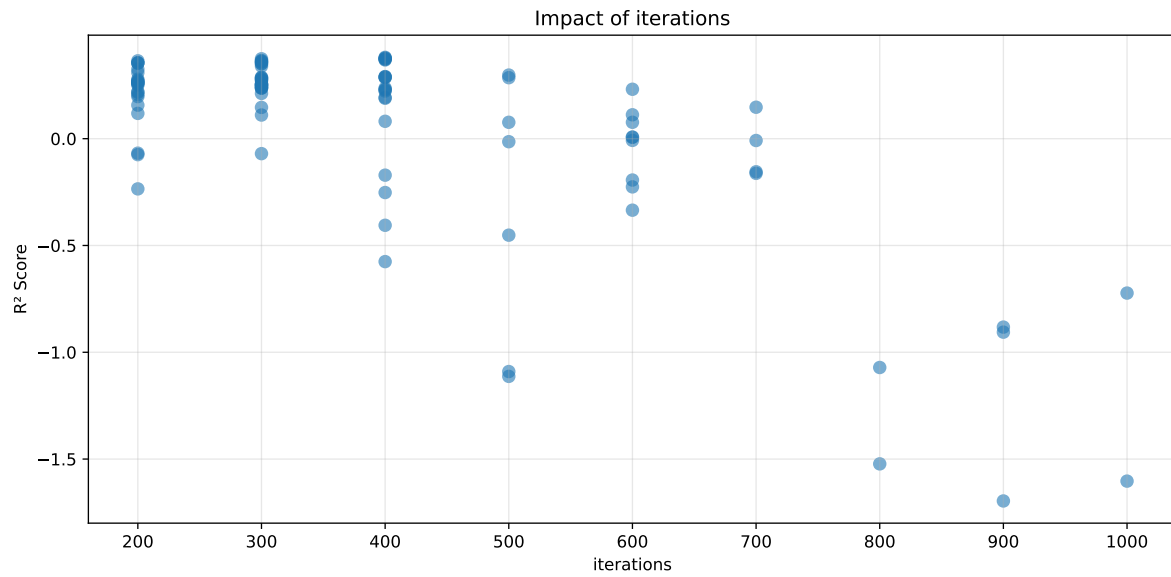
Len(group) : 1587, Len(training) : 1111, Len(test) : 476

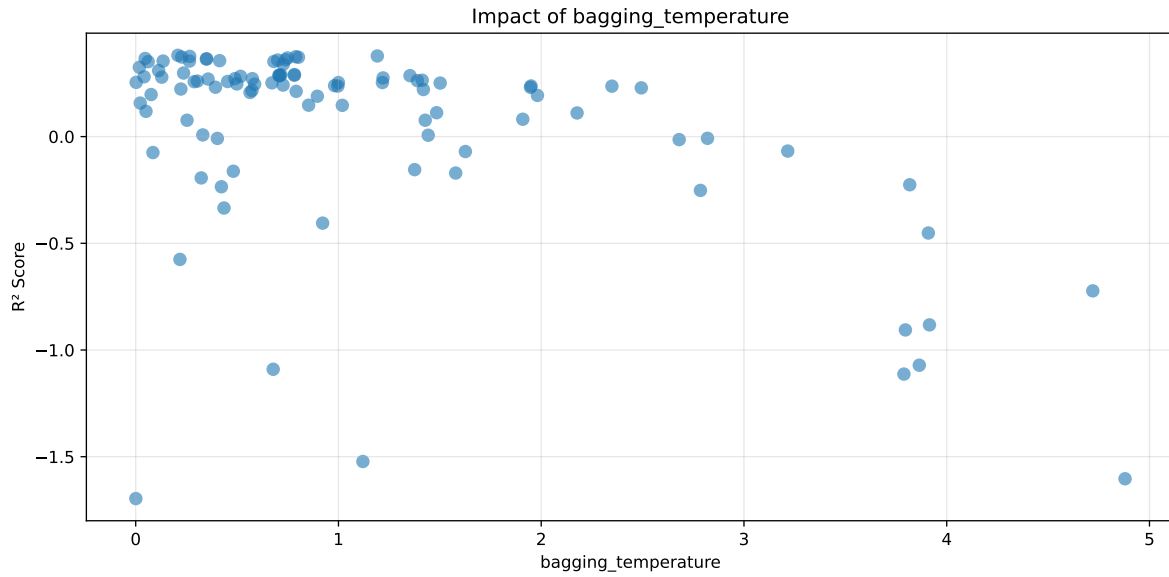
Distribution y_train vs y_test: y_train: mean=2.509, std=0.675, min=1.253, max=6.222
y_test: mean=2.528, std=0.697, min=1.305, max=7.383

===== TOP Importance FEATURES =====
feature importance r_THI 31.223619 r_umidity 29.631623 r_temperature 20.030359 r_NH3
9.762302 r_CO2 9.352098



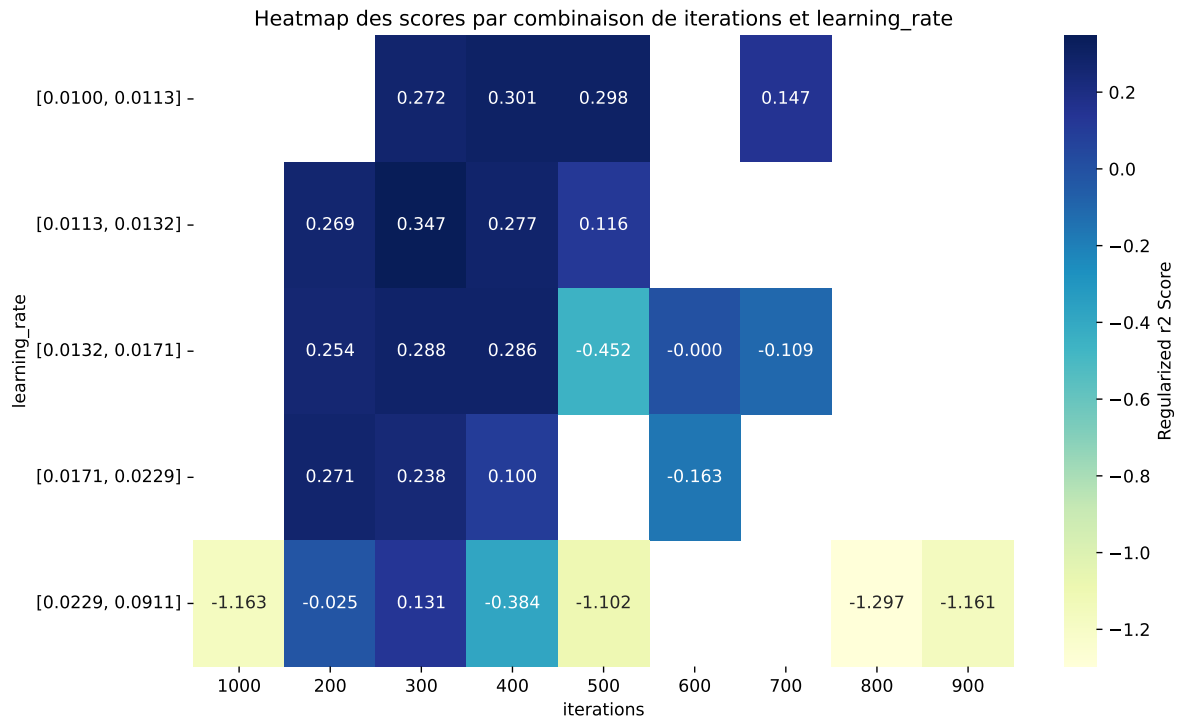
Paramètres avec >5% d'importance : ['iterations', 'learning_rate', 'bagging_temperature']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.3986	0.3808	0.3708	0.0132	0.3808	0.3917	0.0109	1.0287	1.0467
2	0.395	0.3779	0.3681	0.0143	0.3779	0.3881	0.0102	1.0271	1.0452
3	0.3913	0.3759	0.3649	0.0135	0.3759	0.3839	0.008	1.0213	1.041
4	0.3923	0.3742	0.3679	0.0133	0.3742	0.3773	0.003	1.0081	1.0484
5	0.3891	0.3727	0.3631	0.0137	0.3727	0.3835	0.0108	1.029	1.0437
6	0.3879	0.3719	0.3616	0.0145	0.3719	0.3848	0.0129	1.0348	1.0431
7	0.3845	0.3686	0.3586	0.0146	0.3686	0.3837	0.0151	1.0411	1.0433
8	0.3827	0.366	0.3548	0.0148	0.366	0.3662	0.0002	1.0005	1.0456
9	0.3798	0.3645	0.3544	0.0148	0.3645	0.3717	0.0072	1.0196	1.042
10	0.3807	0.3633	0.3542	0.0138	0.3633	0.3733	0.01	1.0275	1.0478
---	---	---	---	---	---	---	---	---	---
100	0.9567	0.5146	0.578	0.0215	-1.6962	0.5885	0.074	1.1438	1.8593

Hyperparameters:

Rank 1: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.01096819901217428, 'reg__l2_leaf_reg': 30.30501751643138, 'reg__bagging_temperature': 0.2081352633006939, 'reg__random_strength': 4.0510558066875735, 'reg__subsample': 0.638809509818466, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6826886428557615, 'reg__border_count': 88, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5528.242901123689}

Rank 2: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.011310194209839778, 'reg__l2_leaf_reg': 30.311186276164168, 'reg__bagging_temperature': 1.1917048467413602, 'reg__random_strength': 4.985664123477463, 'reg__subsample': 0.5225481719784075, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6809572332820627, 'reg__border_count': 81, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5577.238402409568}

Rank 3: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.0107435511277469, 'reg__l2_leaf_reg': 30.75959049918238, 'reg__bagging_temperature': 0.2664678449348765, 'reg__random_strength': 4.980235787572986, 'reg__subsample': 0.6359604403969157, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6844059851303902, 'reg__border_count': 88, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5661.996715800382}

Rank 4: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.011242703289514155, 'reg__l2_leaf_reg': 34.46249863109253, 'reg__bagging_temperature': 0.7890525196920901, 'reg__random_strength': 4.302158311655894, 'reg__subsample': 0.5489058812723183, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 2, 'reg__rsm':

0.5942065114287421, 'reg__border_count': 98, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6286.202767292457}

Rank 5: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.011398596566478184, 'reg__l2_leaf_reg': 39.86942511384539, 'reg__bagging_temperature': 0.22712819639810744, 'reg__random_strength': 4.945461729814104, 'reg__subsample': 0.6355340952800705, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6881792978941211, 'reg__border_count': 88, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5631.694282595288}

Rank 6: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.011282751896873344, 'reg__l2_leaf_reg': 41.28274001841125, 'reg__bagging_temperature': 0.8027280732635454, 'reg__random_strength': 4.994890196487465, 'reg__subsample': 0.6381149592577954, 'reg__min_data_in_leaf': 21, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6845239965520796, 'reg__border_count': 92, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5705.714826328553}

Rank 7: {'reg__iterations': 400, 'reg__depth': 3, 'reg__learning_rate': 0.011296225733823544, 'reg__l2_leaf_reg': 42.70993337288889, 'reg__bagging_temperature': 0.7485126528524968, 'reg__random_strength': 4.994804544359085, 'reg__subsample': 0.5507382588134659, 'reg__min_data_in_leaf': 25, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6870569045521508, 'reg__border_count': 91, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 4, 'reg__diffusion_temperature': 5687.90281116282}

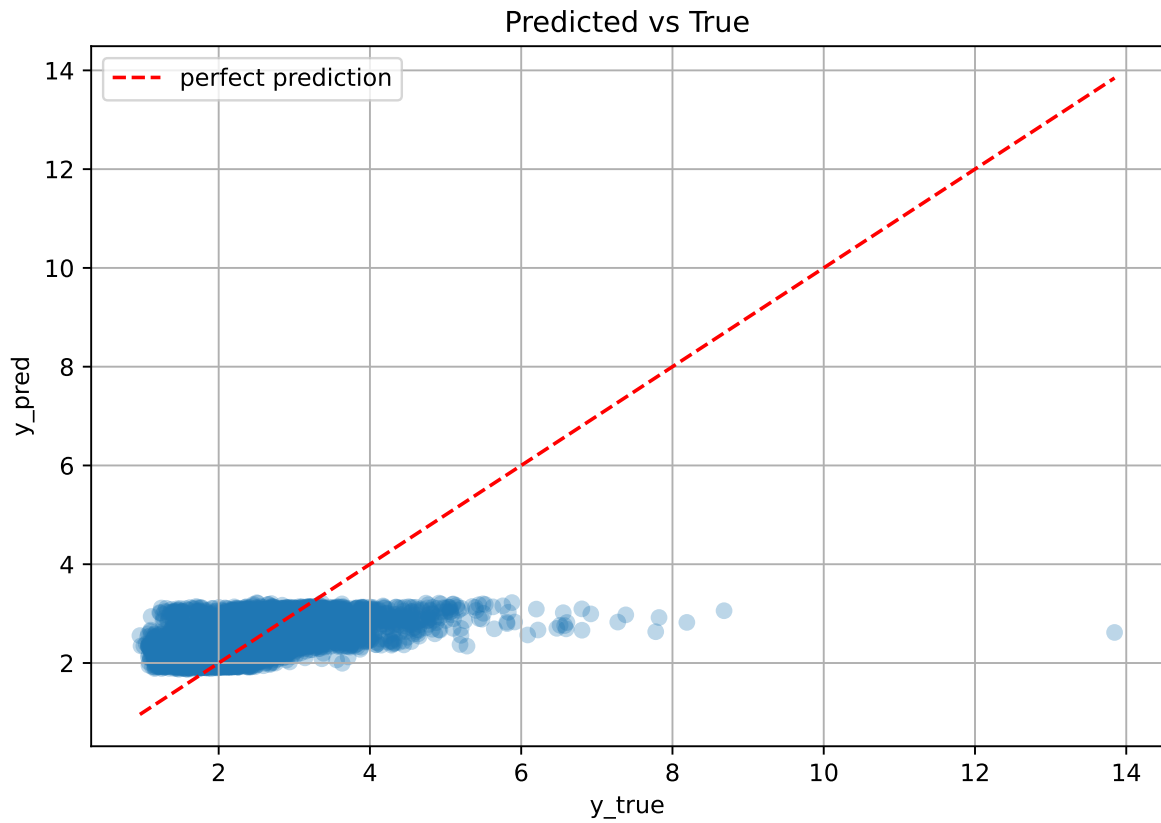
Rank 8: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.012184216669345972, 'reg__l2_leaf_reg': 27.920250578357432, 'reg__bagging_temperature': 0.04669225286066331, 'reg__random_strength': 3.94803091124479, 'reg__subsample': 0.6026651860089847, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6966991780395024, 'reg__border_count': 83, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6779.54037028158}

Rank 9: {'reg__iterations': 200, 'reg__depth': 3, 'reg__learning_rate': 0.017507904075918287, 'reg__l2_leaf_reg': 22.38190828123649, 'reg__bagging_temperature': 0.34972773592294126, 'reg__random_strength': 4.459892349769269, 'reg__subsample': 0.6241719265293063, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6457417802339516, 'reg__border_count': 86, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6708.868147438536}

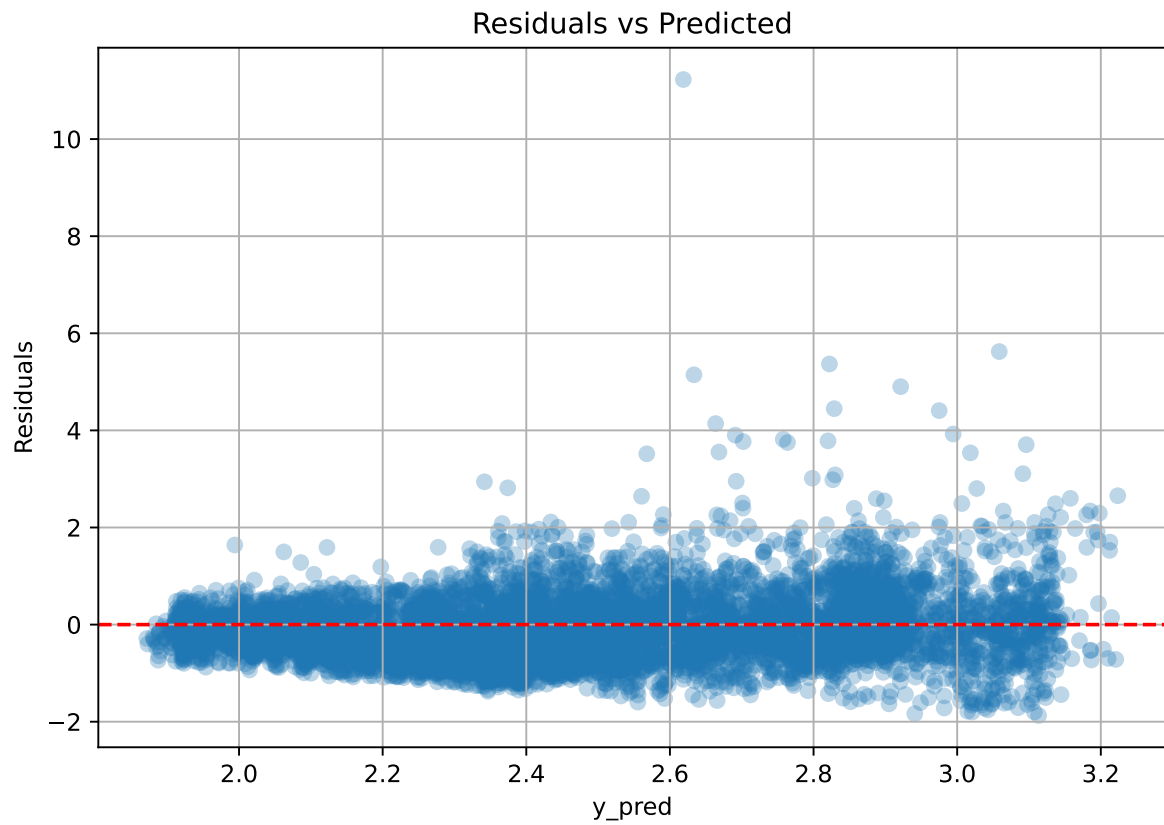
Rank 10: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.011862410051708176, 'reg__l2_leaf_reg': 23.380159912275627, 'reg__bagging_temperature': 0.34852131472056824,

```
'reg__random_strength': 4.546711892141483, 'reg__subsample': 0.7105844411166122,  
'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 1, 'reg__rsm':  
0.6506647913284225, 'reg__border_count': 102, 'reg__leaf_estimation_method': 'Gradient',  
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_tempera-  
ture': 5925.594131249492}
```

2.5.6 Scatter



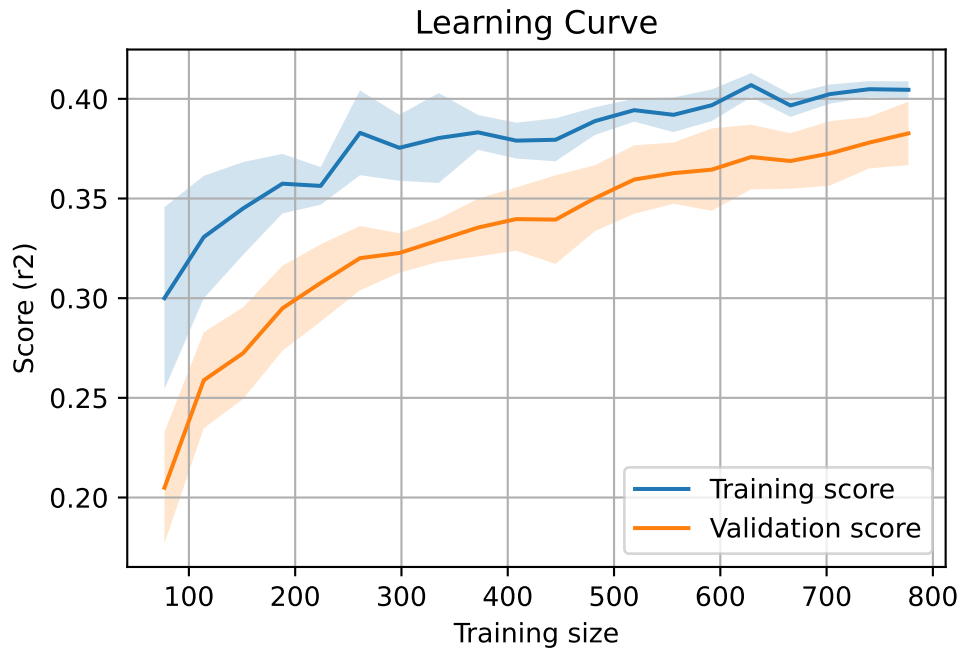
2.5.7 Scatter residuals



2.5.8 Learning curve — Group: 2025_Jun_2

`Len(group)` : 1587, `Len(training)` : 1111,

`Len(training_real)` : 778, `Len(test_of_training)` : 333, `Len(test)` : 476

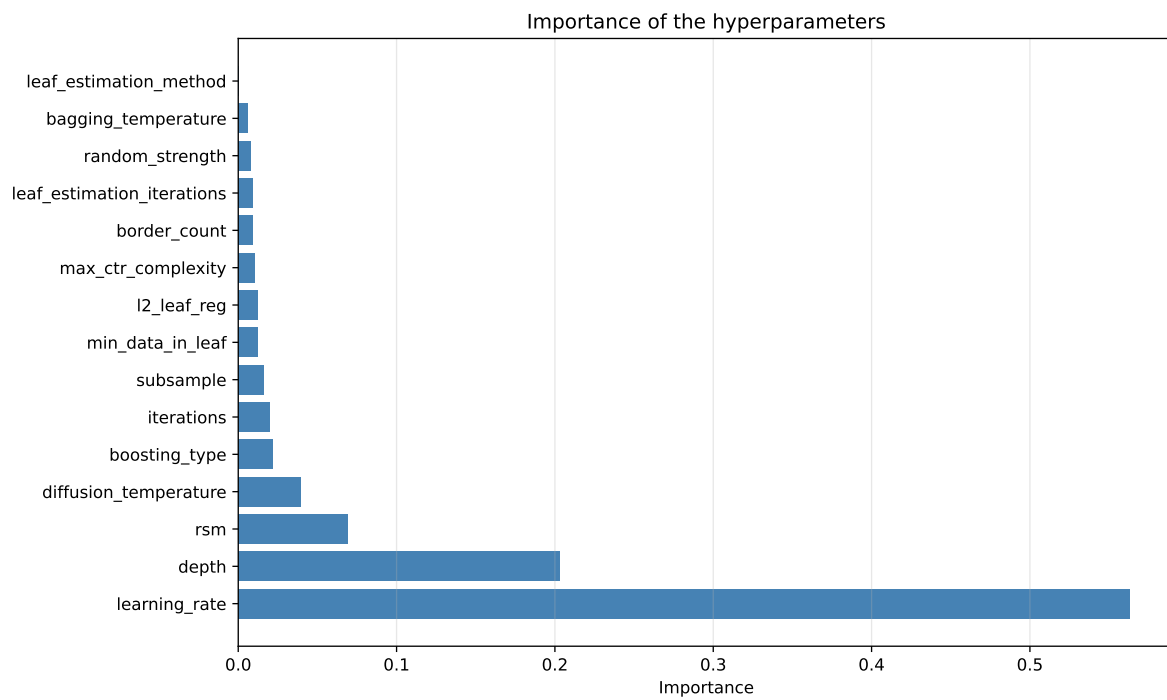
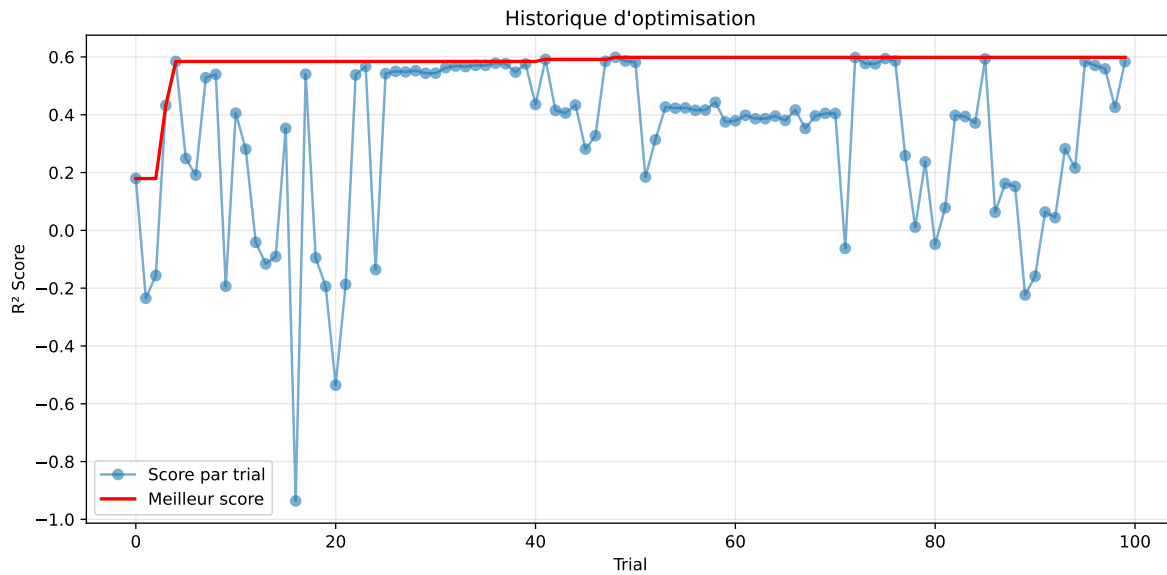


2.5.9 Gridsearch — Group: 2025_Sep_1

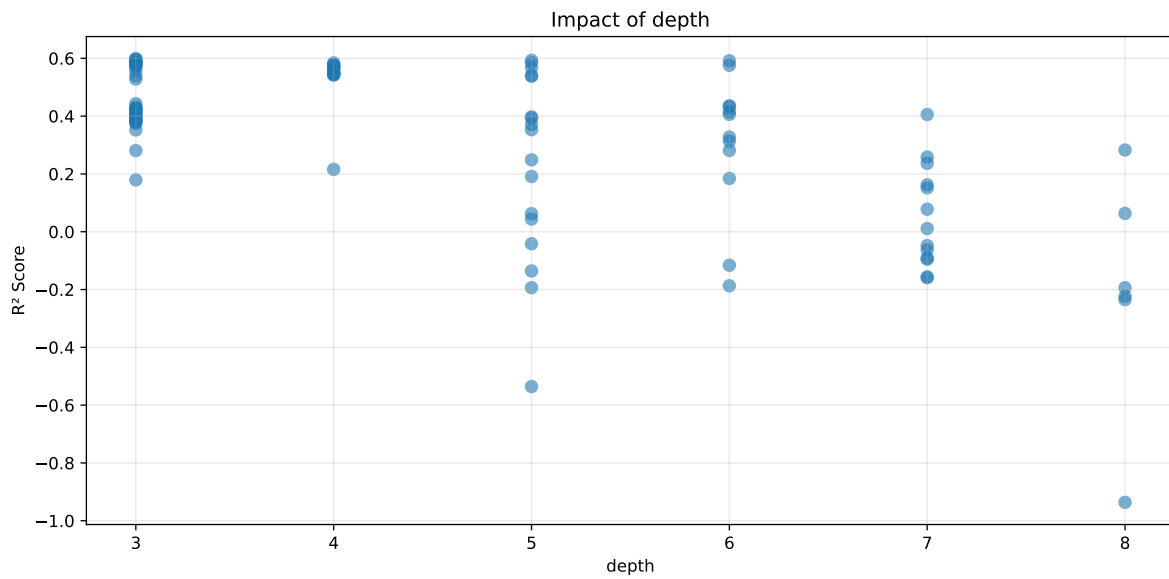
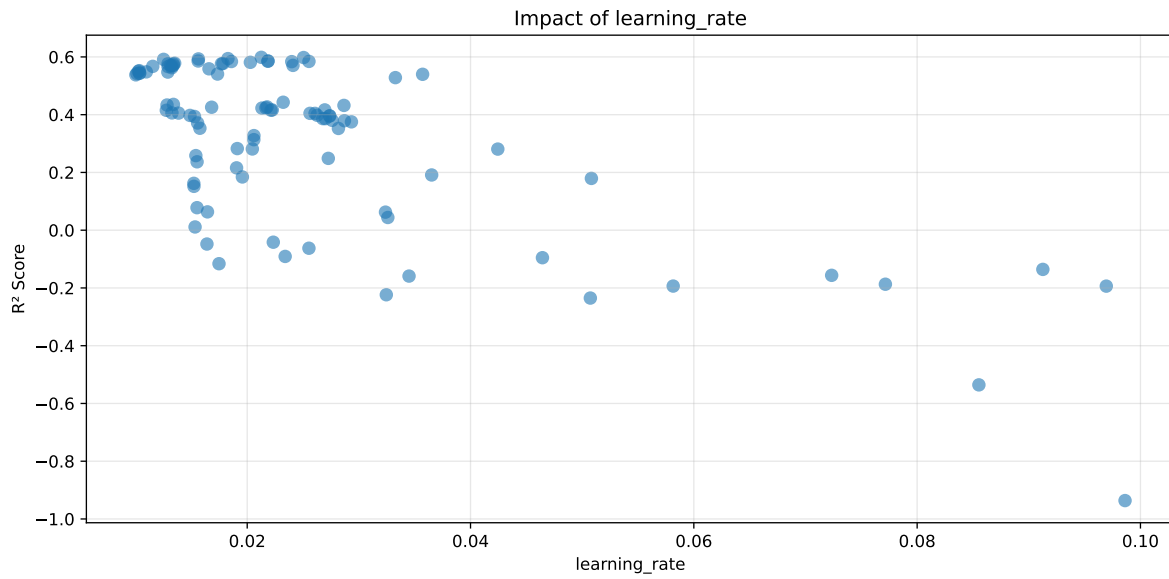
Len(group) : 3867, Len(training) : 2707, Len(test) : 1160

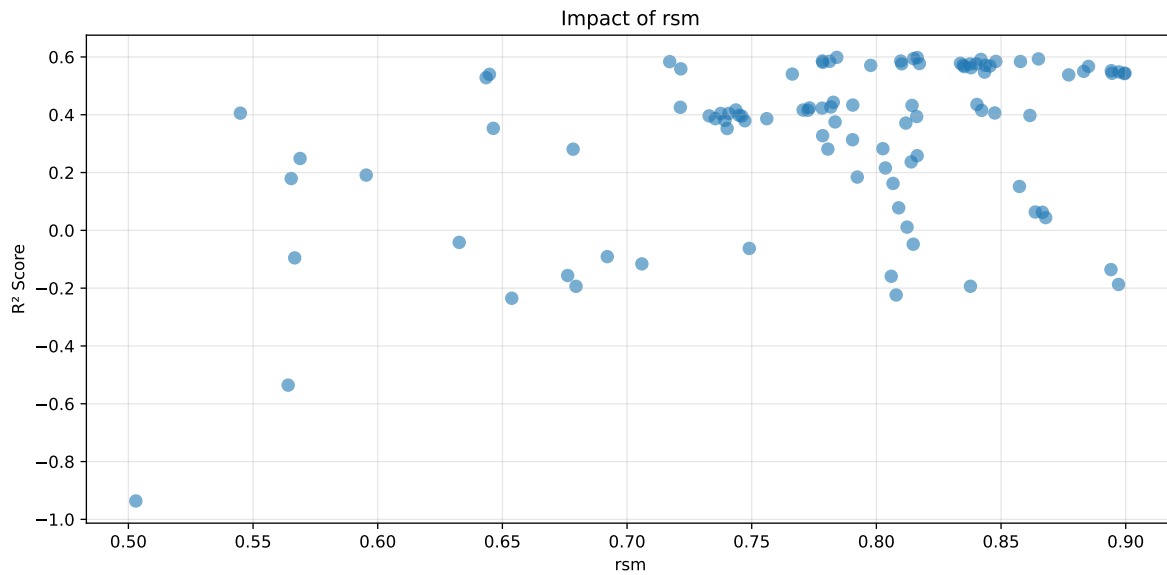
Distribution y_train vs y_test: y_train: mean=2.231, std=0.761, min=0.973, max=8.190
y_test: mean=2.216, std=0.723, min=1.110, max=6.920

===== TOP Importance FEATURES =====
feature importance r_umidity 36.394748 r_NH3 31.798523 r_temperature 12.112525 r_CO2
11.944713 r_THI 7.749491



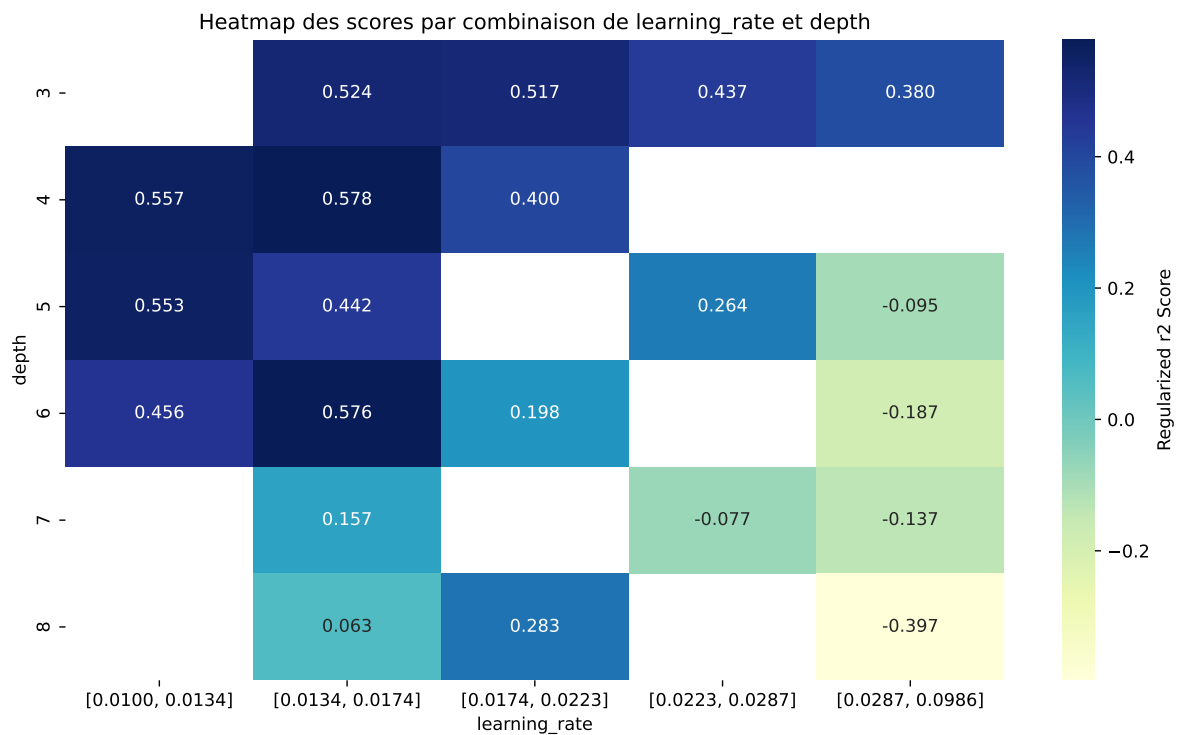
Paramètres avec >5% d'importance : ['learning_rate', 'depth', 'rsm']





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The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.6274	0.5986	0.603	0.0264	0.5986	0.6068	0.0082	1.0138	1.0482
2	0.6261	0.5979	0.6004	0.0257	0.5979	0.6053	0.0074	1.0124	1.0472
3	0.6214	0.5942	0.5949	0.0302	0.5942	0.5974	0.0032	1.0053	1.0458
4	0.6203	0.5932	0.6012	0.0288	0.5932	0.6066	0.0135	1.0227	1.0458
5	0.6184	0.5916	0.6012	0.0295	0.5916	0.6099	0.0183	1.031	1.0454
6	0.6058	0.586	0.5898	0.0271	0.586	0.5939	0.0079	1.0134	1.0337
7	0.6072	0.5859	0.591	0.0284	0.5859	0.593	0.007	1.012	1.0363
8	0.6076	0.585	0.59	0.0295	0.585	0.5968	0.0118	1.0202	1.0387
9	0.6095	0.5845	0.5909	0.0302	0.5845	0.5995	0.015	1.0256	1.0428
10	0.608	0.5841	0.588	0.0285	0.5841	0.5938	0.0097	1.0166	1.041
---	---	---	---	---	---	---	---	---	---
100	0.9428	0.6296	0.6226	0.0204	-0.9363	0.6386	0.0091	1.0144	1.4975

Hyperparameters:

Rank 1: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.02126262116870507, 'reg__l2_leaf_reg': 30.010976783594668, 'reg__bagging_temperature': 3.071332897383669, 'reg__random_strength': 3.676530568433411, 'reg__subsample': 0.6959907049957075, 'reg__min_data_in_leaf': 36, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.7841125564705628, 'reg__border_count': 113, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7908.329617431041}

Rank 2: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.02504992083107909, 'reg__l2_leaf_reg': 43.002182666500424, 'reg__bagging_temperature': 2.675289904011315, 'reg__random_strength': 4.11577320968653, 'reg__subsample': 0.6834590515908542, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8163851792415946, 'reg__border_count': 103, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 8588.49272271773}

Rank 3: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.018276059111619322, 'reg__l2_leaf_reg': 26.64801614723194, 'reg__bagging_temperature': 2.2833295140489067, 'reg__random_strength': 0.8615008252533637, 'reg__subsample': 0.7325393444413776, 'reg__min_data_in_leaf': 17, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8149443129383215, 'reg__border_count': 121, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 9438.609371932755}

Rank 4: {'reg__iterations': 500, 'reg__depth': 5, 'reg__learning_rate': 0.015629641558759046, 'reg__l2_leaf_reg': 48.72788667347489, 'reg__bagging_temperature': 2.3278391907429428, 'reg__random_strength': 2.305432174662534, 'reg__subsample': 0.6875109807595913, 'reg__min_data_in_leaf': 15, 'reg__leaf_estimation_iterations': 7, 'reg__rsm':

0.8650005663093113, 'reg__border_count': 94, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 9405.815835385509}

Rank 5: {'reg__iterations': 500, 'reg__depth': 6, 'reg__learning_rate': 0.01251220717872606, 'reg__l2_leaf_reg': 28.7913432814321, 'reg__bagging_temperature': 3.4394585419128094, 'reg__random_strength': 3.3986480016075973, 'reg__subsample': 0.6316019613655594, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 6, 'reg__rsm': 0.8419270508762398, 'reg__border_count': 110, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7398.809016193159}

Rank 6: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.021872092054418736, 'reg__l2_leaf_reg': 35.09213948188294, 'reg__bagging_temperature': 1.1966536540931039, 'reg__random_strength': 1.6930159147593036, 'reg__subsample': 0.702244696342641, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.7783451886779079, 'reg__border_count': 105, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7394.851680256665}

Rank 7: {'reg__iterations': 600, 'reg__depth': 3, 'reg__learning_rate': 0.015609077985825717, 'reg__l2_leaf_reg': 25.47688477650153, 'reg__bagging_temperature': 2.313000909711678, 'reg__random_strength': 0.9477230811662416, 'reg__subsample': 0.7431003368555487, 'reg__min_data_in_leaf': 16, 'reg__leaf_estimation_iterations': 7, 'reg__rsm': 0.8097779618460297, 'reg__border_count': 98, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 8360.58632523055}

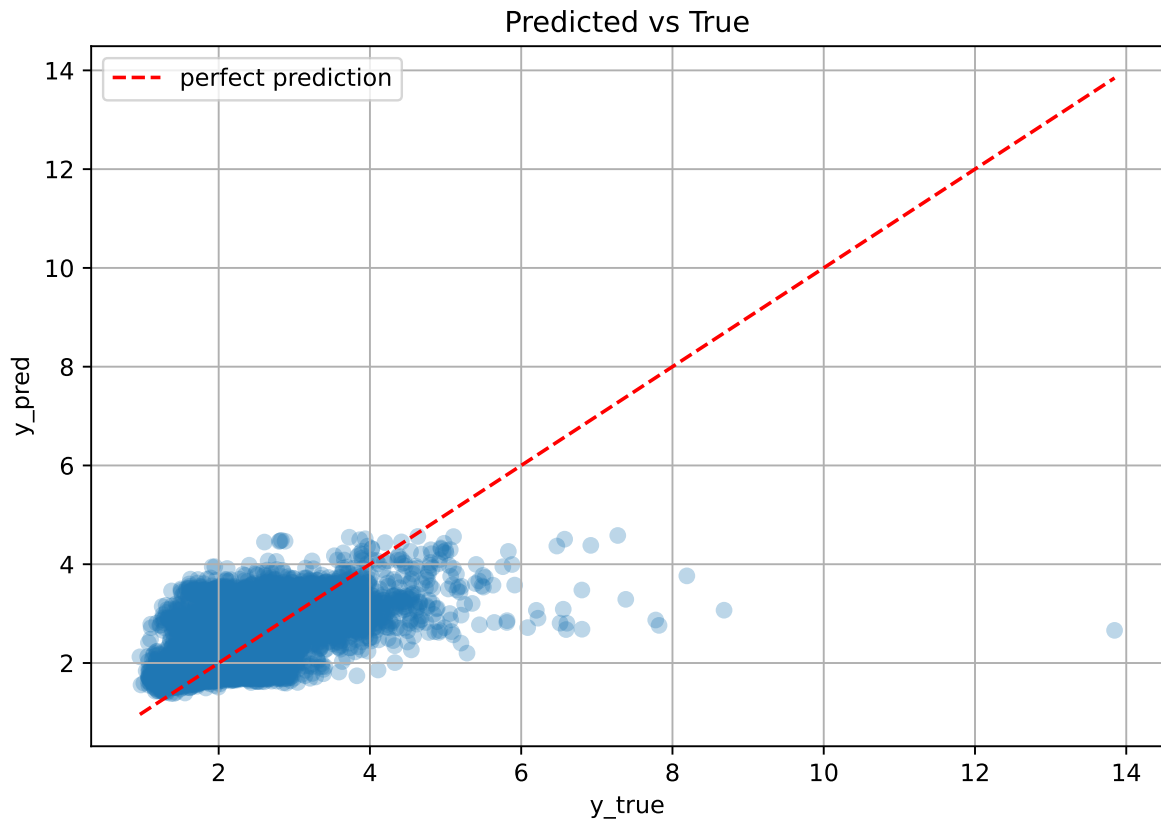
Rank 8: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.021849291604885218, 'reg__l2_leaf_reg': 29.039795497611546, 'reg__bagging_temperature': 3.0489417583815195, 'reg__random_strength': 1.6149131496225029, 'reg__subsample': 0.5513342303738457, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 5, 'reg__rsm': 0.7812308648519032, 'reg__border_count': 112, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 2, 'reg__diffusion_temperature': 7950.055623859613}

Rank 9: {'reg__iterations': 400, 'reg__depth': 5, 'reg__learning_rate': 0.025530288534820415, 'reg__l2_leaf_reg': 42.293430283746524, 'reg__bagging_temperature': 3.1604254227432698, 'reg__random_strength': 2.0623264712748863, 'reg__subsample': 0.6634140309713372, 'reg__min_data_in_leaf': 22, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.8479222058216522, 'reg__border_count': 72, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_temperature': 3789.525323534747}

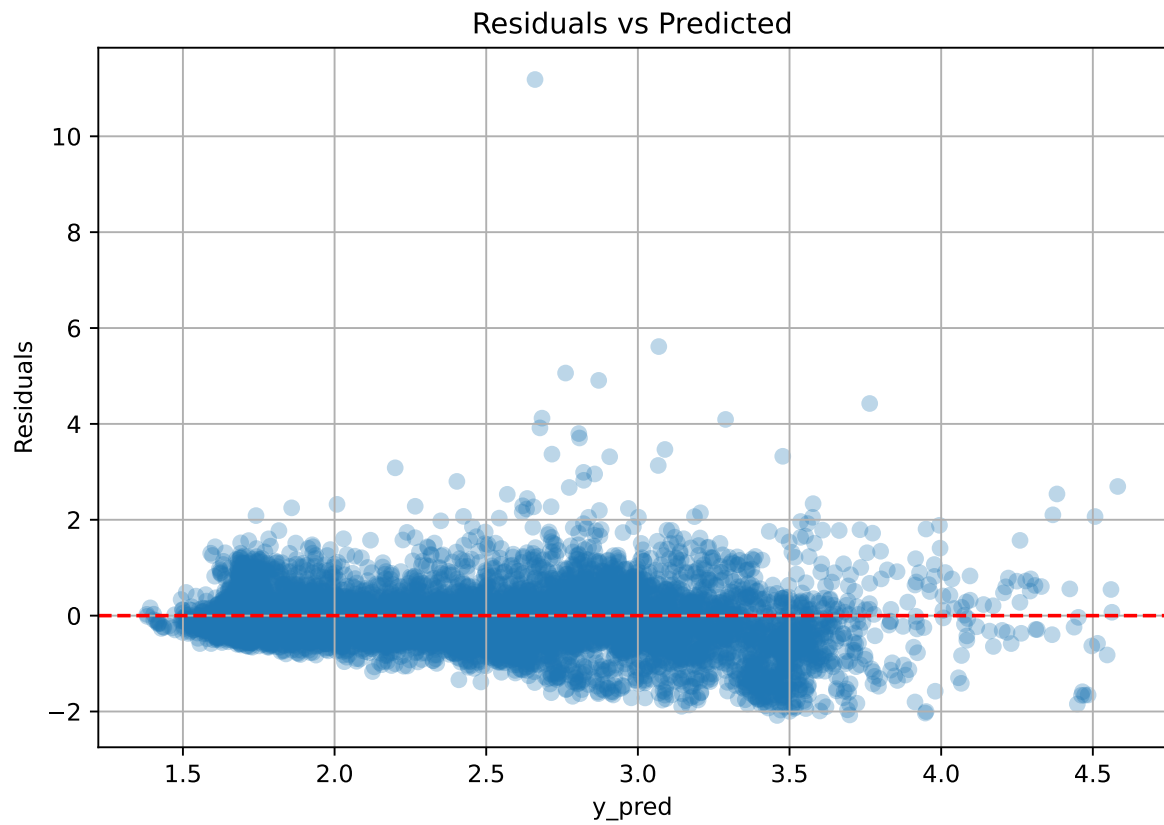
Rank 10: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.01858762271635421, 'reg__l2_leaf_reg': 22.59872095823297, 'reg__bagging_temperature': 3.2721262051042084,

```
'reg__random_strength': 0.579835342563974, 'reg__subsample': 0.6443432844164002,  
'reg__min_data_in_leaf': 10, 'reg__leaf_estimation_iterations': 9, 'reg__rsm':  
0.8577853100433198, 'reg__border_count': 91, 'reg__leaf_estimation_method': 'Newton',  
'reg__boosting_type': 'Ordered', 'reg__max_ctr_complexity': 3, 'reg__diffusion_tempera-  
ture': 9710.913575160946}
```

2.5.10 Scatter



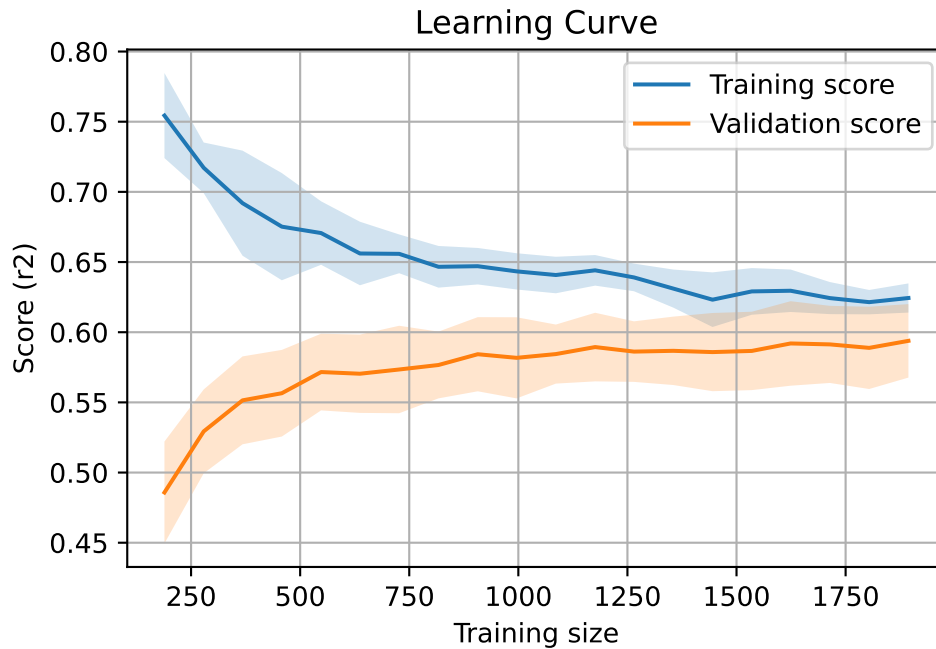
2.5.11 Scatter residuals



2.5.12 Learning curve — Group: 2025_Sep_1

`Len(group)` : 3867, `Len(training)` : 2707,

`Len(training_real)` : 1895, `Len(test_of_training)` : 812, `Len(test)` : 1160

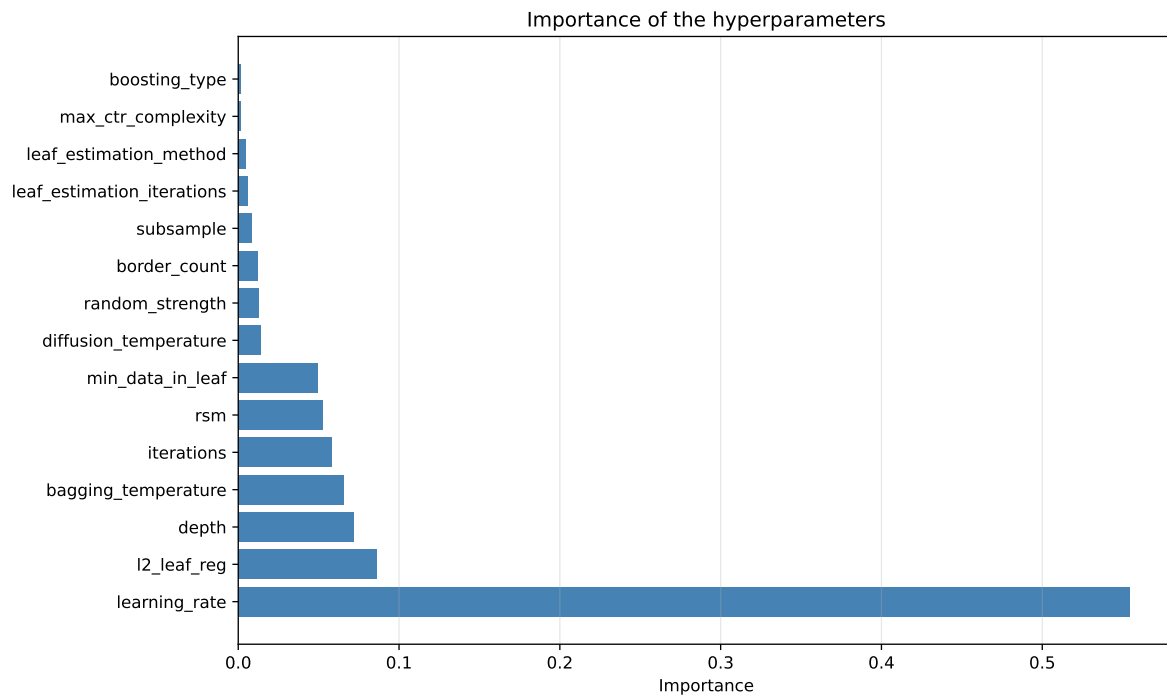
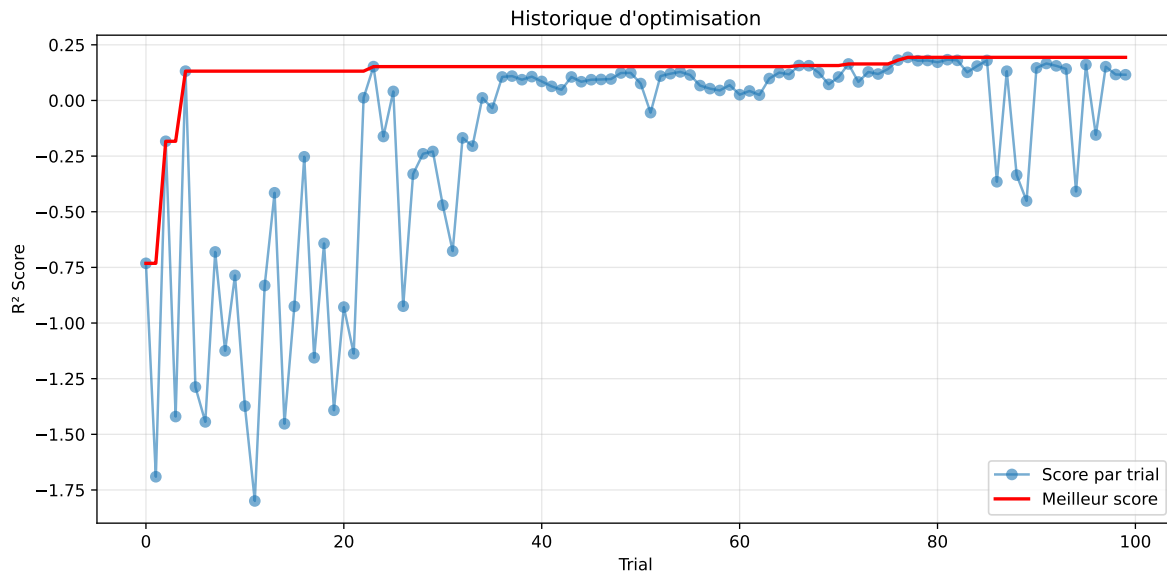


2.5.13 Gridsearch — Group: 2025_Sep_2

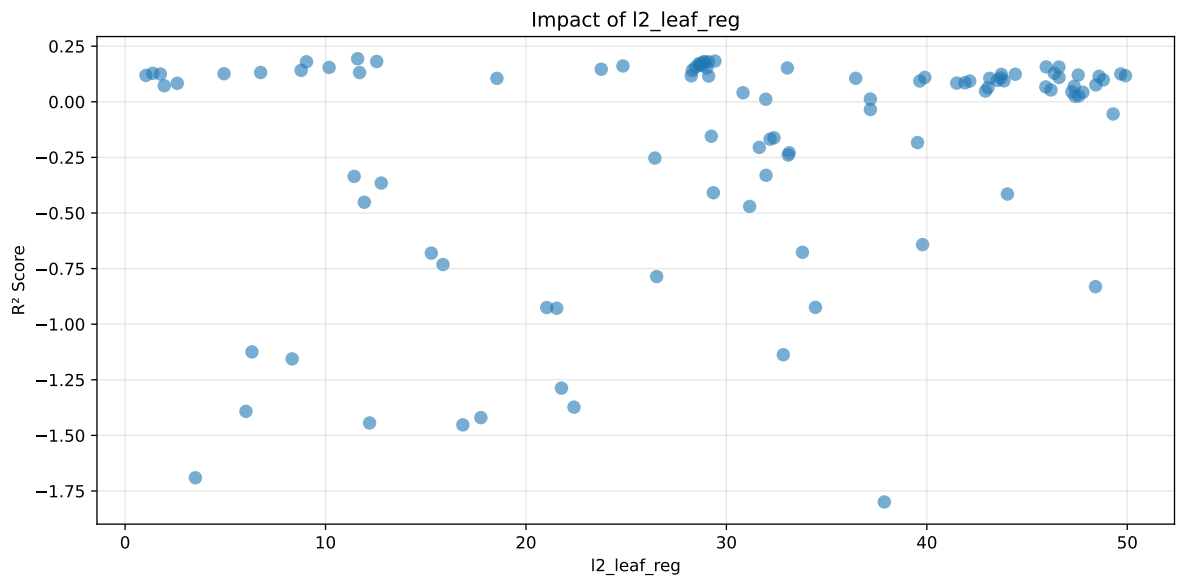
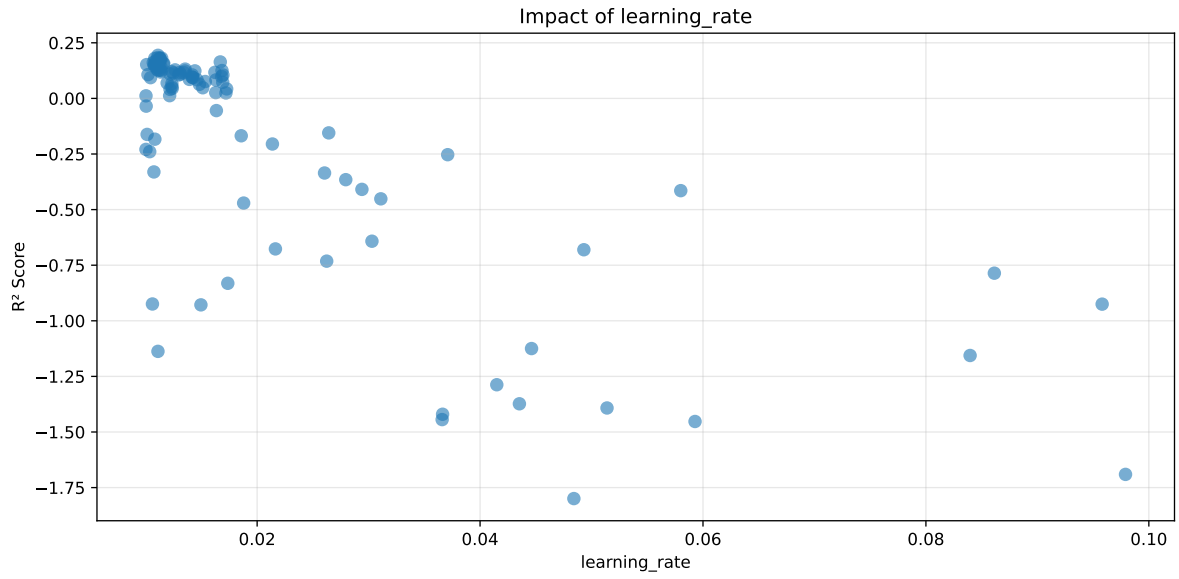
Len(group) : 1129, Len(training) : 791, Len(test) : 338

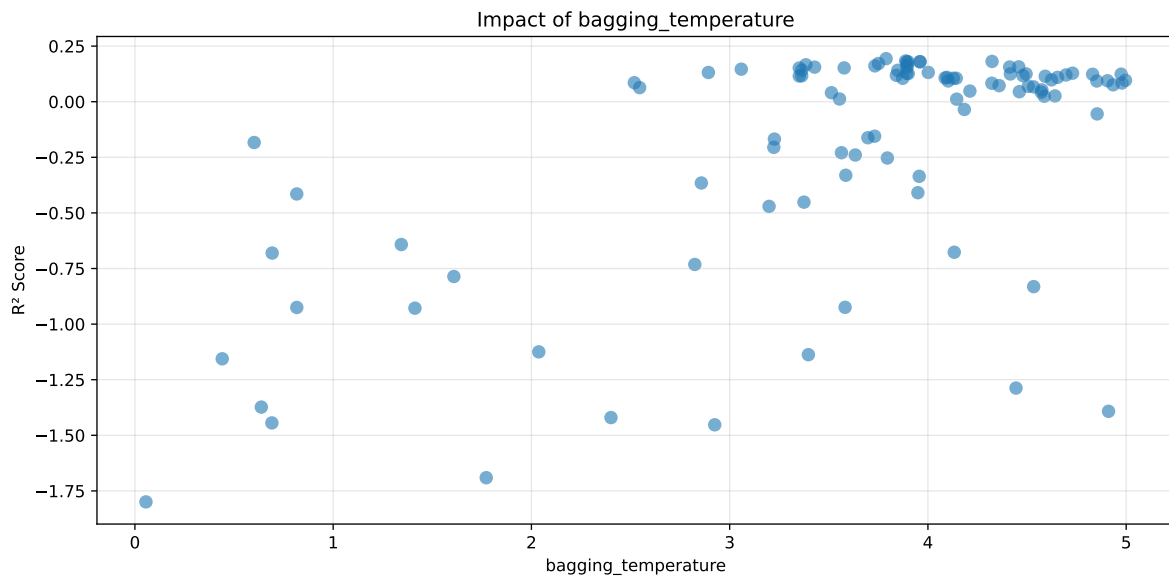
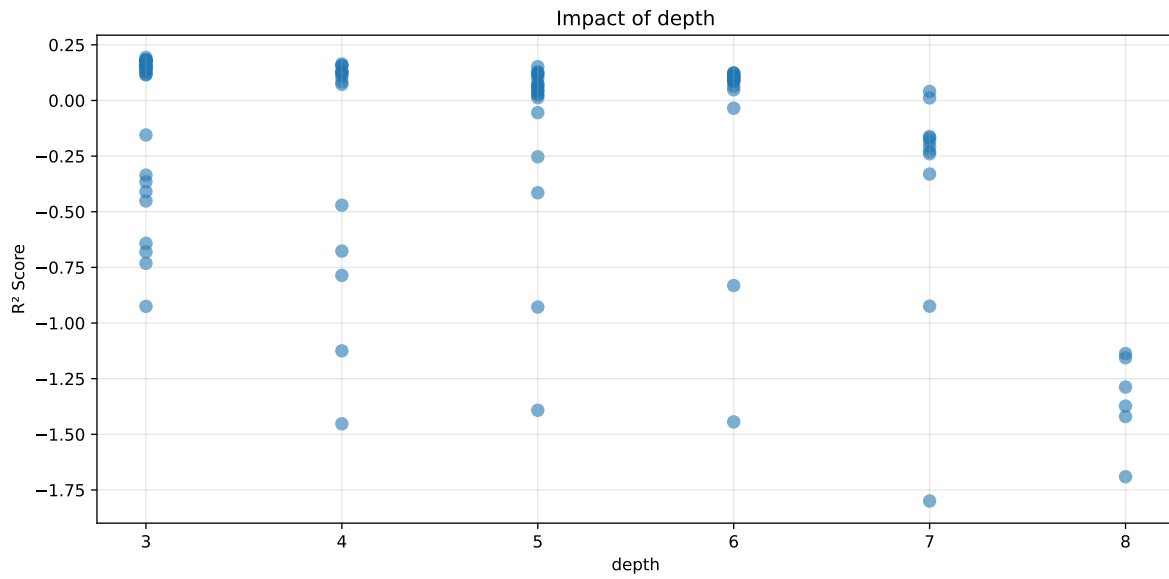
Distribution y_train vs y_test: y_train: mean=1.926, std=0.742, min=0.959, max=6.804
y_test: mean=1.945, std=0.805, min=1.070, max=7.822

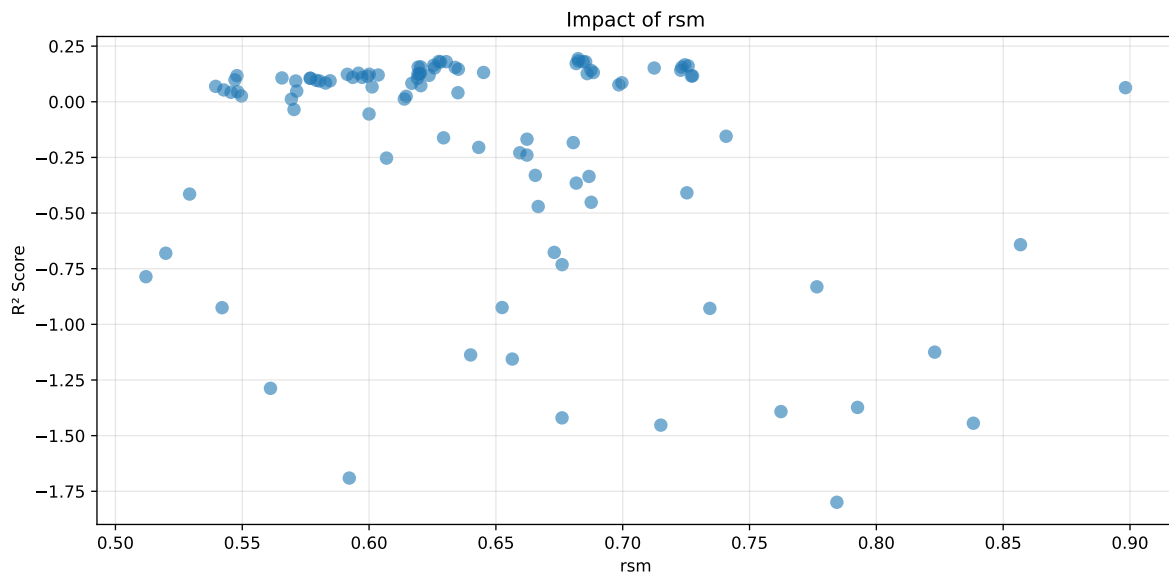
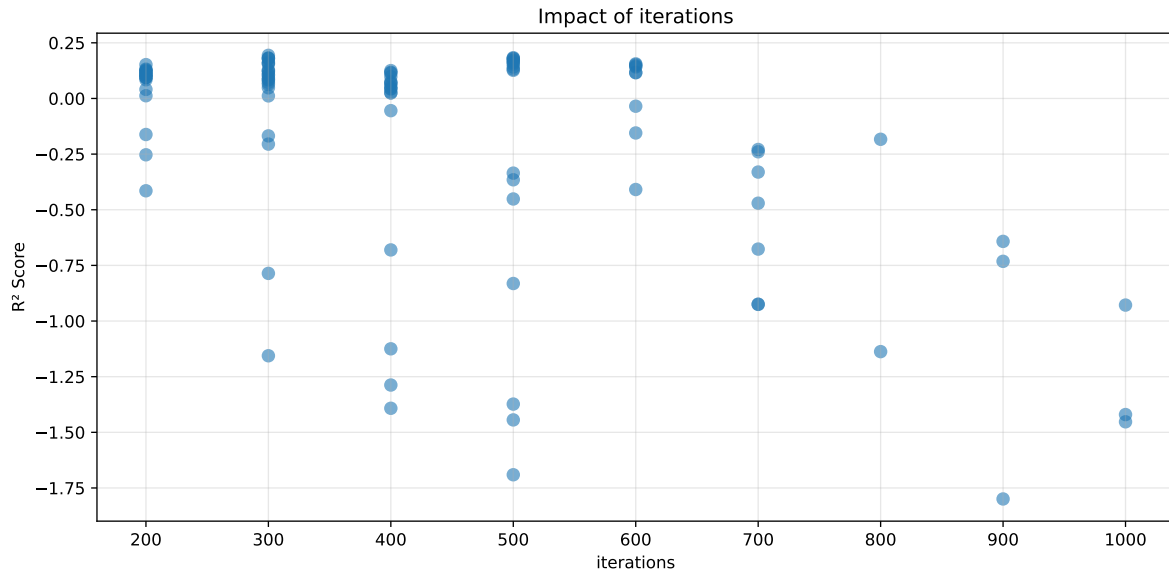
===== TOP Importance FEATURES =====
feature importance r_umidity 55.608696 r_THI 12.745803 r_CO2 11.967292 r_temperature
10.889823 r_NH3 8.788386



Paramètres avec >5% d'importance : ['learning_rate', 'l2_leaf_reg', 'depth', 'bagging_temperature', 'iterations', 'rsm']

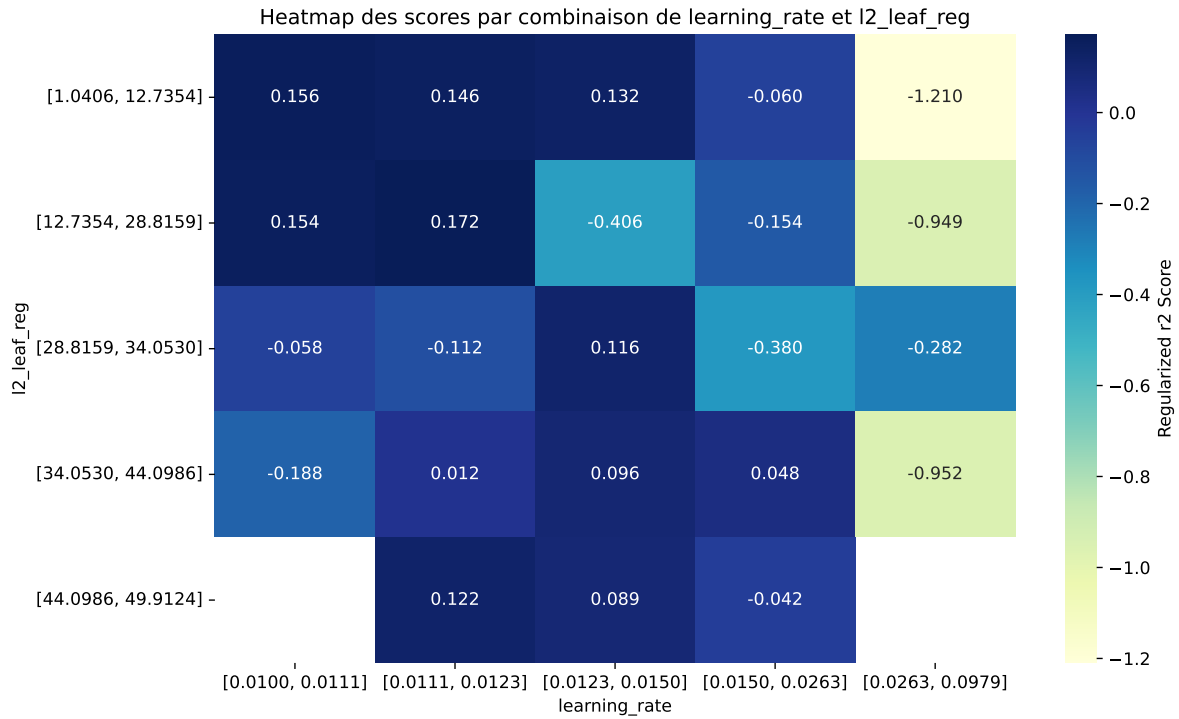






C:\Users\titou\Desktop\UNICT\1- Gas sensor calibration\Python\machine_learning\workflow_main

The default value of observed=False is deprecated and will change to observed=True in a future



rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4223	0.3842	0.4114	0.036	0.1937	0.4238	0.0396	1.1032	1.0992
2	0.483	0.433	0.4645	0.0393	0.1832	0.4747	0.0417	1.0964	1.1154
3	0.4249	0.3843	0.4072	0.0376	0.1812	0.4138	0.0295	1.0769	1.1057
4	0.4277	0.3864	0.4141	0.0381	0.1803	0.4243	0.0379	1.0981	1.1067
5	0.4449	0.4007	0.4298	0.0371	0.1798	0.4376	0.0369	1.092	1.1102
6	0.4574	0.4111	0.4403	0.0361	0.1797	0.4567	0.0456	1.111	1.1126
7	0.4547	0.4087	0.4348	0.04	0.1786	0.4468	0.0381	1.0932	1.1126
8	0.4855	0.4332	0.4668	0.0399	0.1717	0.4778	0.0446	1.1029	1.1207
9	0.4808	0.4284	0.4632	0.0423	0.1662	0.4733	0.0449	1.1049	1.1224
10	0.4666	0.4161	0.441	0.0407	0.164	0.4526	0.0365	1.0877	1.1212
---	---	---	---	---	---	---	---	---	---
100	0.9179	0.465	0.629	0.0642	-1.7994	0.6451	0.18	1.3871	1.9739

Hyperparameters:

Rank 1: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.011104279200328684, 'reg__l2_leaf_reg': 11.60272320909771, 'reg__bagging_temperature': 3.7890463441214752, 'reg__random_strength': 4.219716767034171, 'reg__subsample': 0.6986349979358872, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6824050749300039, 'reg__border_count': 46, 'reg__leaf_estimation_method': 'Gradient',

'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7337.537067744508}

Rank 2: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.011100409543453597, 'reg__l2_leaf_reg': 29.42987860679243, 'reg__bagging_temperature': 3.8876541351103264, 'reg__random_strength': 3.7904097534882735, 'reg__subsample': 0.7442435387429678, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6825536115287945, 'reg__border_count': 45, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7174.837339874058}

Rank 3: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.011432832642583168, 'reg__l2_leaf_reg': 12.555037394363339, 'reg__bagging_temperature': 4.323449147184571, 'reg__random_strength': 3.7513718076381504, 'reg__subsample': 0.6970576733205514, 'reg__min_data_in_leaf': 35, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6274965371292175, 'reg__border_count': 45, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7175.257200889078}

Rank 4: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.011251494441439653, 'reg__l2_leaf_reg': 28.924351367181334, 'reg__bagging_temperature': 3.899056774577345, 'reg__random_strength': 3.7867365612585377, 'reg__subsample': 0.7490801901236255, 'reg__min_data_in_leaf': 28, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6304202062058536, 'reg__border_count': 45, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6234.329411319324}

Rank 5: {'reg__iterations': 300, 'reg__depth': 3, 'reg__learning_rate': 0.010811271120866975, 'reg__l2_leaf_reg': 9.050002534274551, 'reg__bagging_temperature': 3.9605795693821215, 'reg__random_strength': 3.7713073948510916, 'reg__subsample': 0.743085124525136, 'reg__min_data_in_leaf': 27, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6843856478328982, 'reg__border_count': 45, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7347.91445005314}

Rank 6: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.011281461506449863, 'reg__l2_leaf_reg': 29.10968180239222, 'reg__bagging_temperature': 3.95914423516338, 'reg__random_strength': 3.7859819344070185, 'reg__subsample': 0.698529850418205, 'reg__min_data_in_leaf': 28, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6853792291412906, 'reg__border_count': 46, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7287.060783148658}

Rank 7: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.011293479109828865, 'reg__l2_leaf_reg': 28.841430030275973, 'reg__bagging_temperature': 3.8931524313642445, 'reg__random_strength': 3.7223726015262137, 'reg__subsample': 0.6996981870424146,

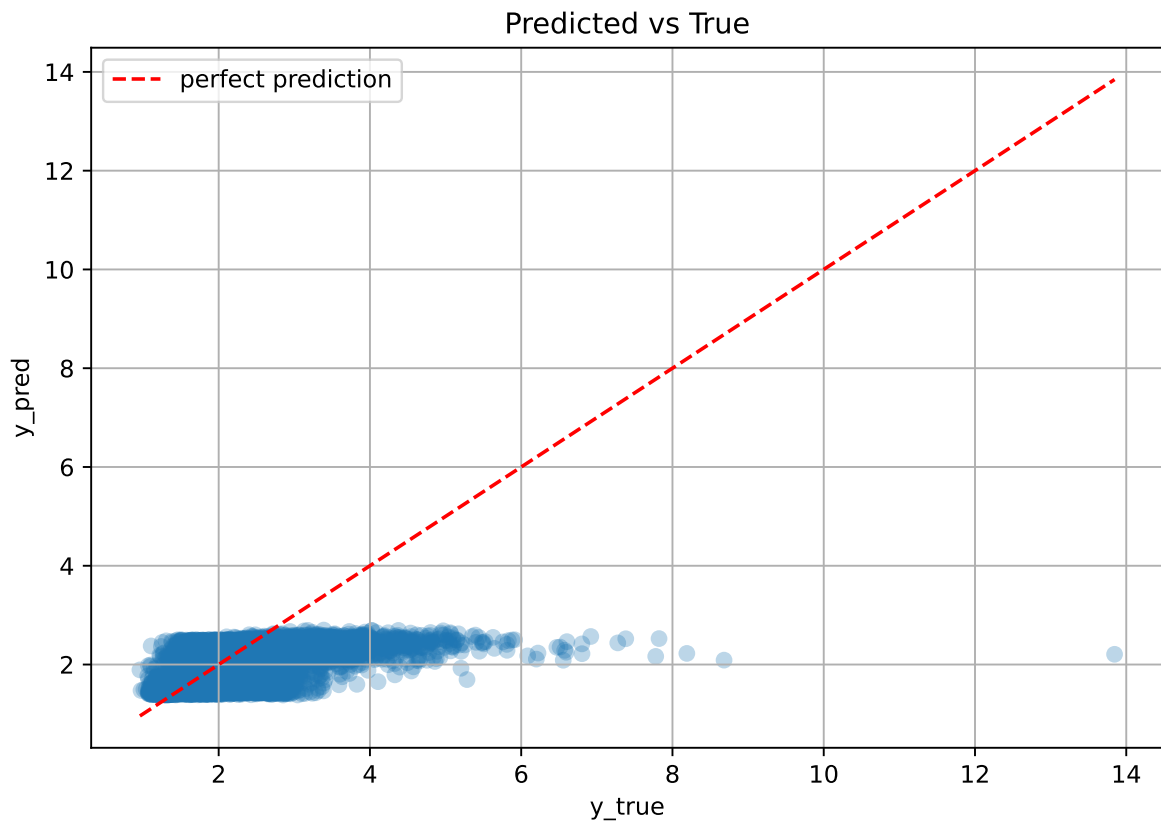
'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6281904404143477, 'reg__border_count': 43, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7301.151555959163}

Rank 8: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.0112950522086458, 'reg__l2_leaf_reg': 28.64173547354376, 'reg__bagging_temperature': 3.7505265944440804, 'reg__random_strength': 3.787582508955974, 'reg__subsample': 0.7420594711090303, 'reg__min_data_in_leaf': 30, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.6816525731419003, 'reg__border_count': 45, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 5625.525054291541}

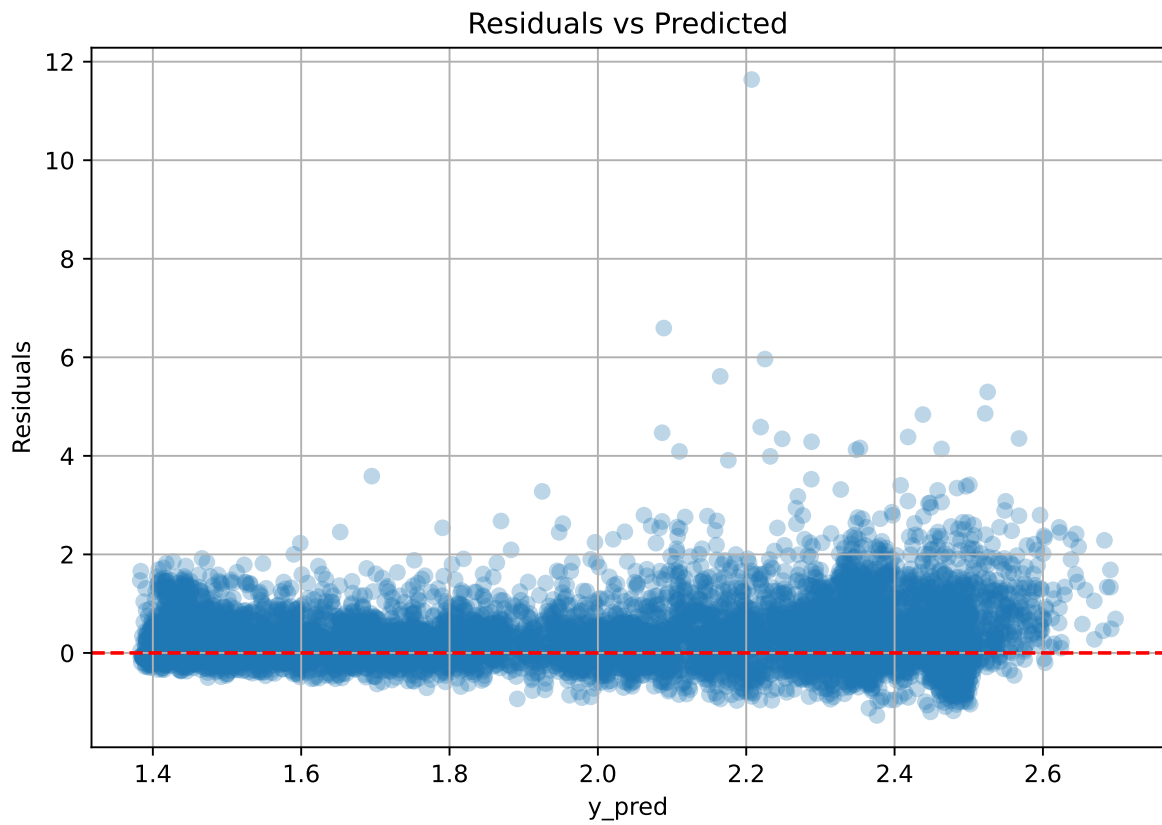
Rank 9: {'reg__iterations': 500, 'reg__depth': 3, 'reg__learning_rate': 0.010869048903579835, 'reg__l2_leaf_reg': 28.63660731881904, 'reg__bagging_temperature': 3.3843520827496323, 'reg__random_strength': 3.9523308312125085, 'reg__subsample': 0.7376876865885468, 'reg__min_data_in_leaf': 28, 'reg__leaf_estimation_iterations': 2, 'reg__rsm': 0.7245020583242614, 'reg__border_count': 36, 'reg__leaf_estimation_method': 'Gradient', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 7858.069060728793}

Rank 10: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.0166933220510748, 'reg__l2_leaf_reg': 28.777669852592098, 'reg__bagging_temperature': 3.896368975563547, 'reg__random_strength': 4.117076289924145, 'reg__subsample': 0.684963822840822, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 1, 'reg__rsm': 0.6254500976166193, 'reg__border_count': 51, 'reg__leaf_estimation_method': 'Newton', 'reg__boosting_type': 'Plain', 'reg__max_ctr_complexity': 1, 'reg__diffusion_temperature': 6164.180190478835}

2.5.14 Scatter



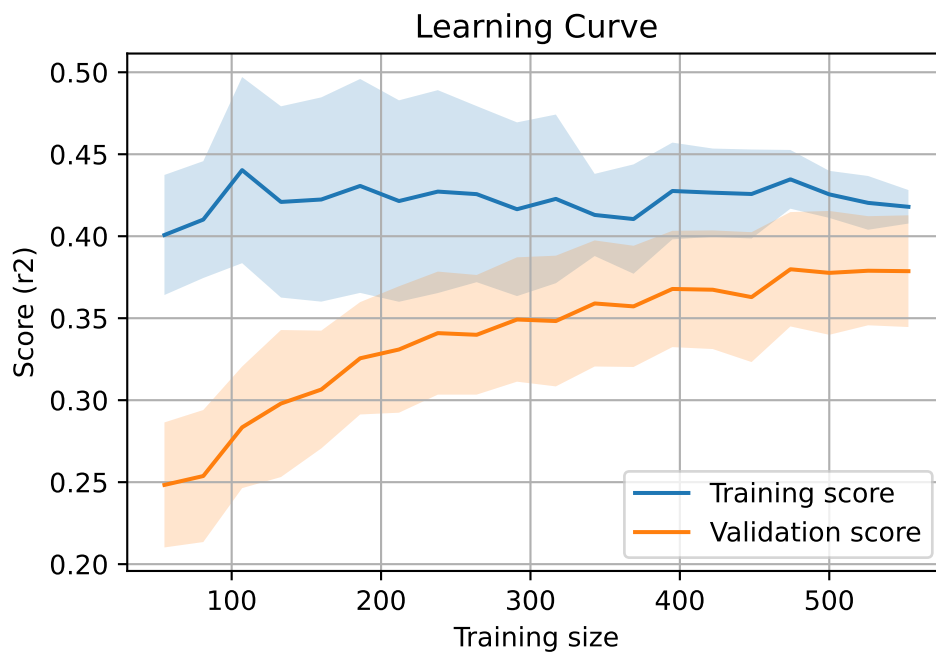
2.5.15 Scatter residuals



2.5.16 Learning curve — Group: 2025_Sep_2

`Len(group)` : 1129, `Len(training)` : 791,

`Len(training_real)` : 554, `Len(test_of_training)` : 237, `Len(test)` : 338



2.6 Sumup-table

model	group	Len training	Len test	Len total	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	2025_Jun_1	1110	475	1585	0.347	0.417	0.042	0.394	0.396	-0.021	0.95	0.833	nan
Linear	2025_Jun_2	1111	476	1587	0.218	0.206	0.022	0.216	0.217	0.011	1.054	1.061	nan
Linear	2025_Sep_1	2707	1160	3867	0.426	0.433	0.022	0.449	0.45	0.017	1.039	0.984	nan
Linear	2025_Sep_2	791	338	1129	0.377	0.351	0.022	0.374	0.374	0.024	1.067	1.075	nan
Linear	MEAN	5719	2449	8168	0.342	0.352	0.027	0.358	0.359	0.008	1.028	0.988	nan
LGBM	2025_Jun_1	1110	475	1585	0.587	0.586	0.038	0.572	0.599	0.012	1.021	1.001	0.5863
LGBM	2025_Jun_2	1111	476	1587	0.423	0.394	0.013	0.38	0.405	0.011	1.027	1.074	0.2488
LGBM	2025_Sep_1	2707	1160	3867	0.585	0.559	0.029	0.561	0.579	0.02	1.036	1.046	0.5589
LGBM	2025_Sep_2	791	338	1129	0.364	0.339	0.032	0.346	0.345	0.006	1.018	1.075	0.2124
LGBM	MEAN	5719	2449	8168	0.49	0.47	0.028	0.465	0.482	0.012	1.025	1.049	0.402
CatBoost	2025_Jun_1	1110	475	1585	0.656	0.625	0.041	0.646	0.669	0.044	1.07	1.05	0.6253
CatBoost	2025_Jun_2	1111	476	1587	0.399	0.381	0.013	0.371	0.392	0.011	1.029	1.047	0.3808
CatBoost	2025_Sep_1	2707	1160	3867	0.627	0.599	0.026	0.603	0.607	0.008	1.014	1.048	0.5986
CatBoost	2025_Sep_2	791	338	1129	0.422	0.384	0.036	0.411	0.424	0.04	1.103	1.099	0.1937
CatBoost	MEAN	5719	2449	8168	0.526	0.497	0.029	0.508	0.523	0.026	1.054	1.061	0.45

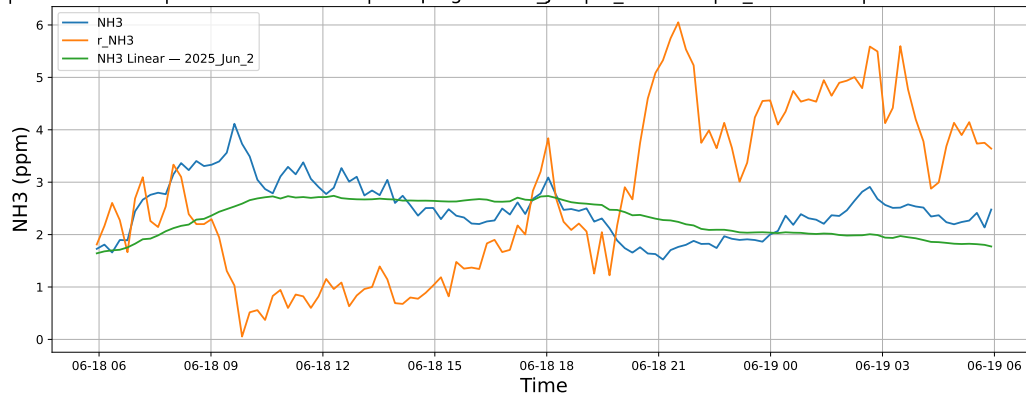
3 Plot

3.1 Standalone plots

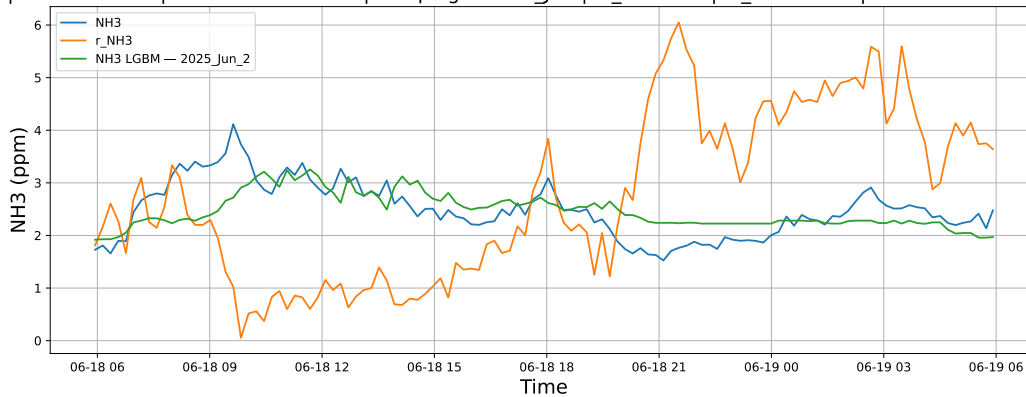
3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

3.1.1.1 Time window : 24

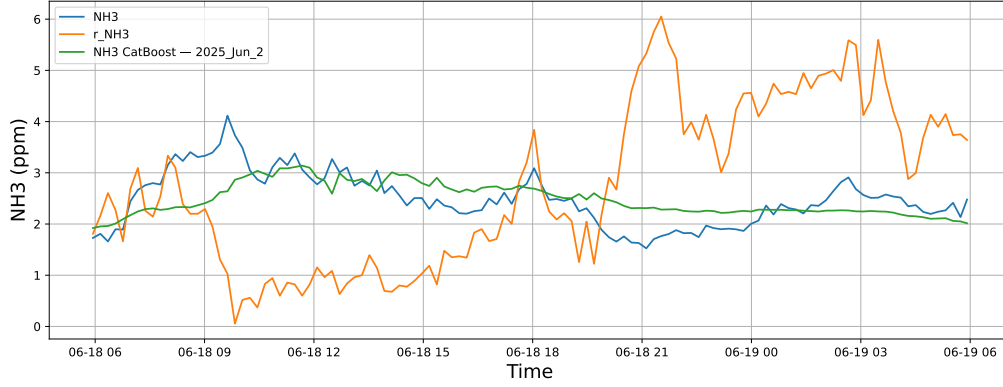
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

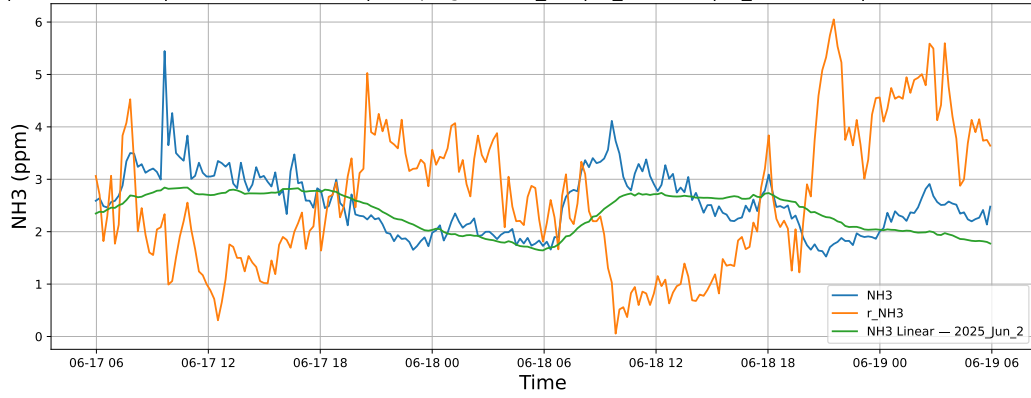


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19

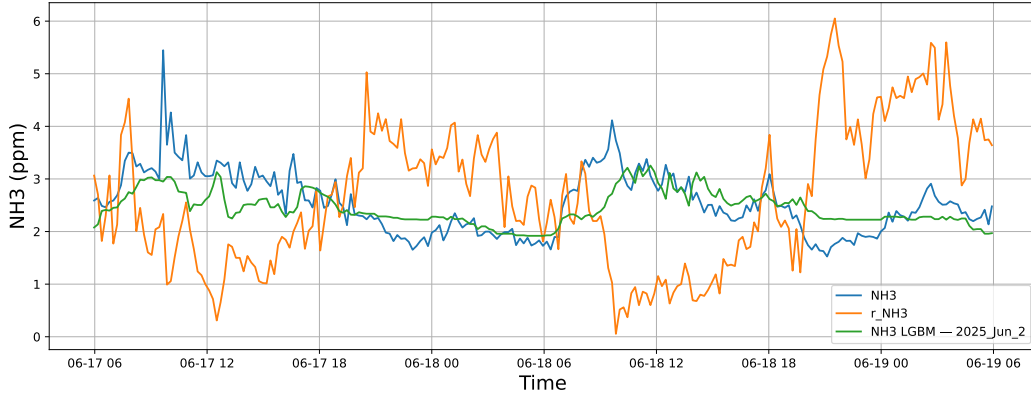


3.1.1.2 Time window : 48

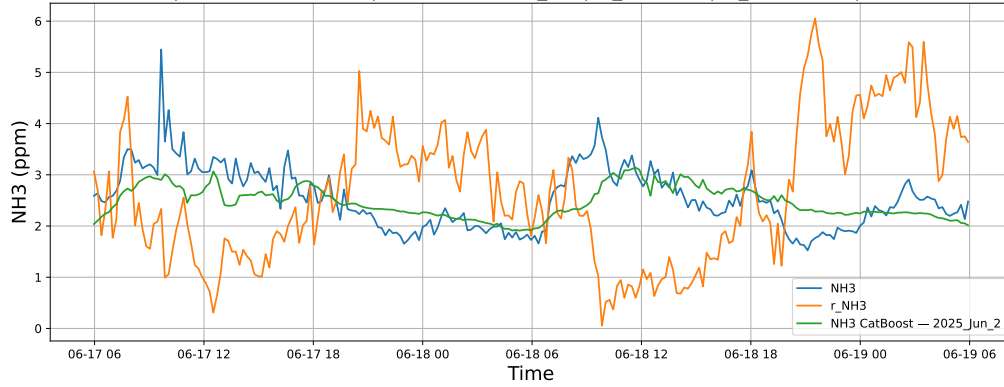
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

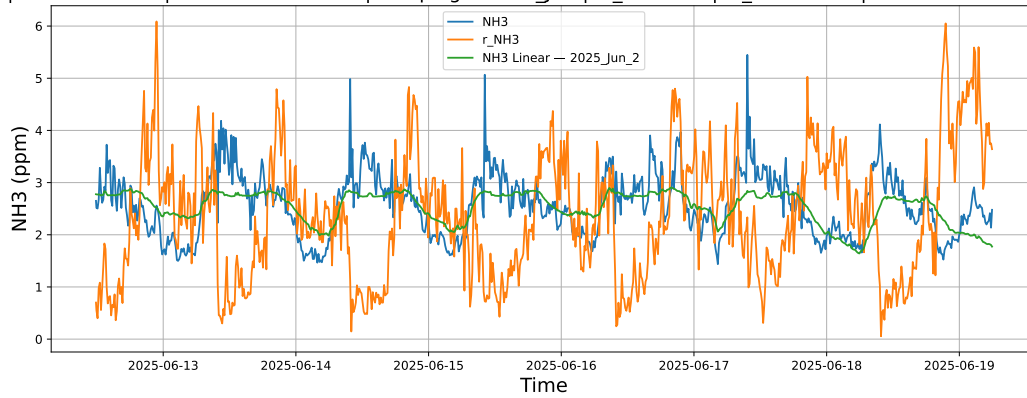


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19

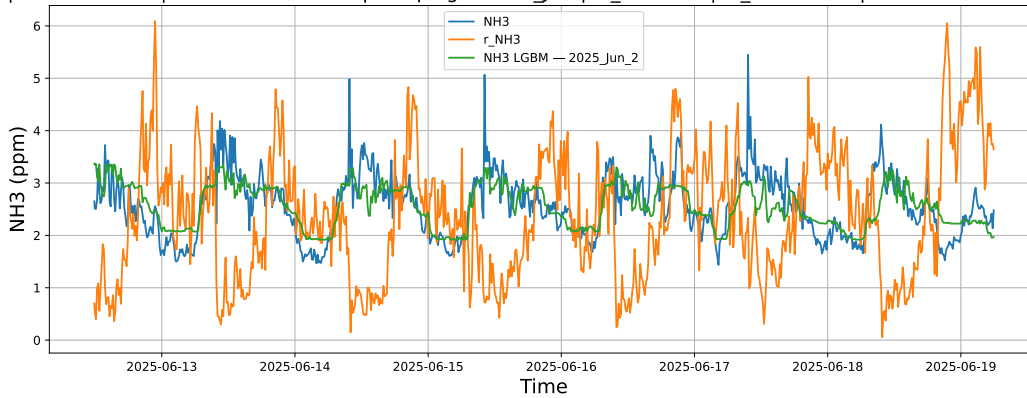


3.1.1.3 Time window : ALL

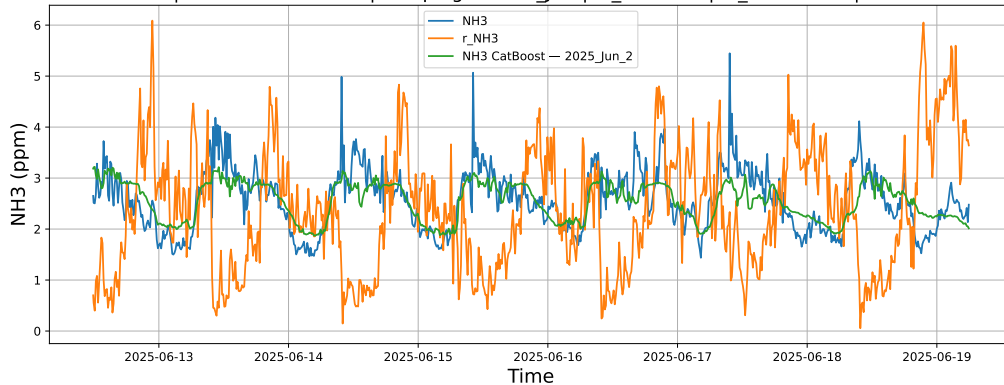
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



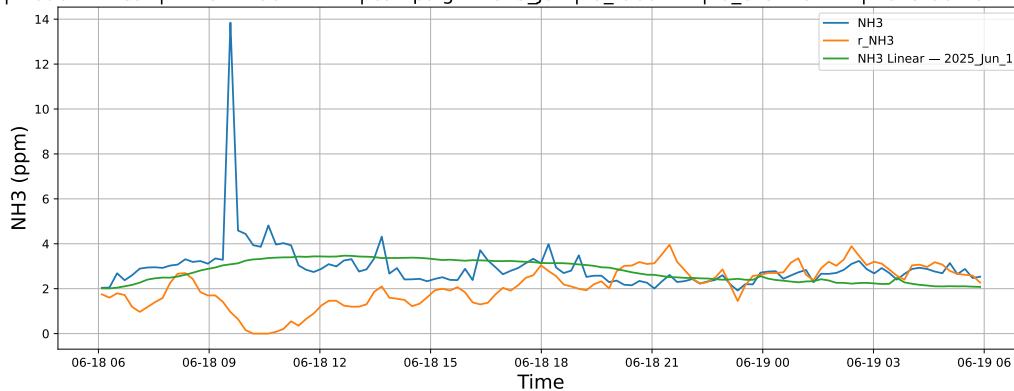
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



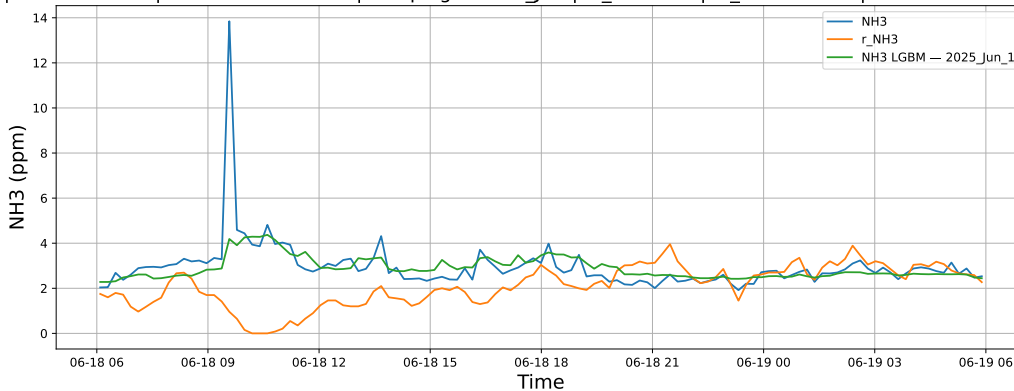
3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

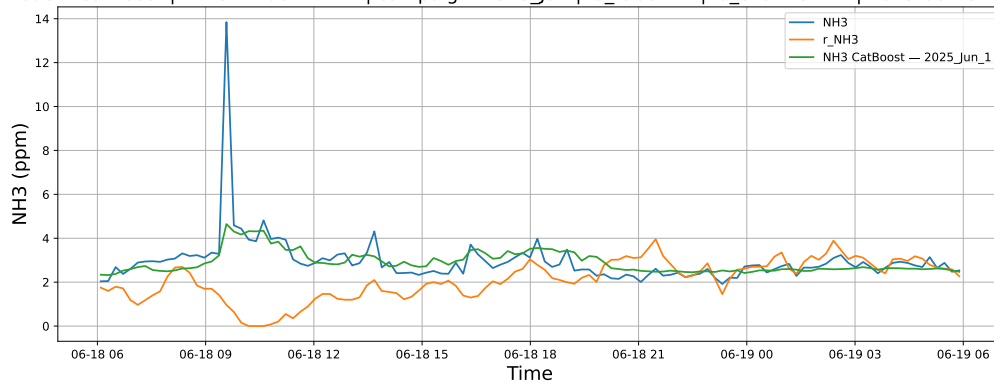
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

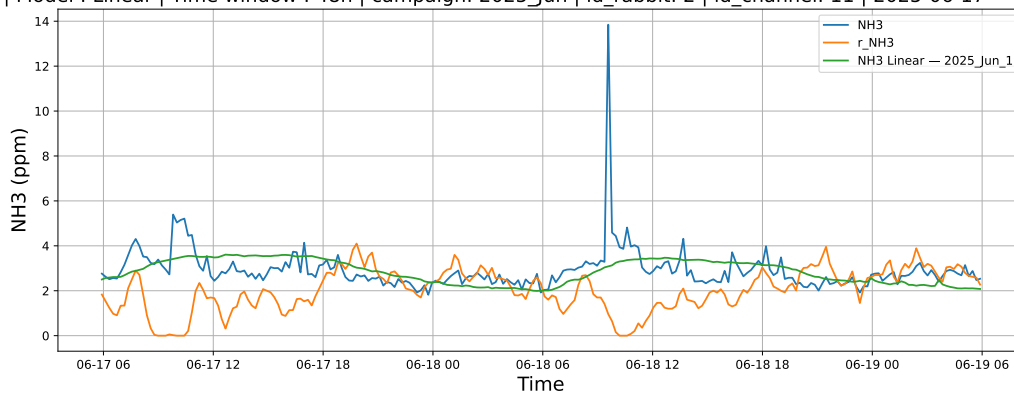


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19

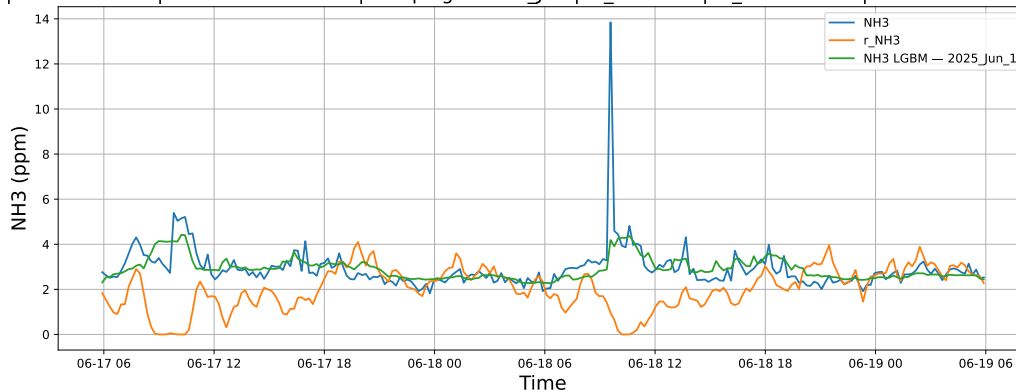


3.1.2.2 Time window : 48

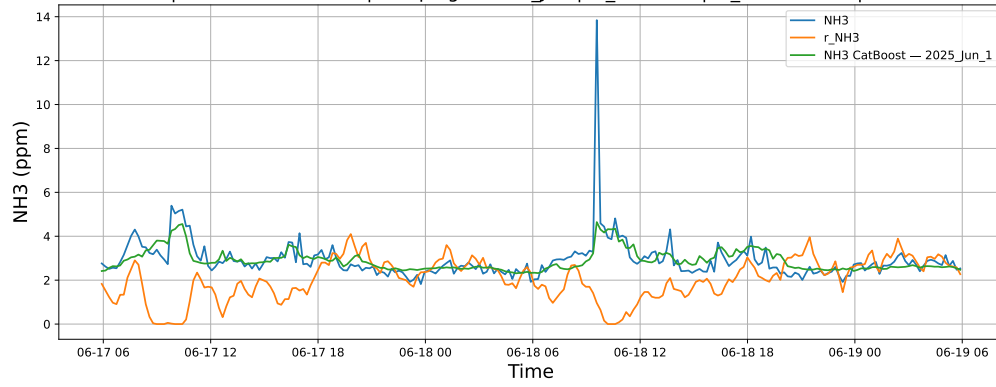
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

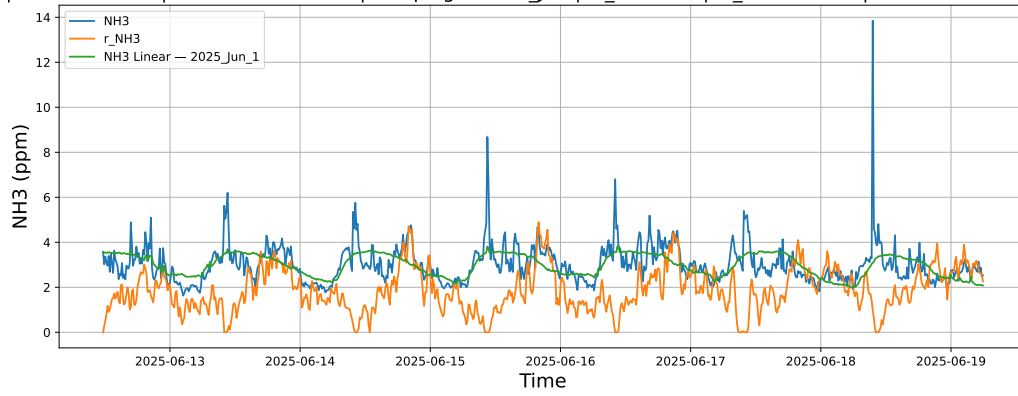


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

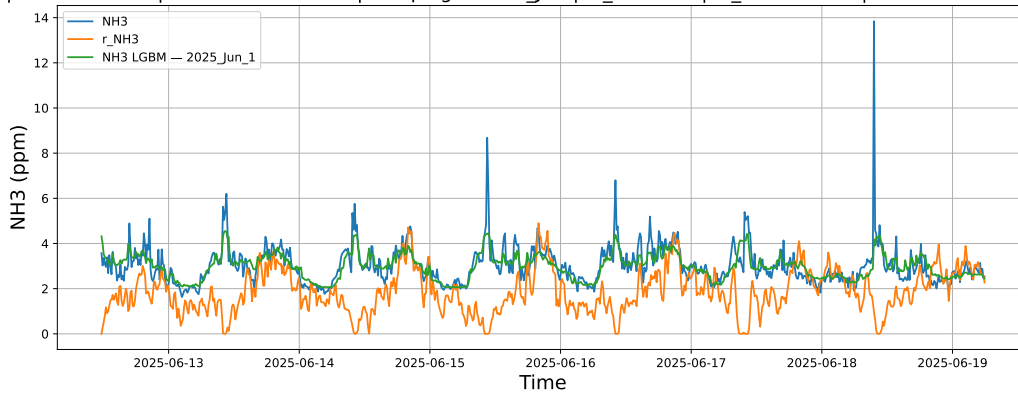


3.1.2.3 Time window : ALL

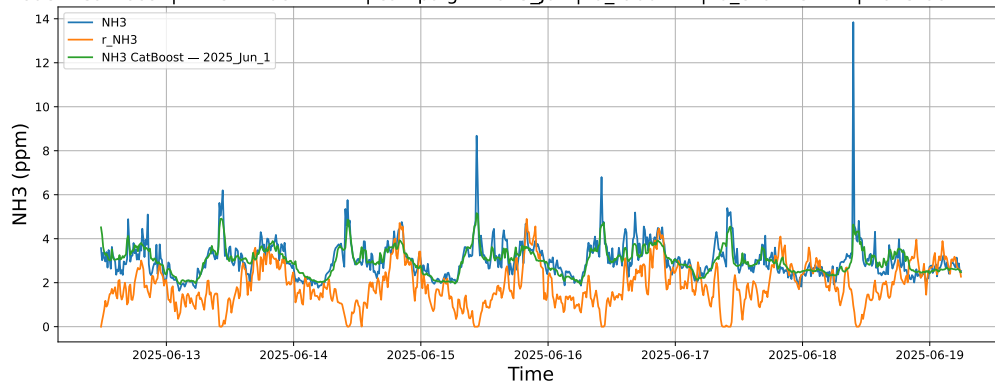
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



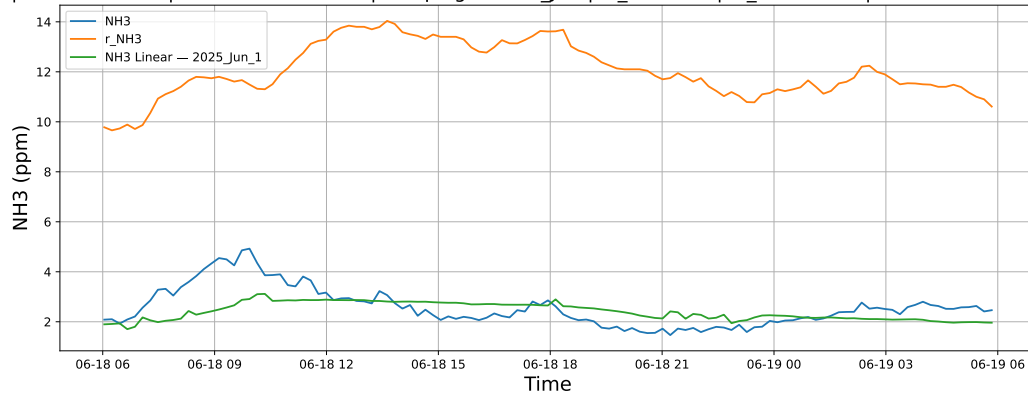
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



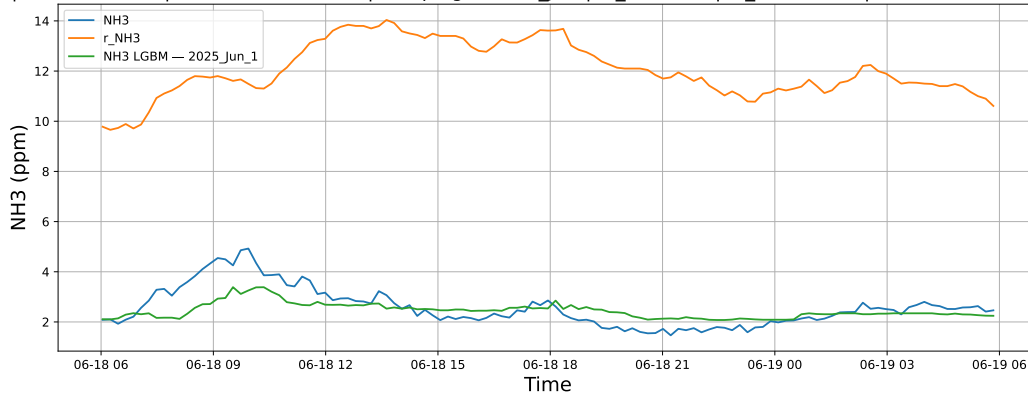
3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

3.1.3.1 Time window : 24

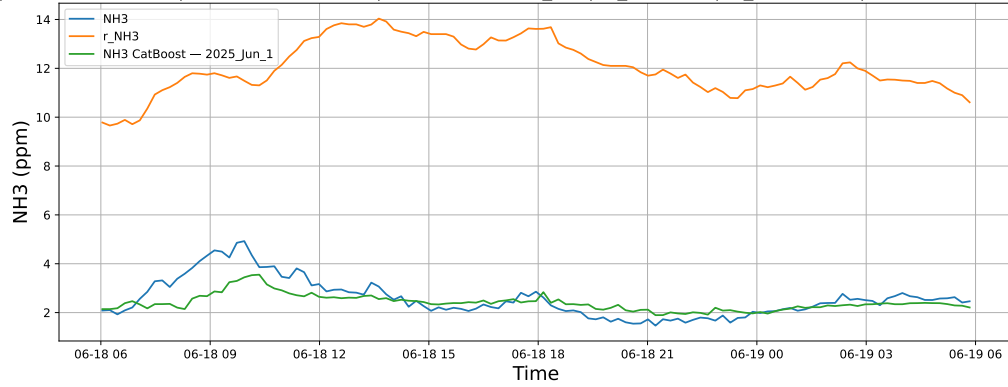
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

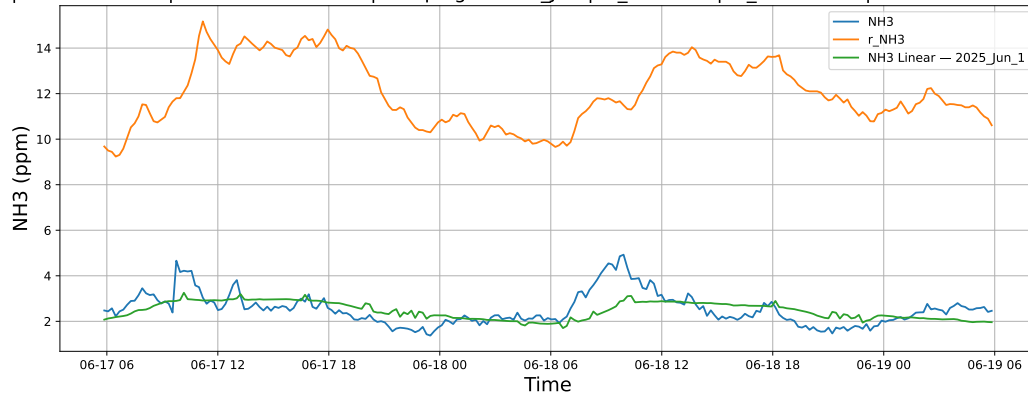


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19

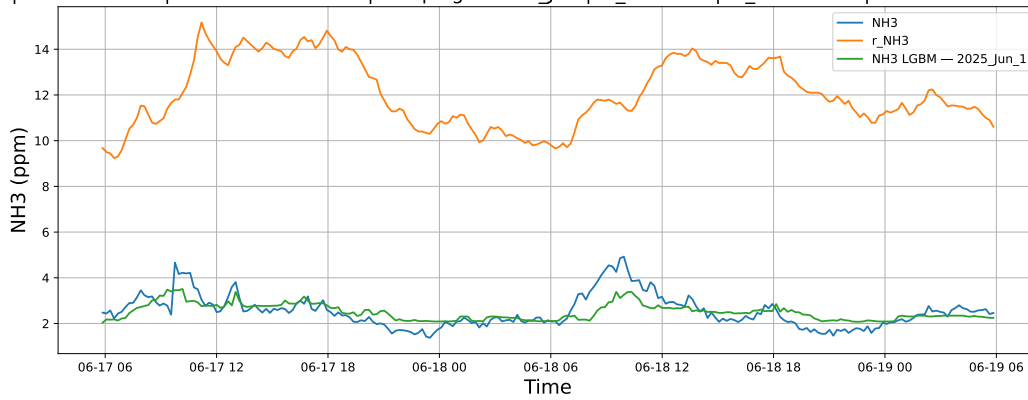


3.1.3.2 Time window : 48

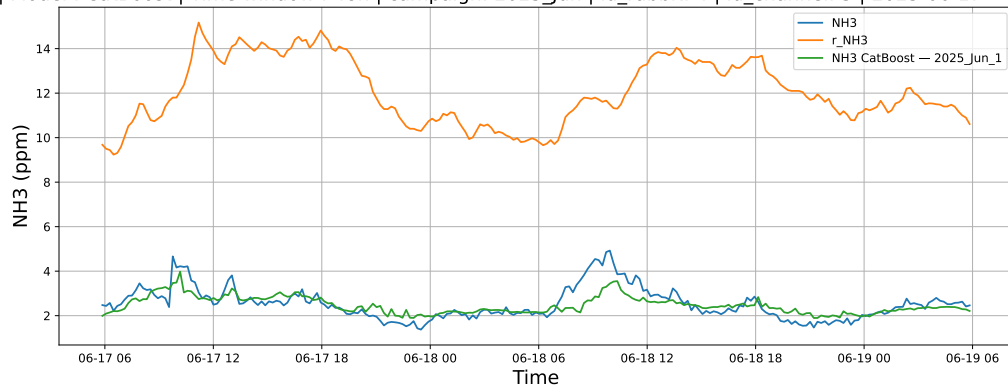
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

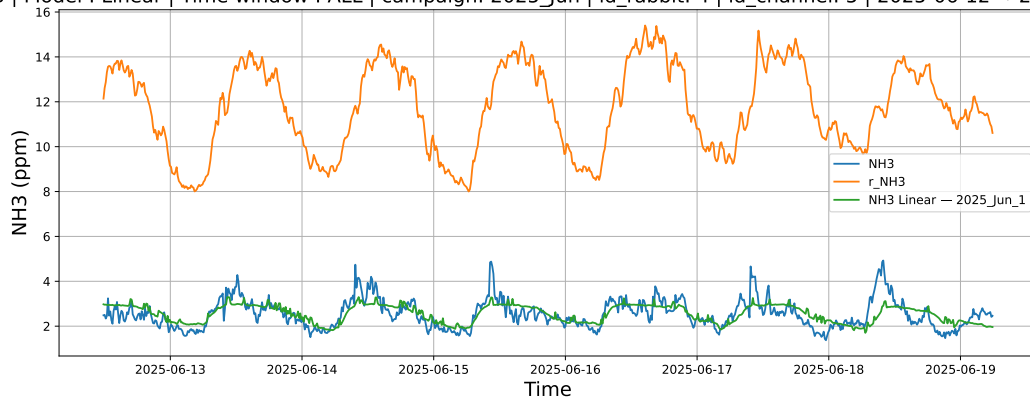


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19

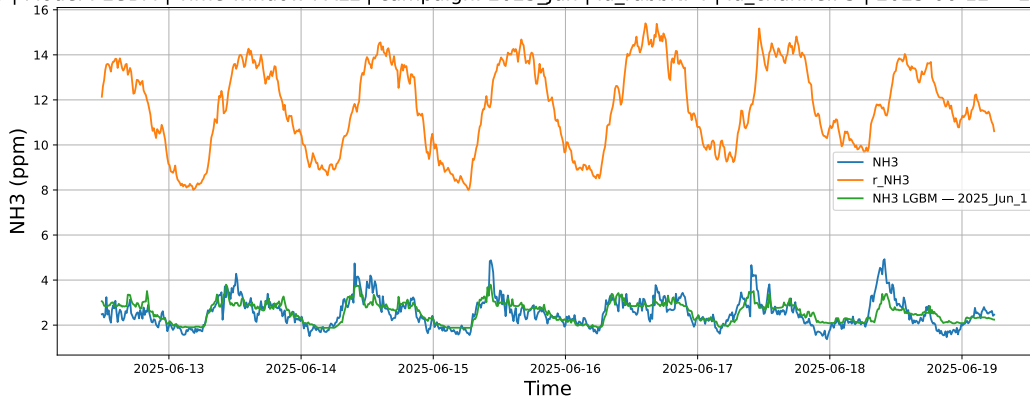


3.1.3.3 Time window : ALL

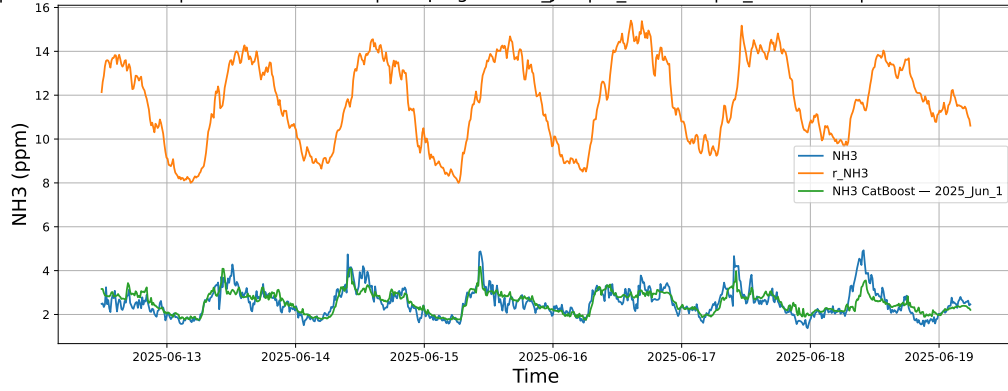
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



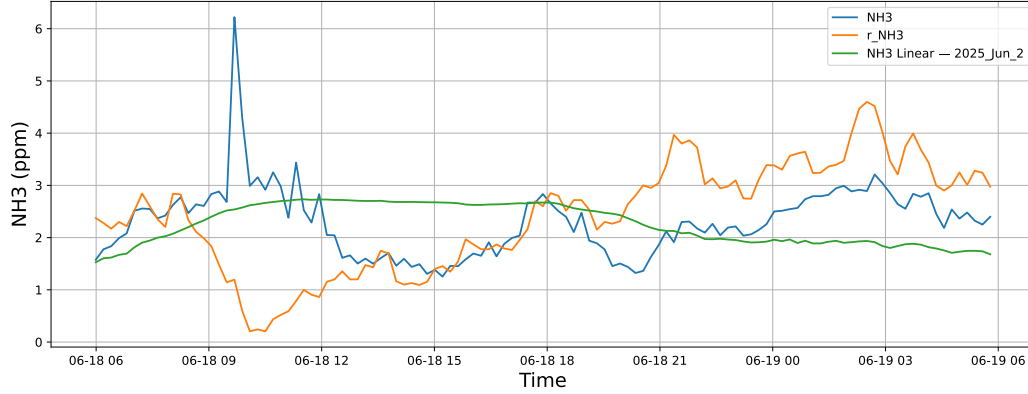
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



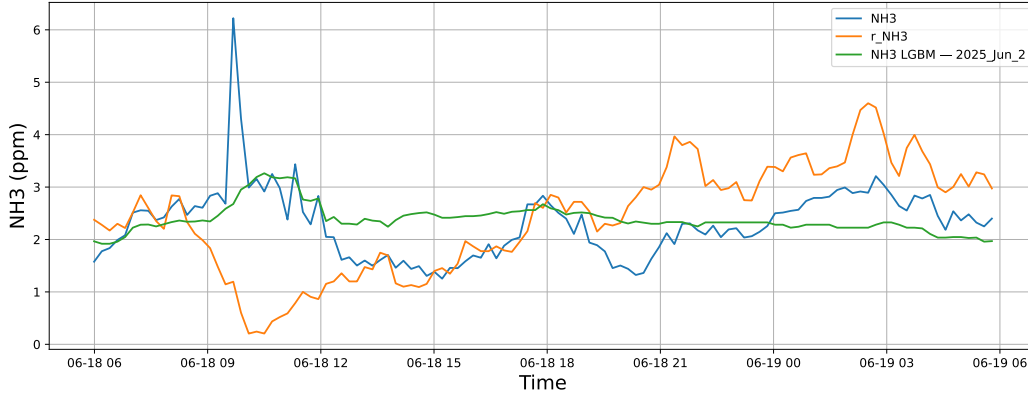
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

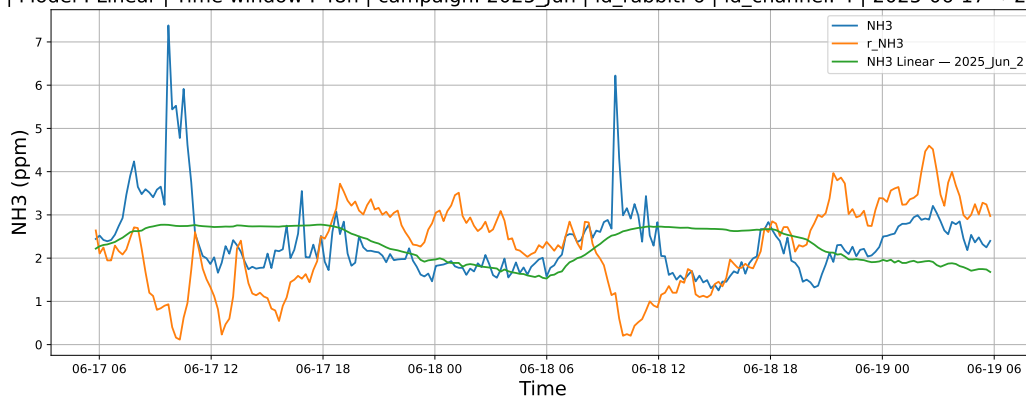


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

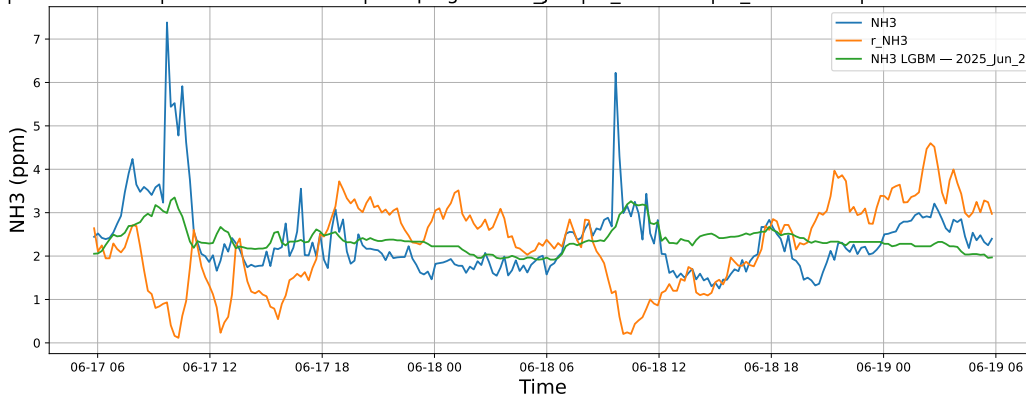


3.1.4.2 Time window : 48

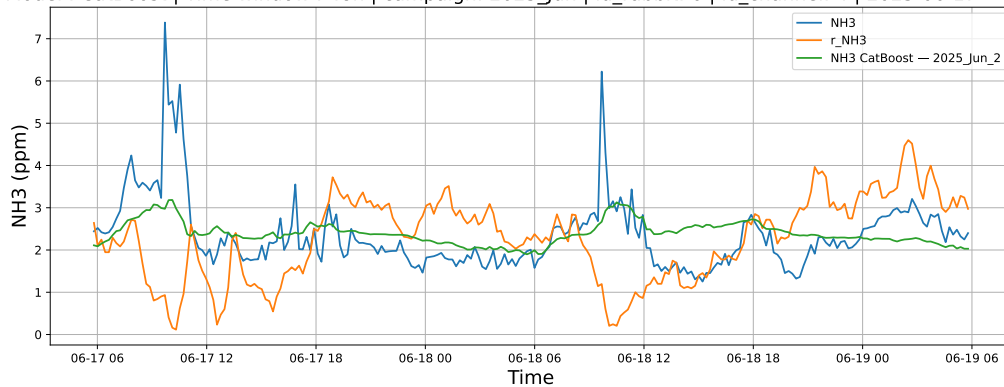
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

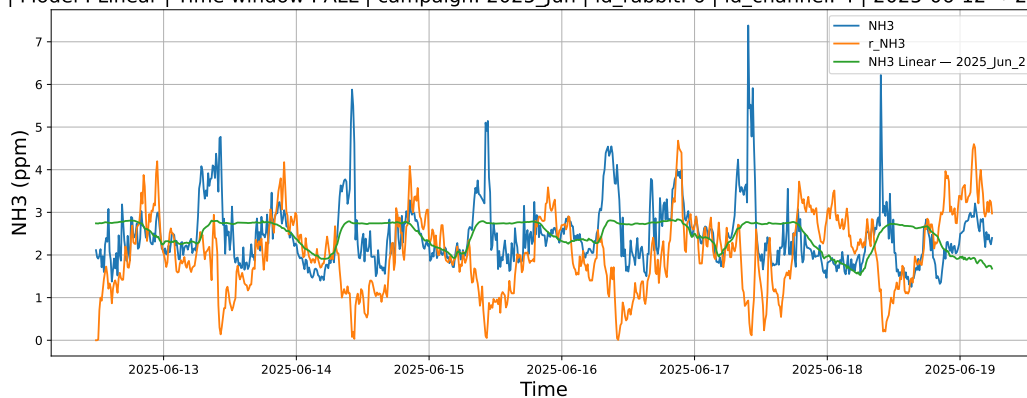


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

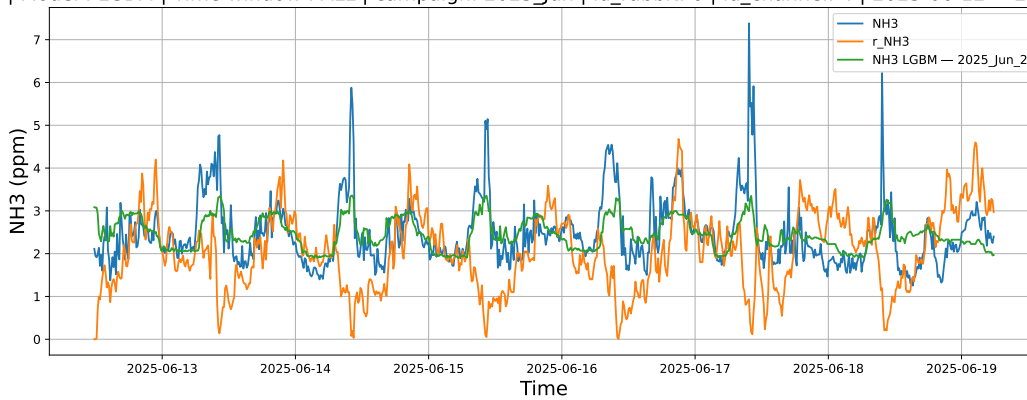


3.1.4.3 Time window : ALL

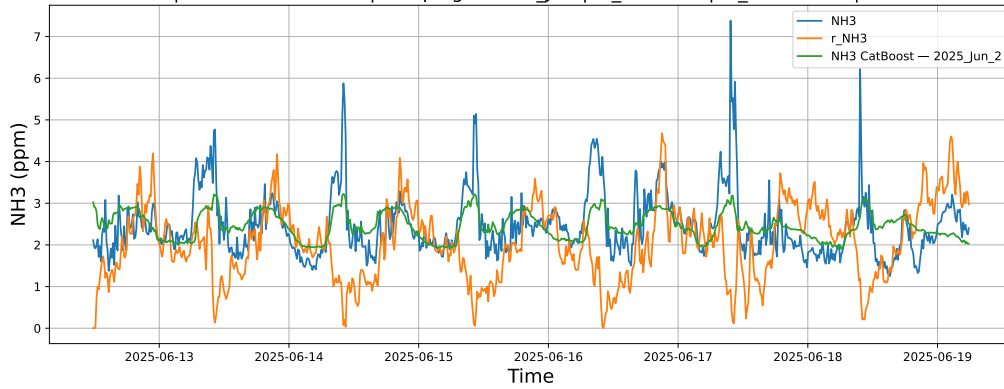
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



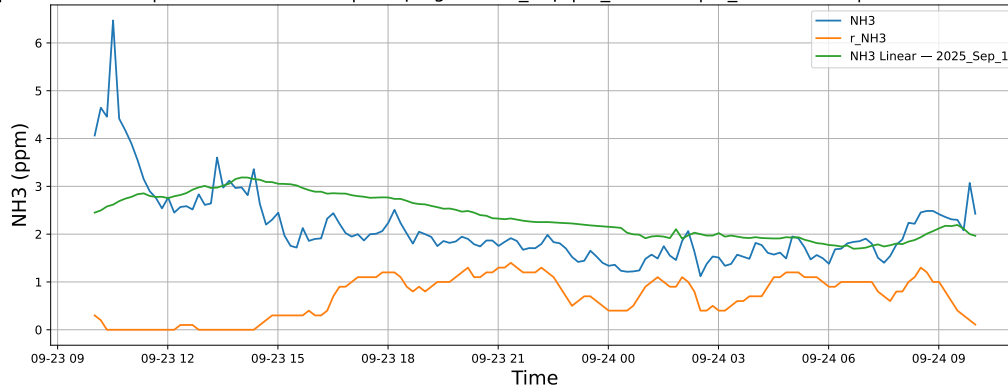
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



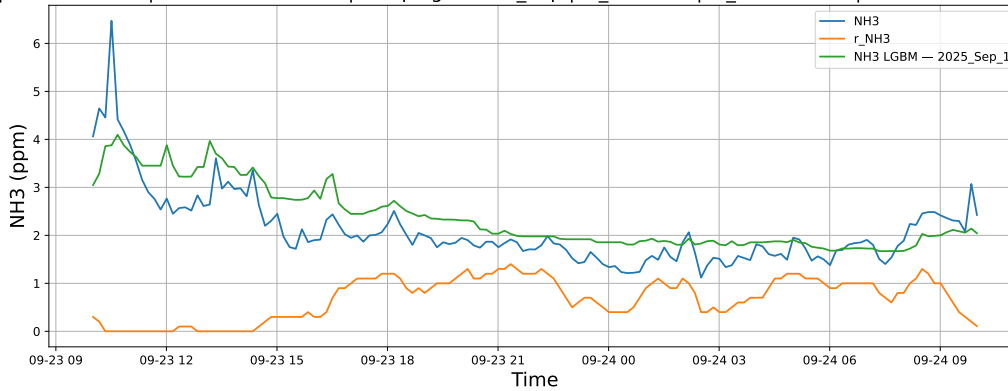
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

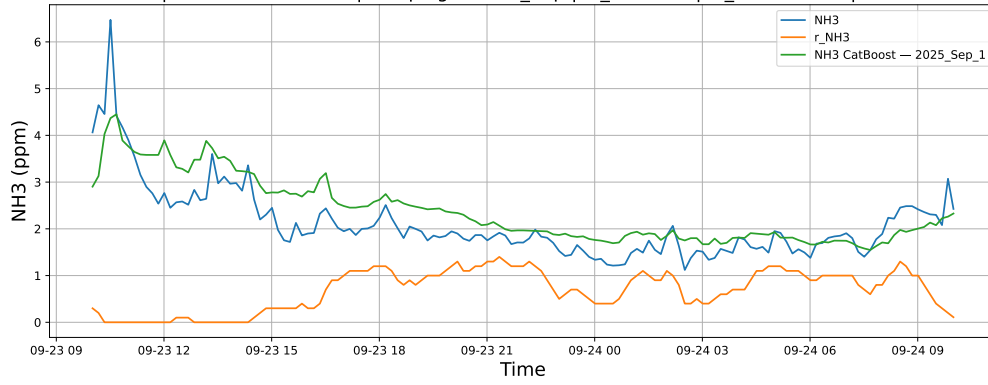
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

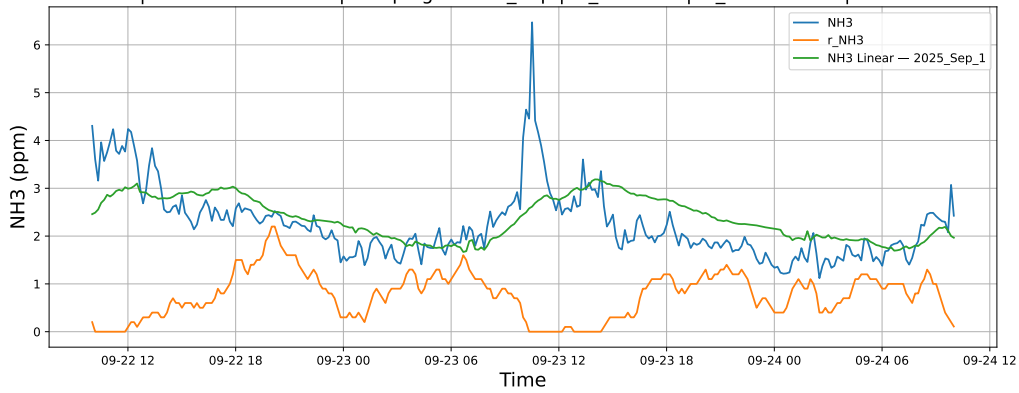


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

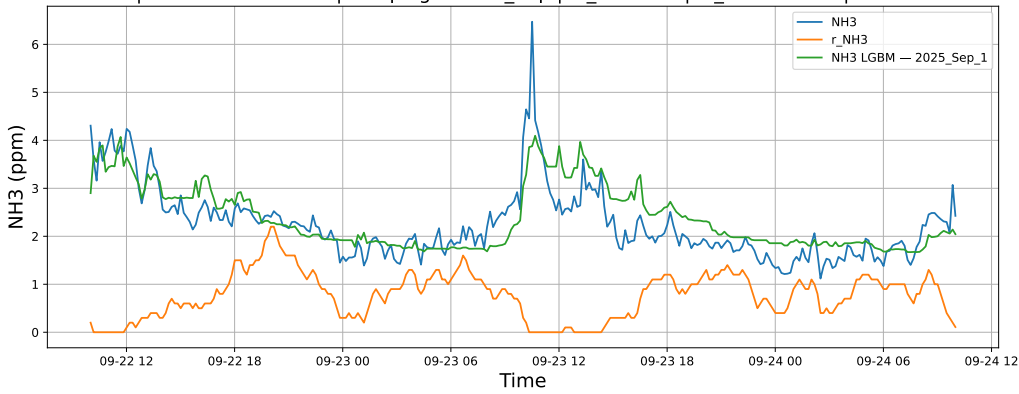


3.1.5.2 Time window : 48

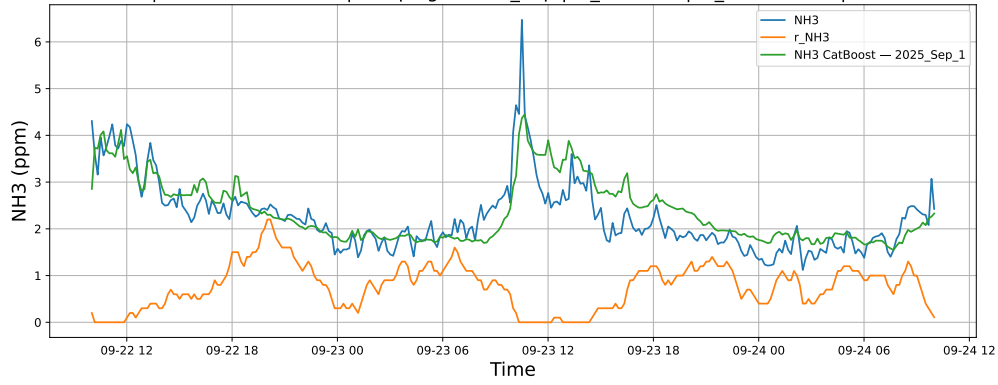
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

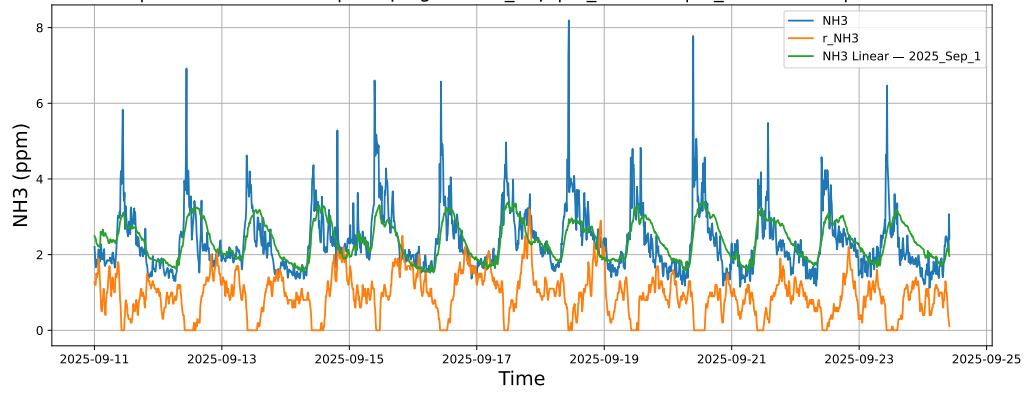


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24

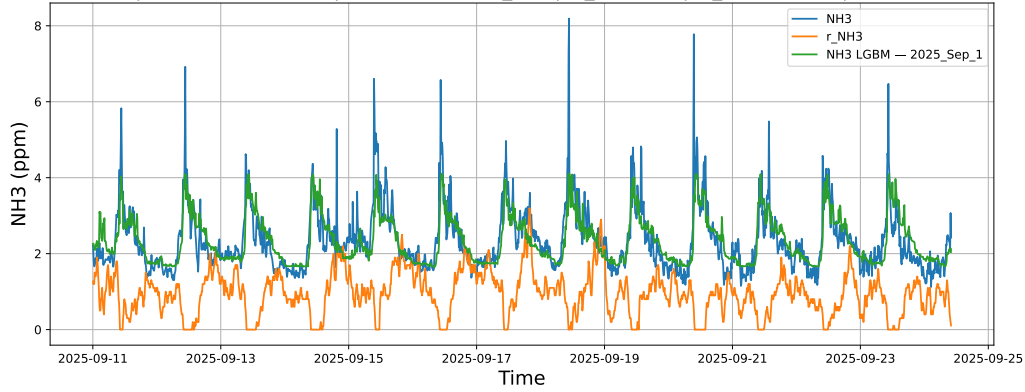


3.1.5.3 Time window : ALL

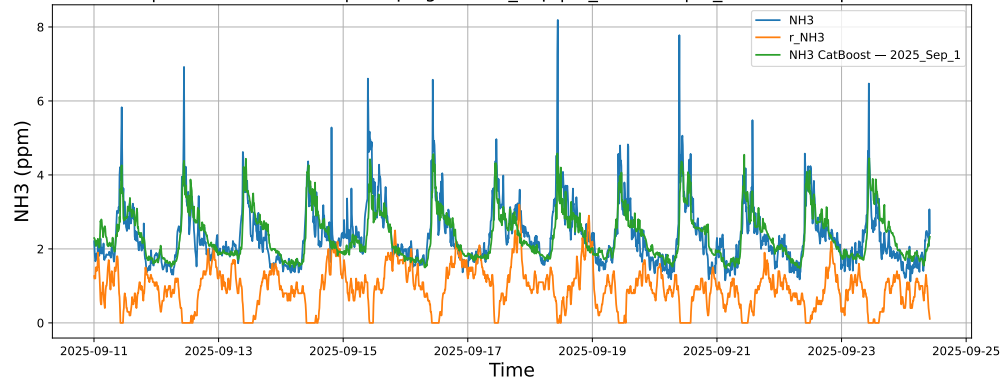
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



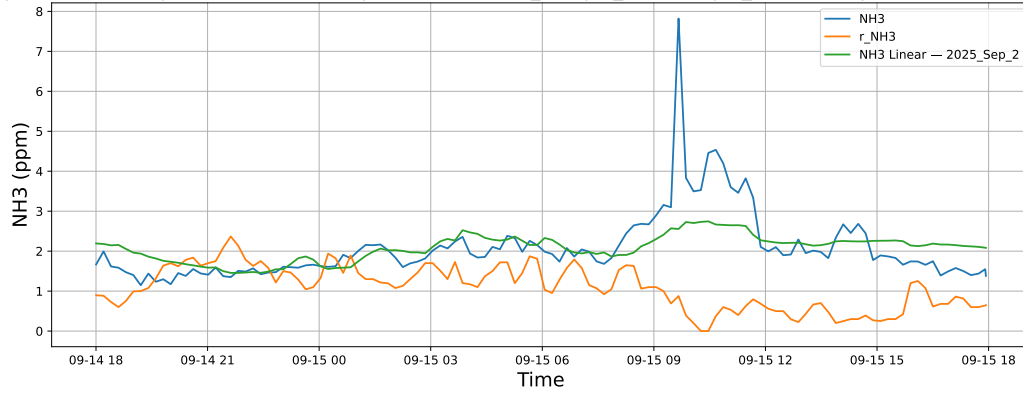
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



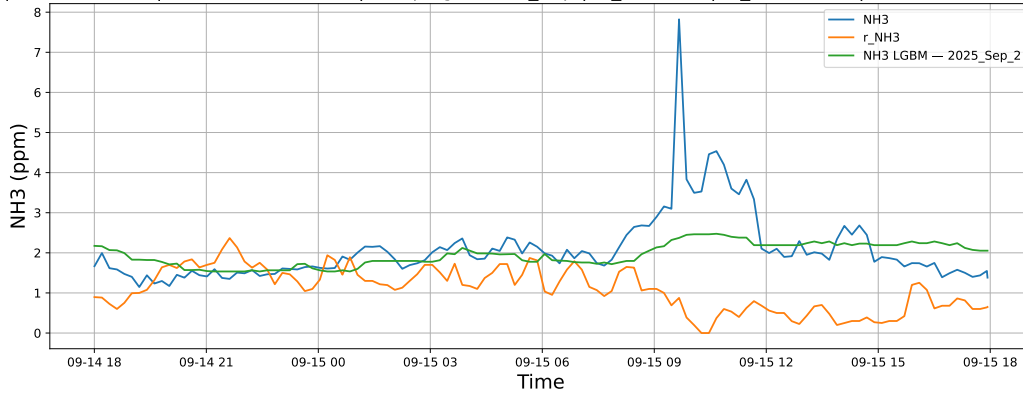
3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

3.1.6.1 Time window : 24

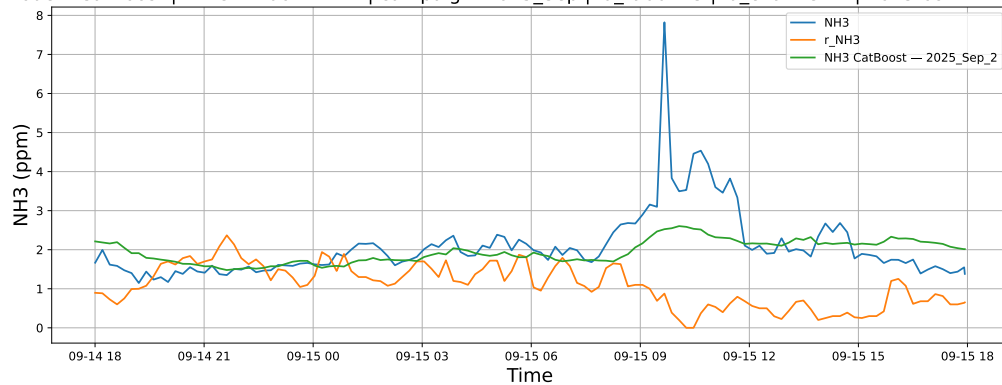
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

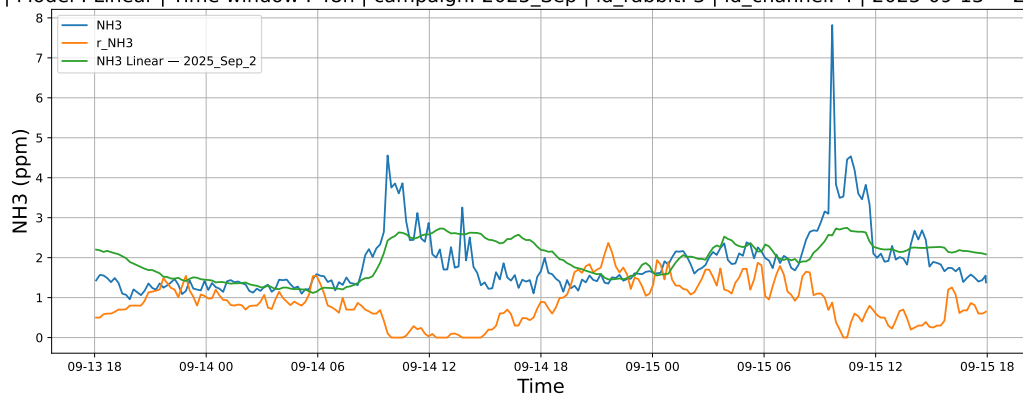


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15

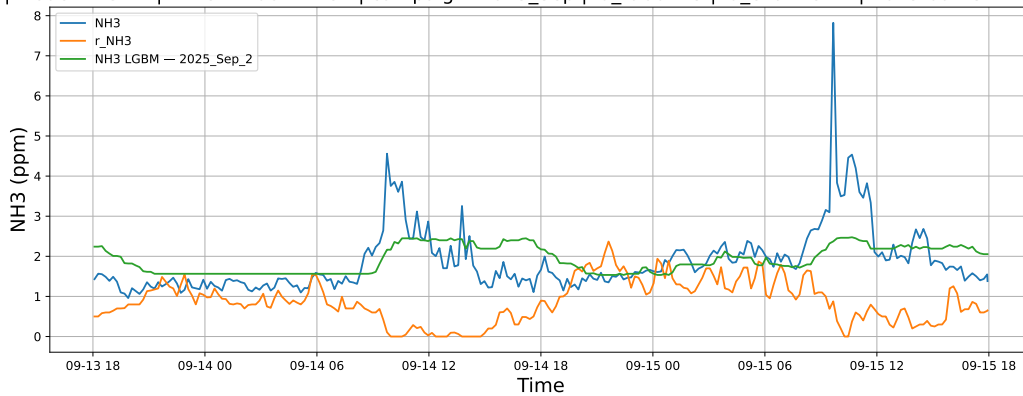


3.1.6.2 Time window : 48

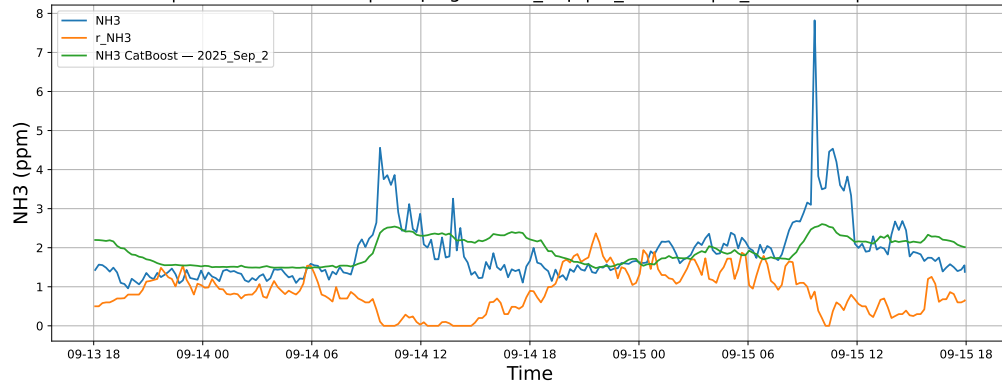
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

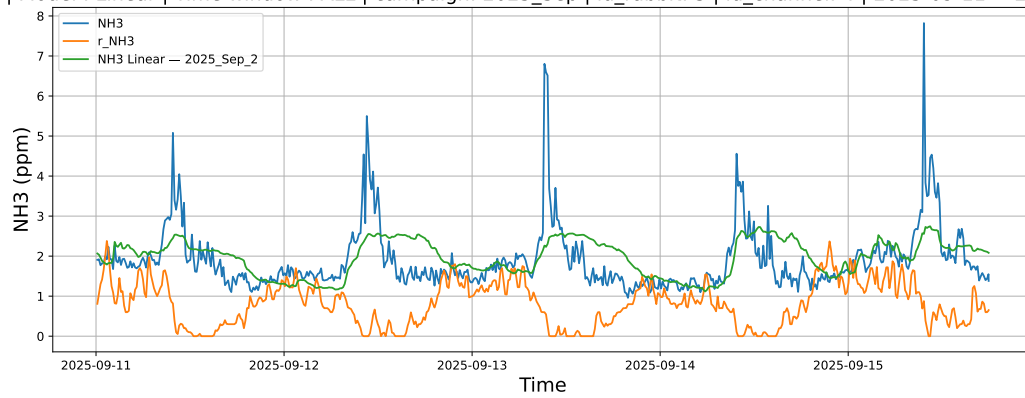


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15

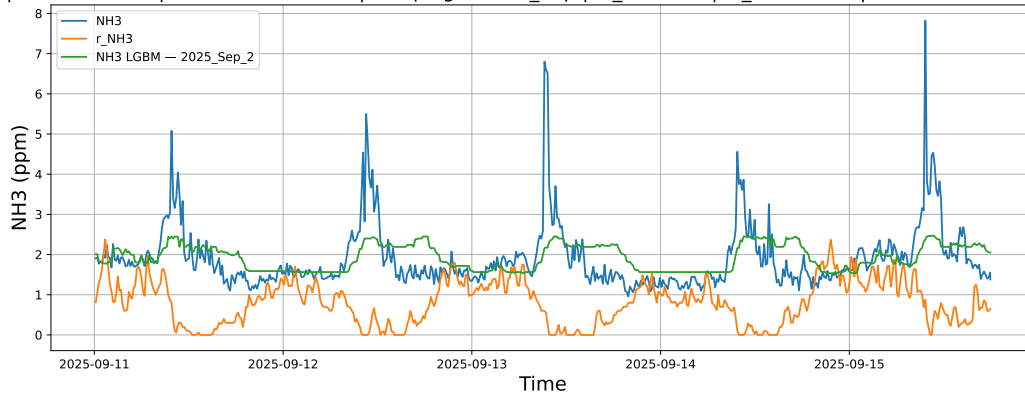


3.1.6.3 Time window : ALL

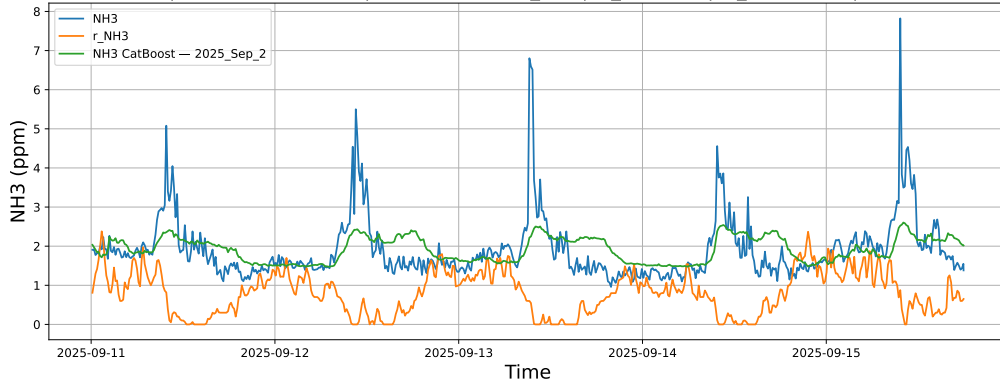
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



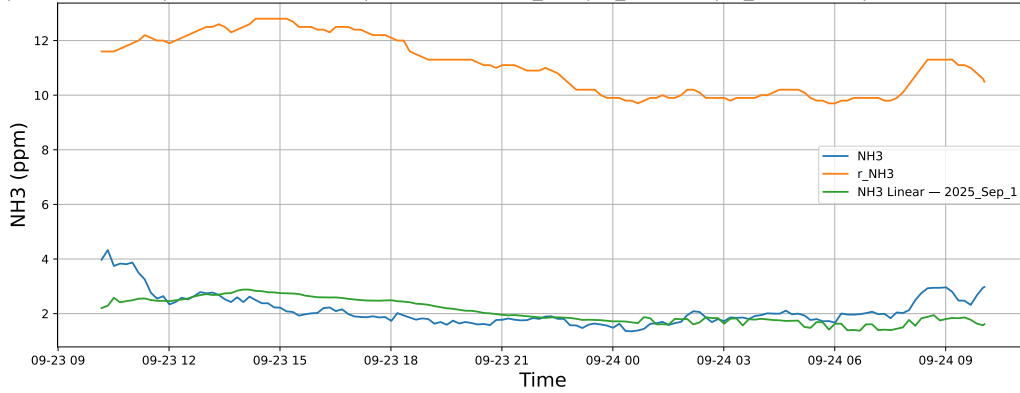
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



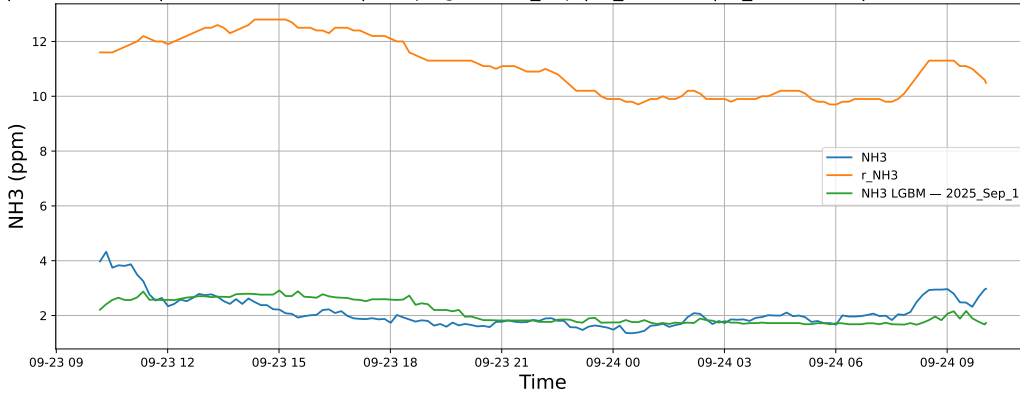
3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

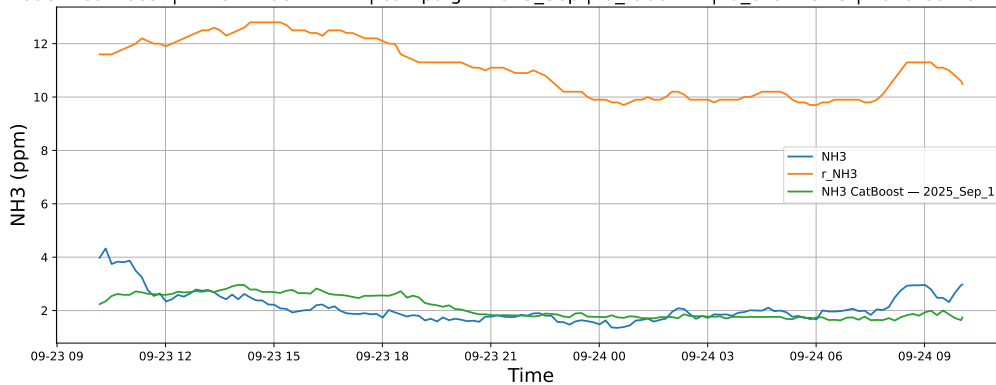
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

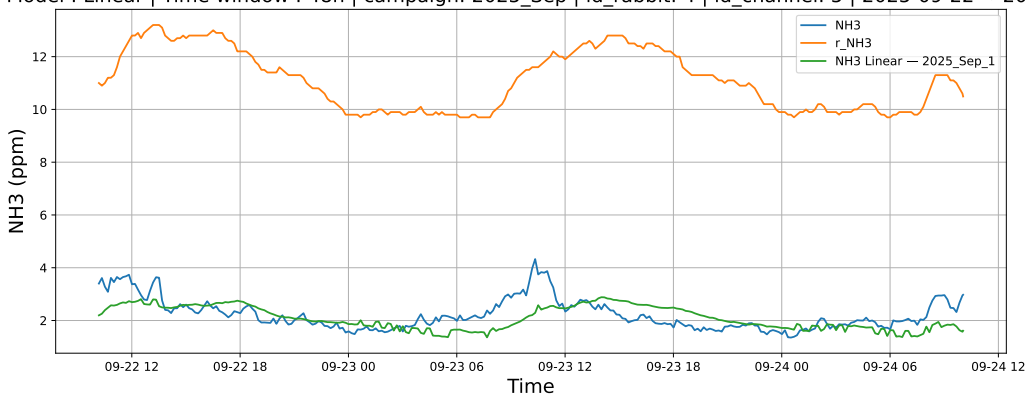


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24

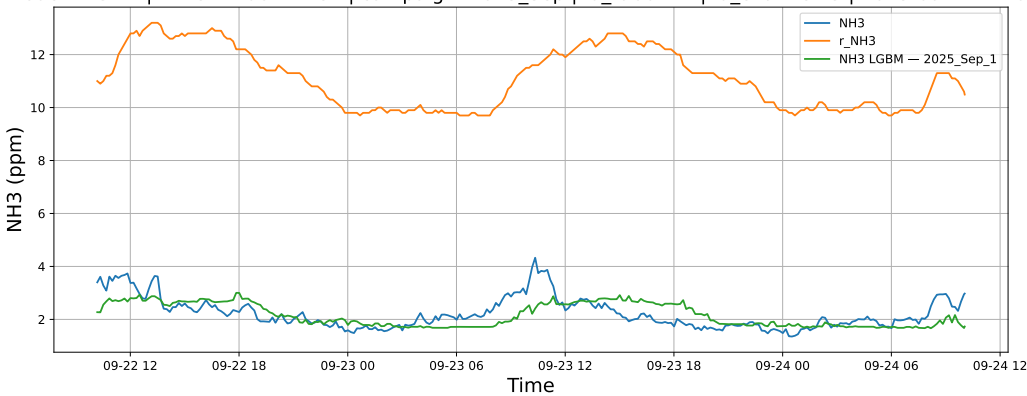


3.1.7.2 Time window : 48

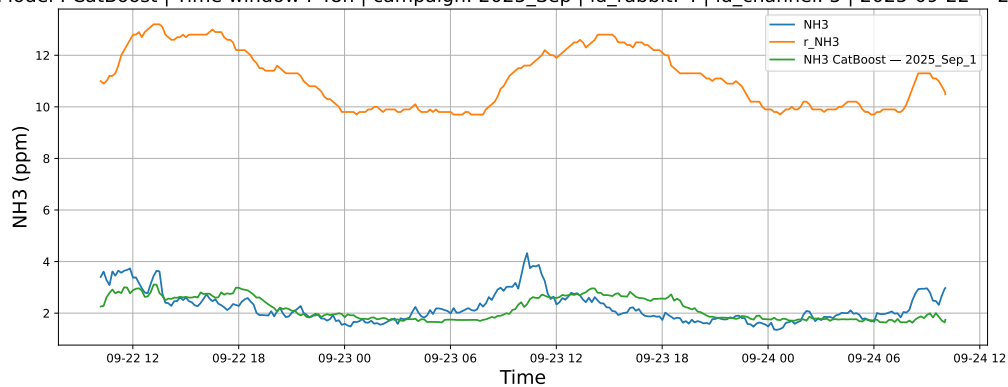
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

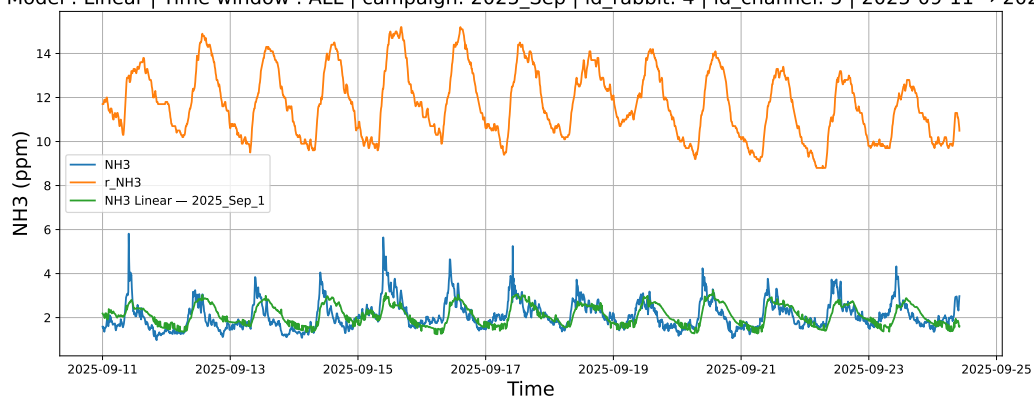


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

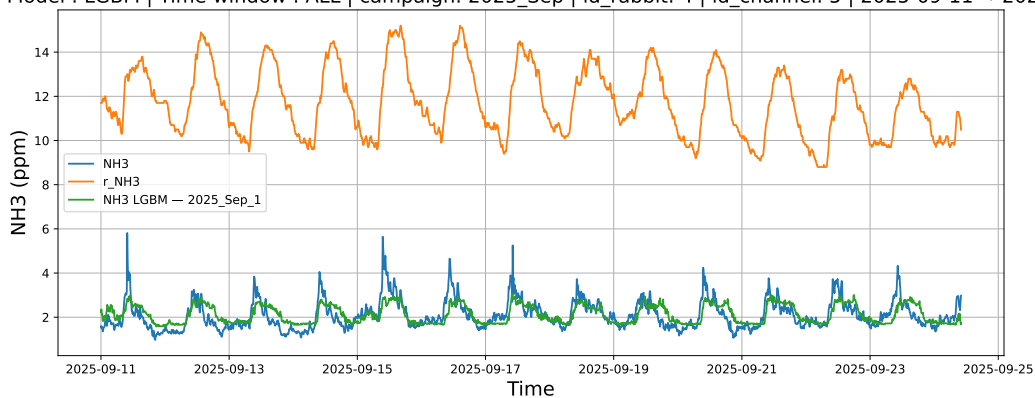


3.1.7.3 Time window : ALL

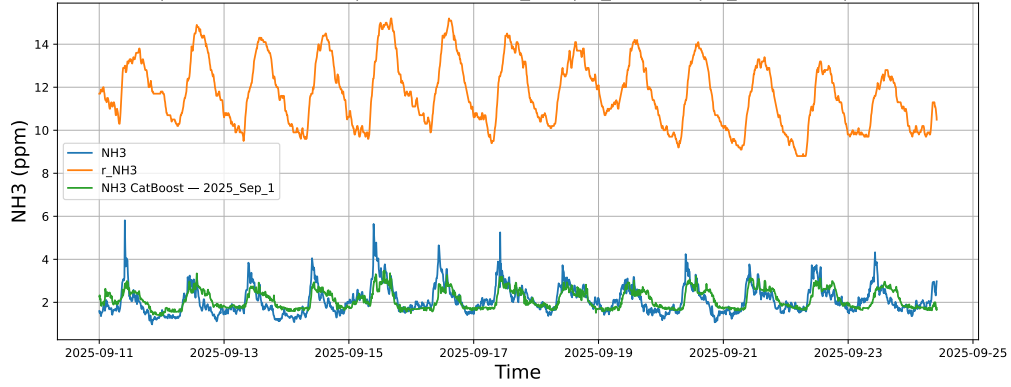
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



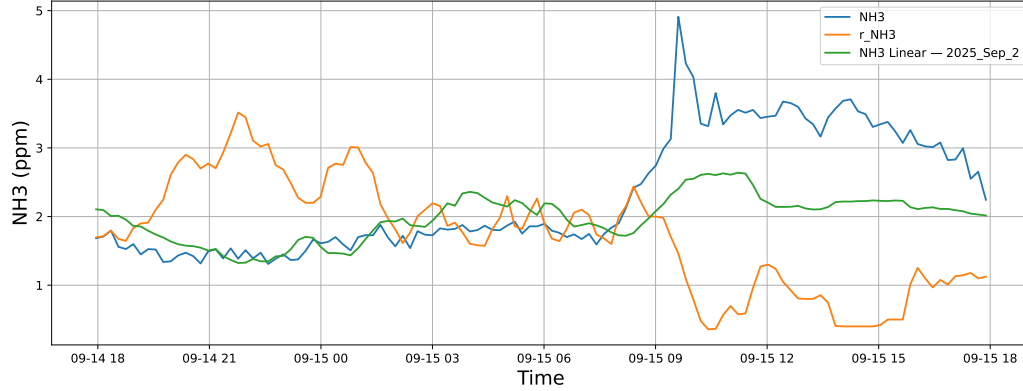
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



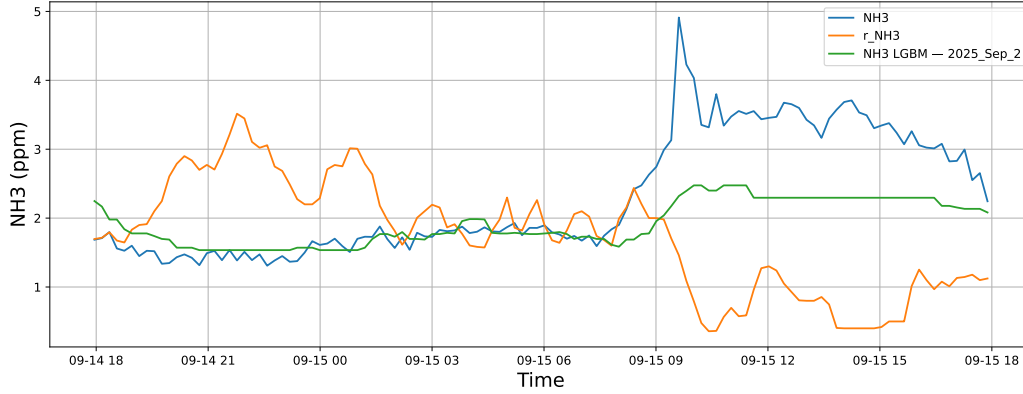
3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

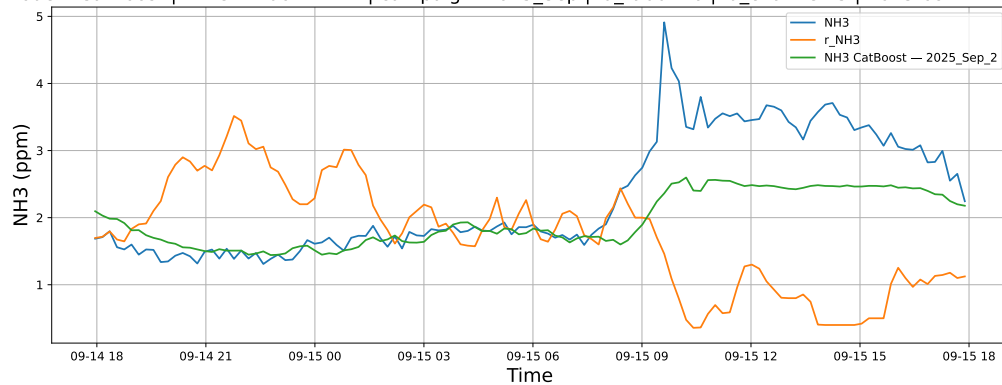
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

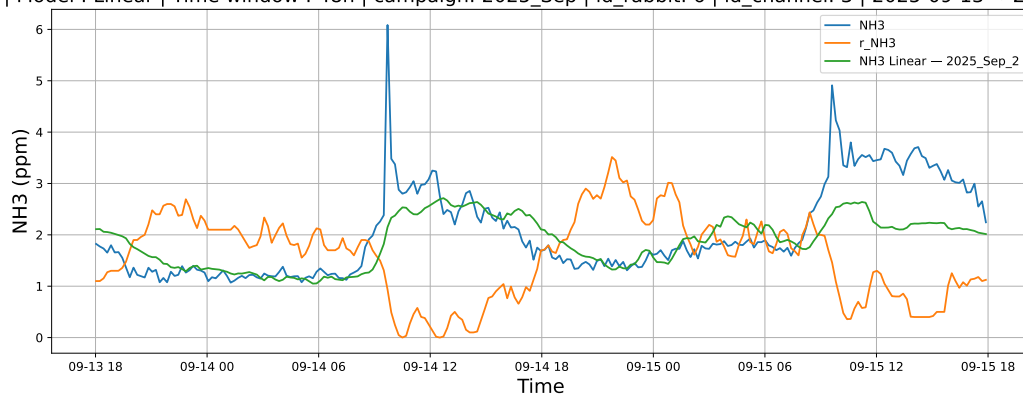


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15

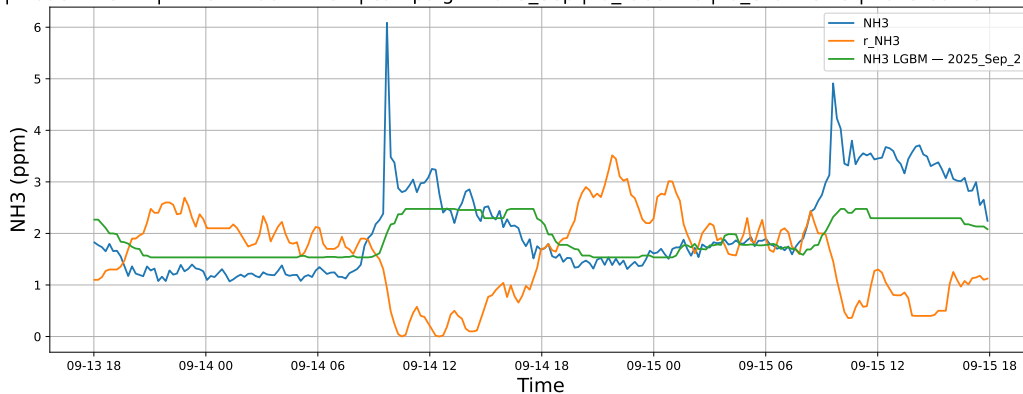


3.1.8.2 Time window : 48

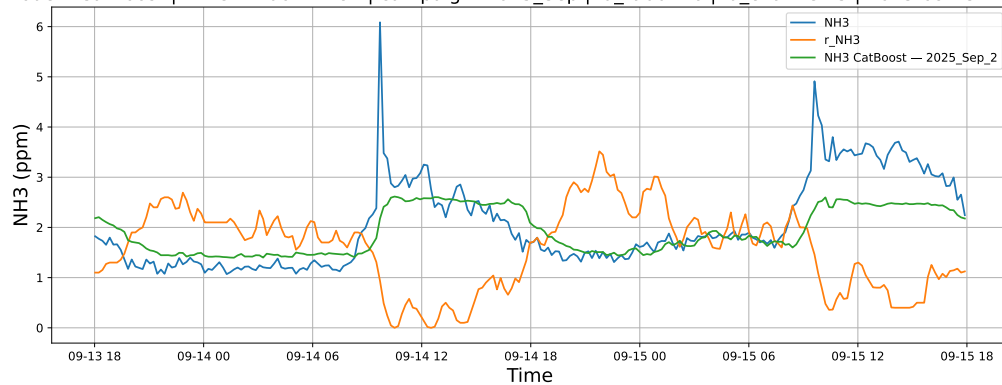
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

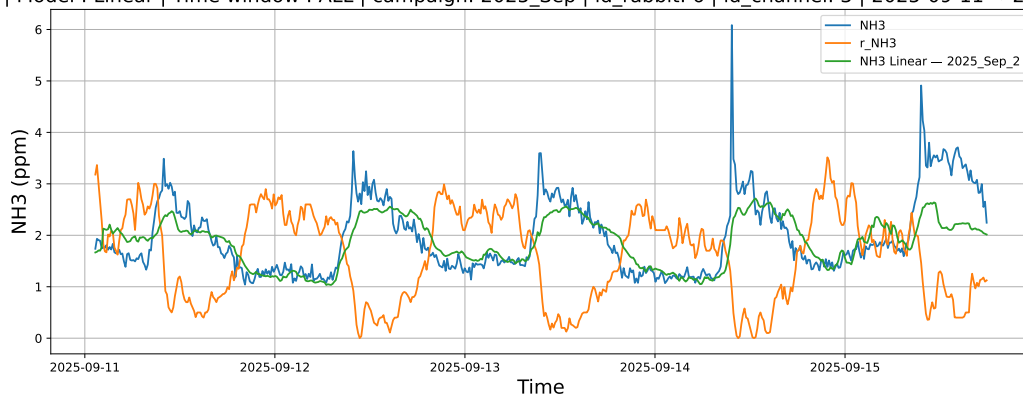


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

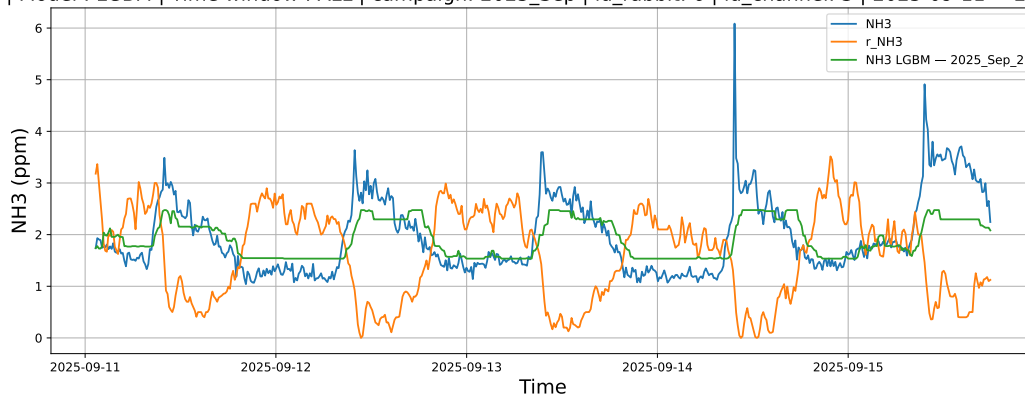


3.1.8.3 Time window : ALL

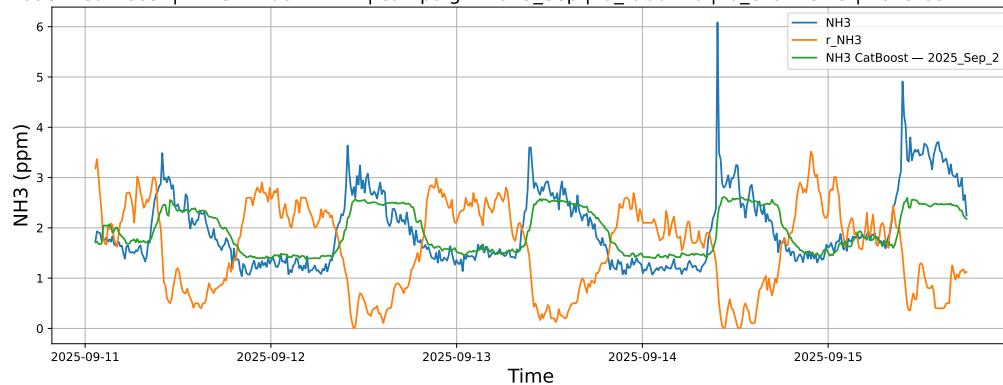
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15



NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

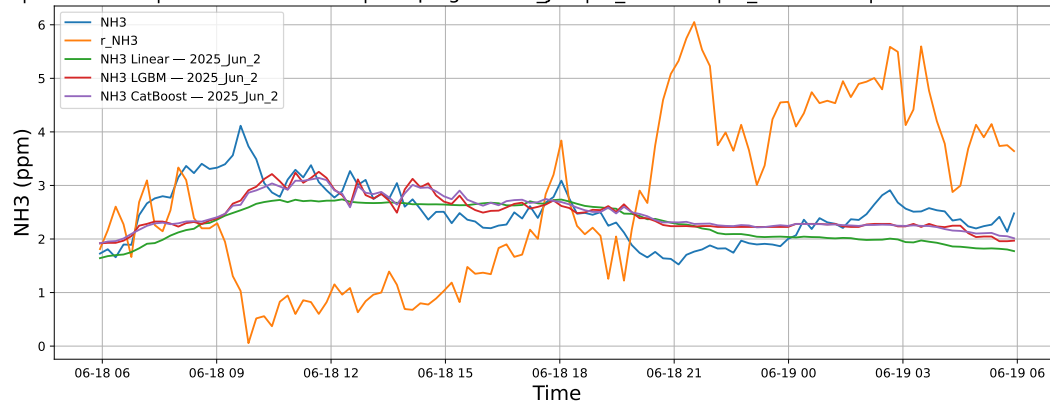


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

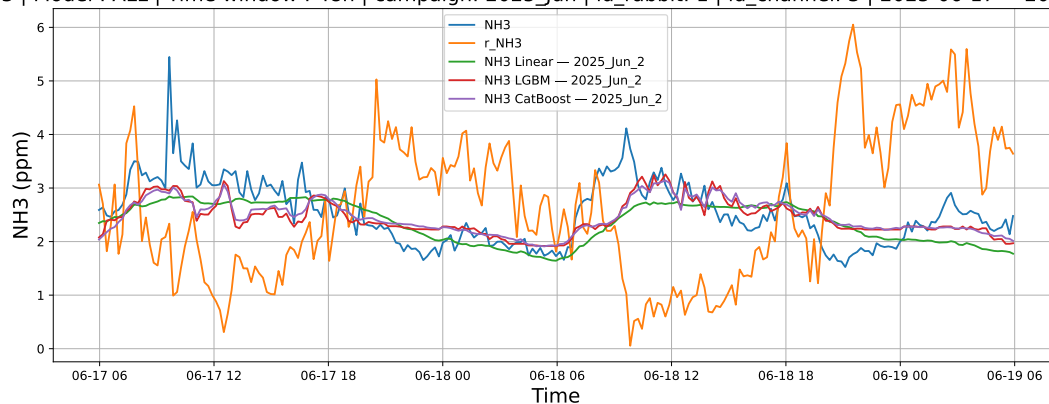
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



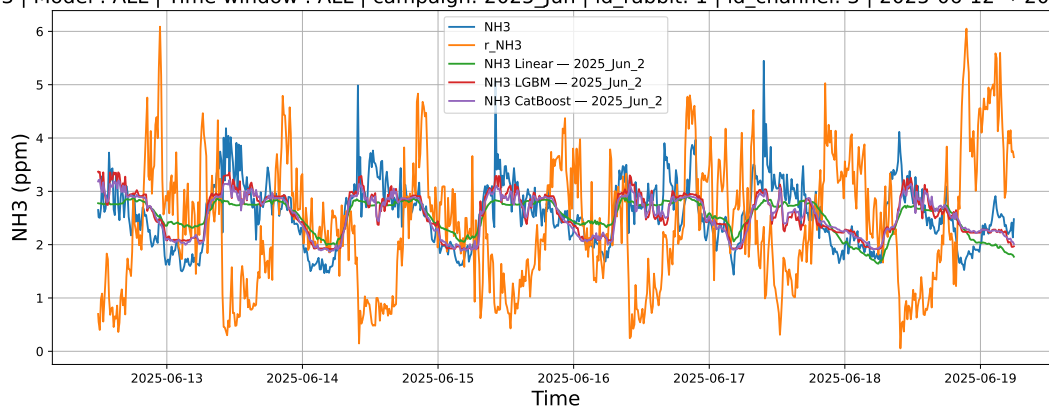
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



3.2.1.3 Time window : ALL

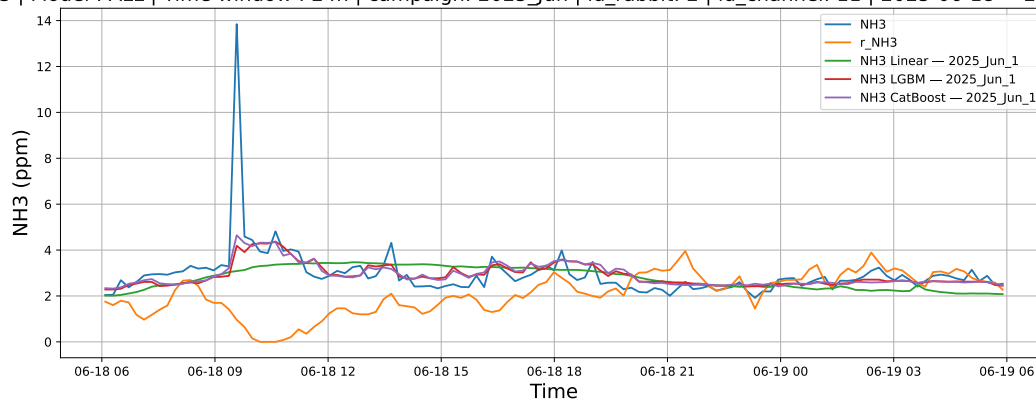
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19



3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

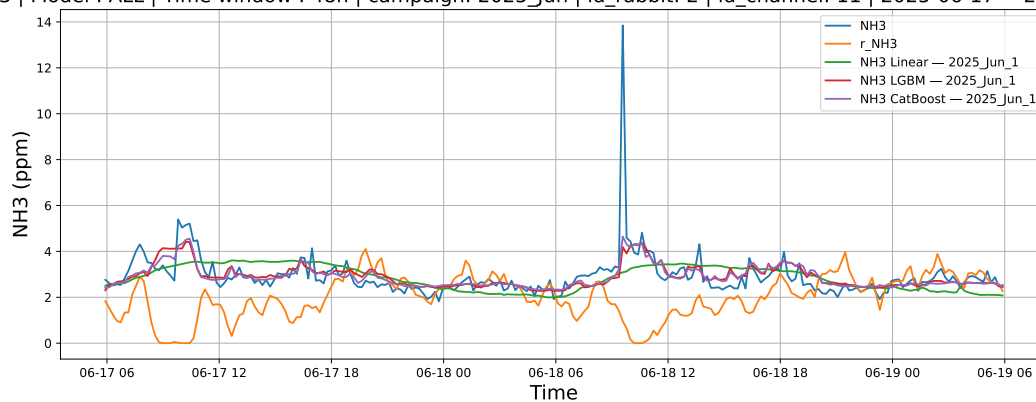
3.2.2.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-18 → 2025-06-19



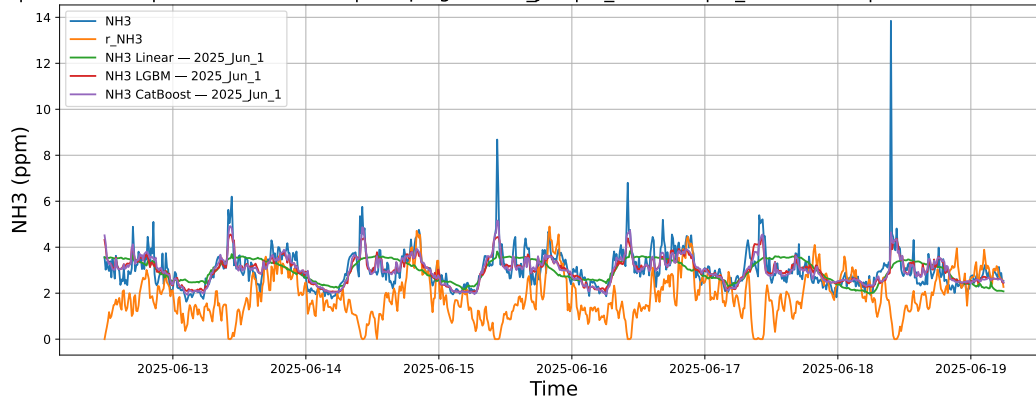
3.2.2.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



3.2.2.3 Time window : ALL

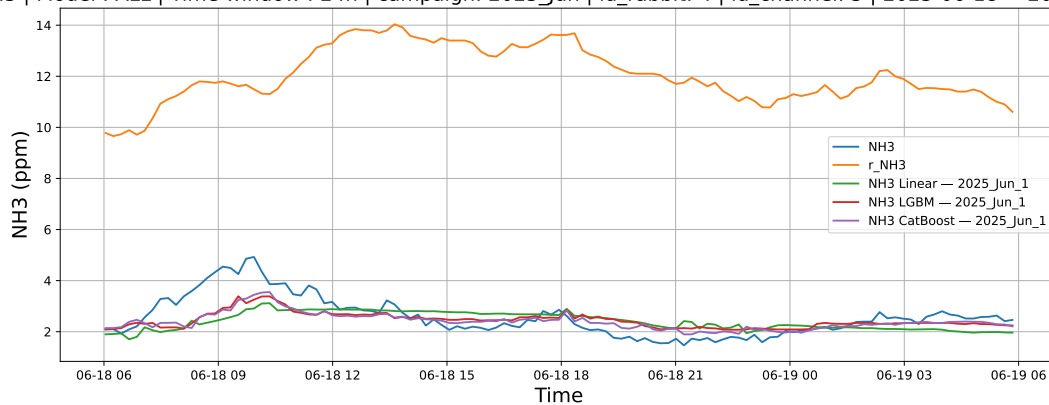
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

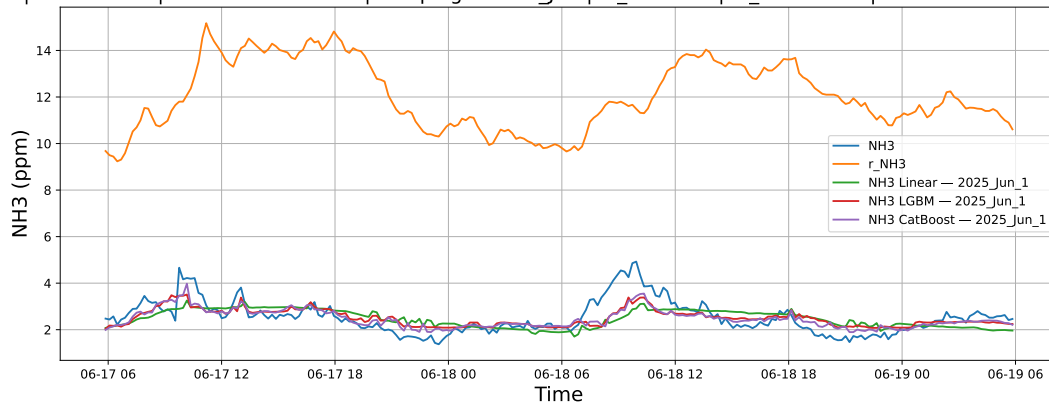
3.2.3.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-18 → 2025-06-19



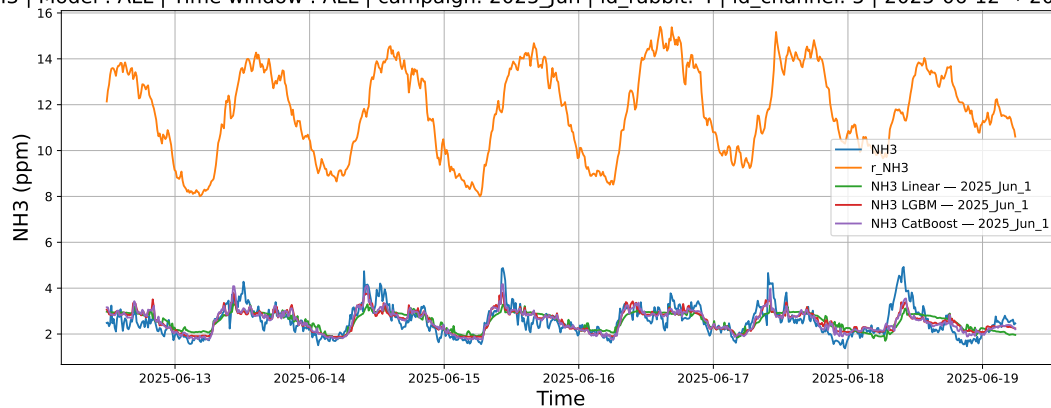
3.2.3.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-17 → 2025-06-19



3.2.3.3 Time window : ALL

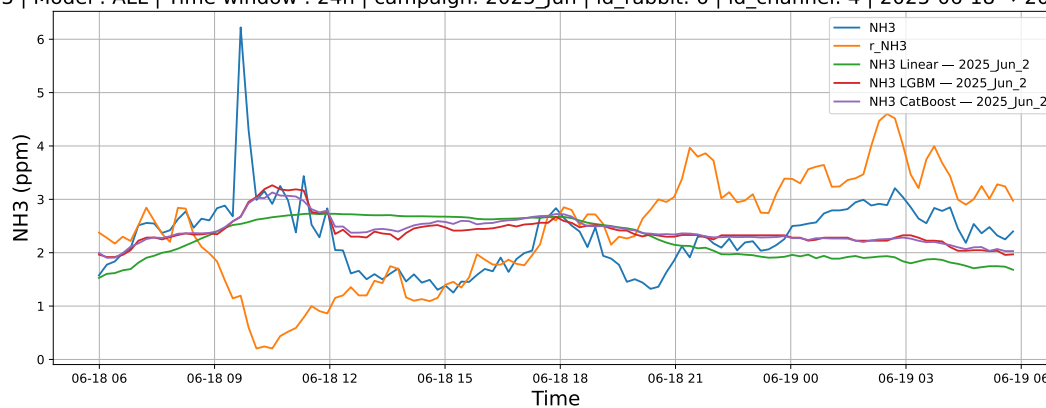
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 4 | id_channel: 5 | 2025-06-12 → 2025-06-19



3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

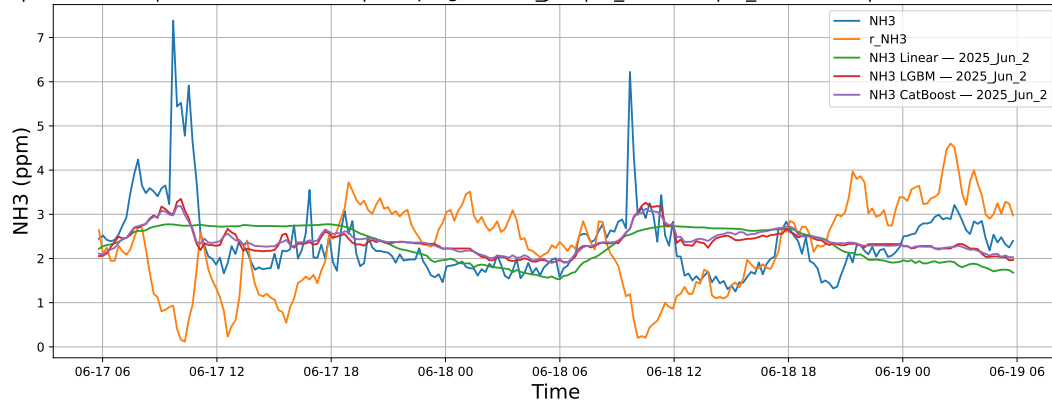
3.2.4.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



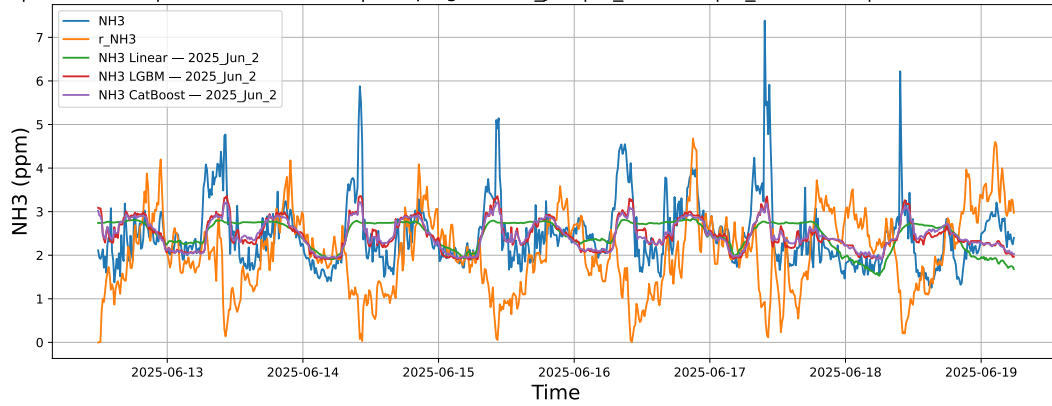
3.2.4.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



3.2.4.3 Time window : ALL

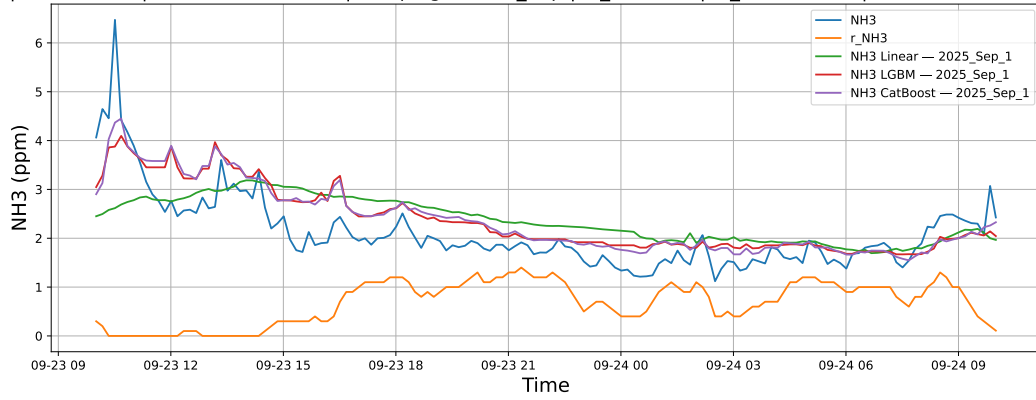
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

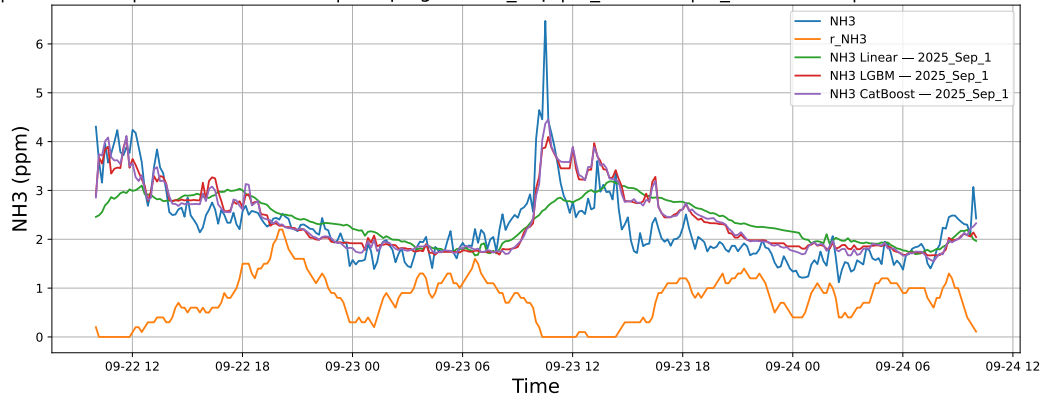
3.2.5.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



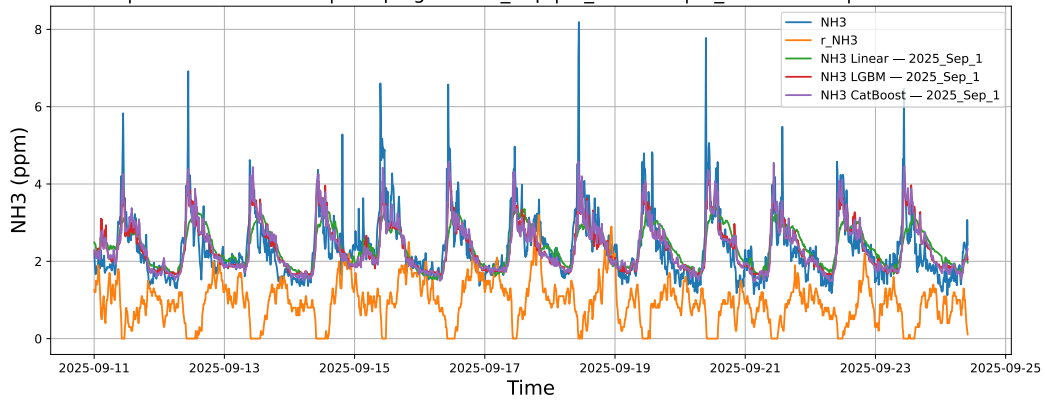
3.2.5.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-22 → 2025-09-24



3.2.5.3 Time window : ALL

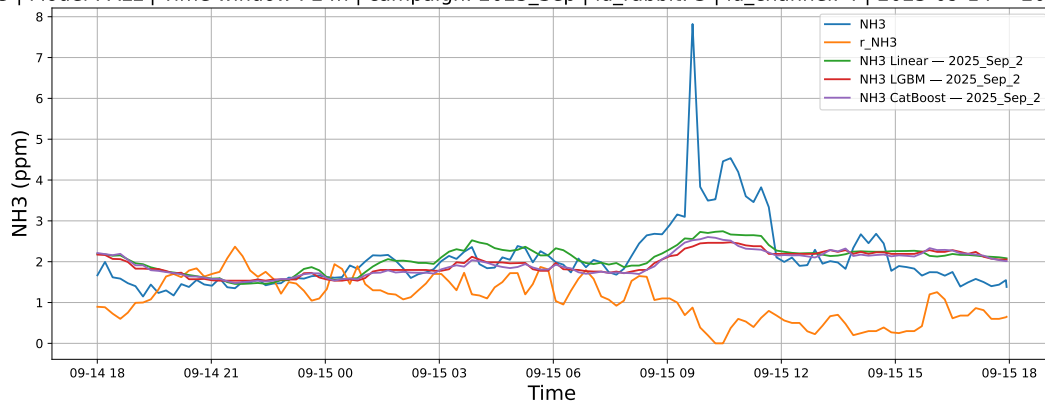
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

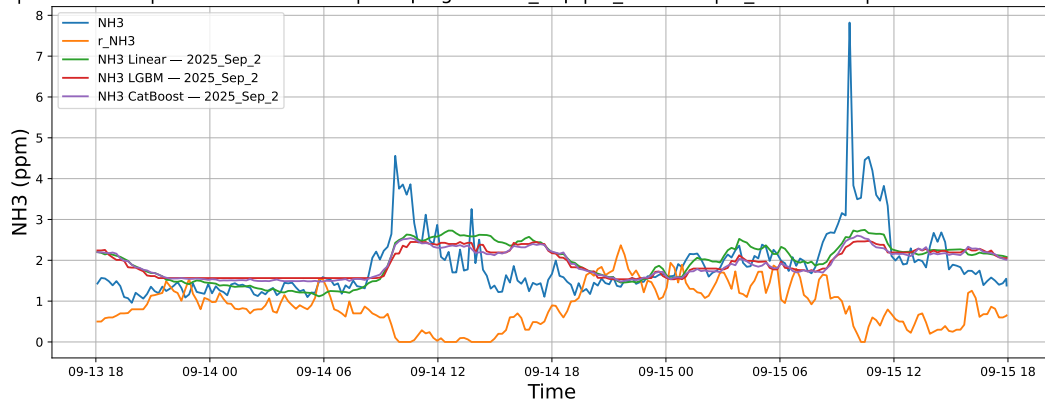
3.2.6.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-14 → 2025-09-15



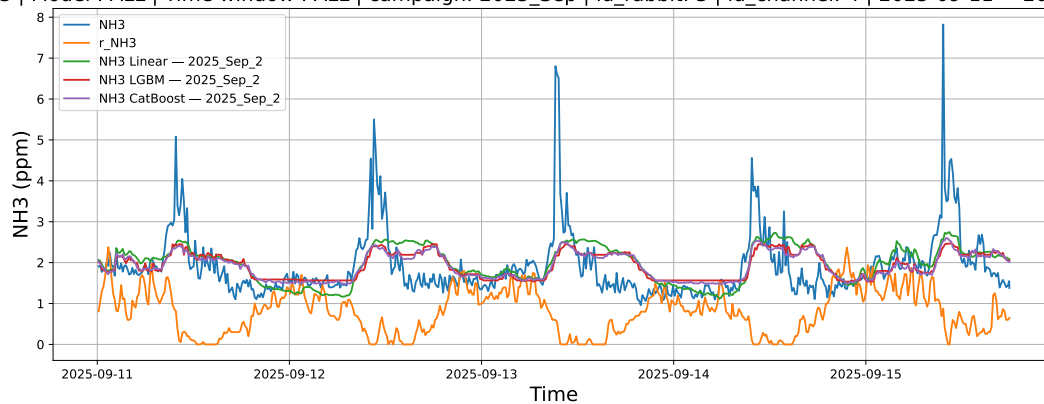
3.2.6.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-13 → 2025-09-15



3.2.6.3 Time window : ALL

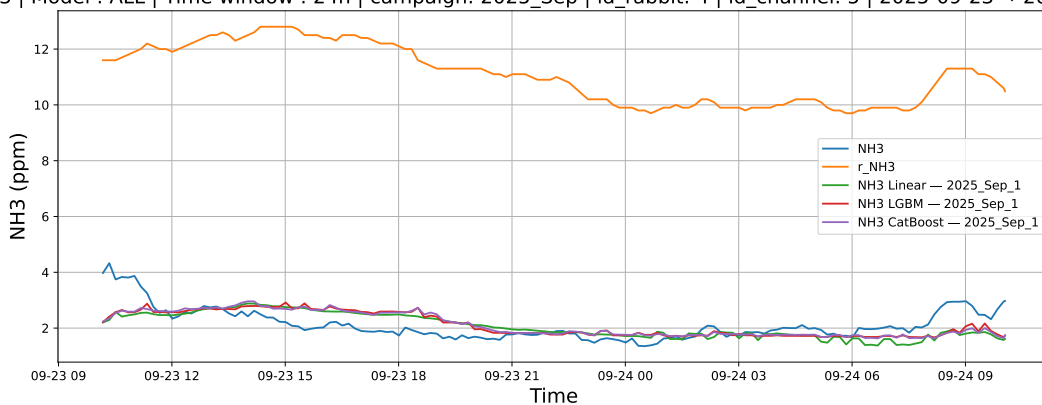
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 3 | id_channel: 4 | 2025-09-11 → 2025-09-15



3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

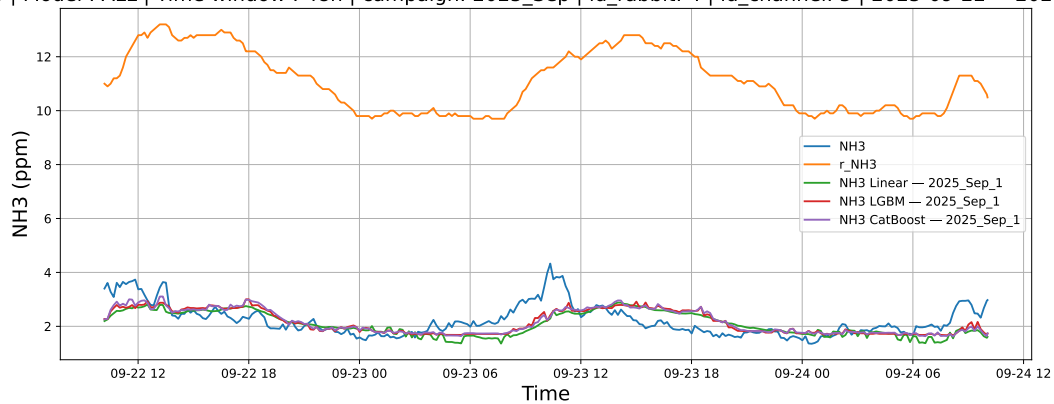
3.2.7.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-23 → 2025-09-24



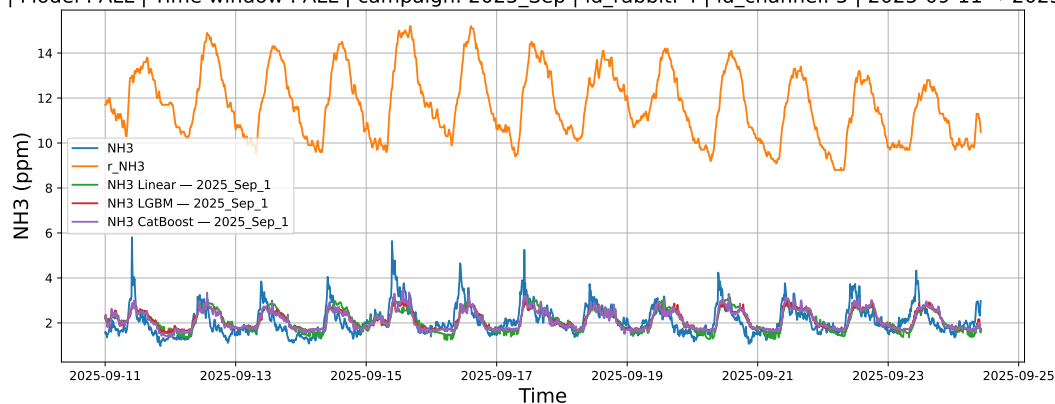
3.2.7.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.2.7.3 Time window : ALL

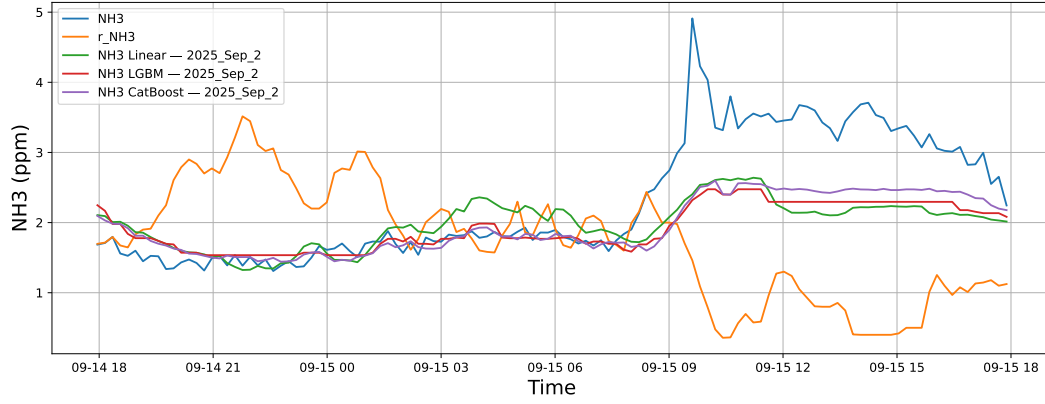
NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-11 → 2025-09-24



3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

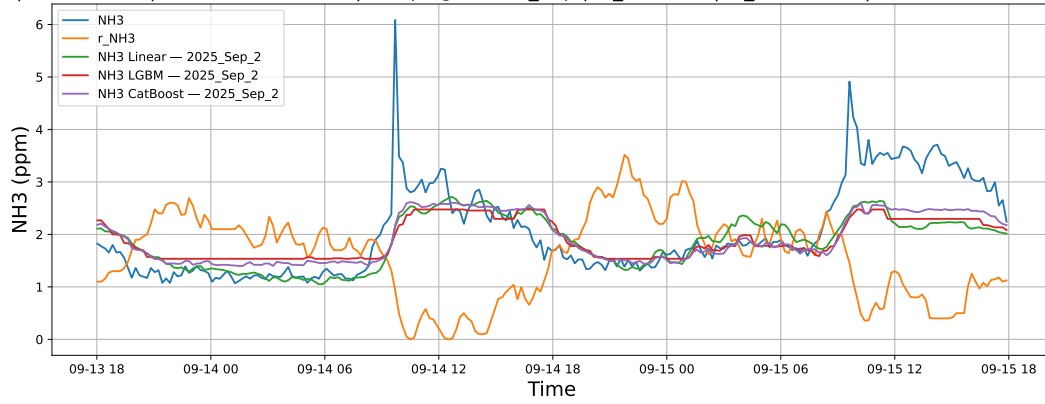
3.2.8.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-14 → 2025-09-15



3.2.8.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15

